



Università degli Studi di Padova

DIPARTIMENTO DI MATEMATICA
Corso di Laurea Magistrale in Matematica

Smooth solutions for the dyadic model on a tree

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Introduction

A classical scheme used to explain energy cascade in turbulence is based on the picture of the fluid as composed of eddies of various sizes. Larger eddies split into smaller ones because of dynamical instabilities and transfer their kinetic energy from their scale to the one of the smaller eddies. One can think of a tree-like structure where nodes are eddies; any substructure father-offsprings, where we denote the father by $j \in J$ (the set of nodes) and the set of offspring by O_j , corresponds to an eddy j and the set O_j of smaller eddies produced by j by instability.

In the simplest possible picture, eddies belong to specified discrete levels, which we call *generations*: level 0 is made of the largest eddy, level 1 of the eddies produced by level zero, and so on. The generation of eddy j may be denoted by $|j|$. Denote also the father of eddy j by $\bar{j}$.

We associate to any eddy j a non-negative intensity $X_j(t)$, at time t , such that the kinetic energy of eddy j is $X_j^2(t)$. We relate intensities by a differential rule, so that the intensity of eddy j increases because of a flux of energy from $\bar{j}$ to j and decreases because of the viscosity and a flux of energy from j to set O_j . We choose the rule

$$\dot{X}_j = -\nu d_j X_j + c_j X_{\bar{j}}^2 - \sum_{k \in O_j} c_k X_j X_k \quad (0.1)$$

where the coefficients c_j are positive.

This model has been introduced by Katz and Pavlovic in [9] as a simplified description of Euler equations and it enjoys the same features of the equations, such as the formal energy conservation, while having a simpler mathematical structure. The coefficients $c_j \sim 2^{\beta|j|}$ represent in our model the speed of the energy flow from an eddy to its children, and β is an approximation, averaged in time and space, of the rate of this speed, while $d_j \sim 2^{2|j|}$

The tree dyadic model is a more structured version of the so called dyadic model of turbulence. The latter is based on variables X_n which represent a cumulative intensity of shell n , and follow the equations

$$\dot{X}_n = -\nu d_n X_n + c_n X_{n-1}^2 - c_{n+1} X_n X_{n+1} \quad (0.2)$$

while in the former the shell n is described by a set of variables, all X_j s with $|j| = n$, the different intensities of eddies of generation n .

In chapter 1, we study the standard dyadic model and we rediscover the results presented in [4] about the regularity of solutions for positive initial conditions, which is also sufficient for the solution to remain positive at all times. Our main focus lies on the invariant region lemma, which allows us to give a boundness condition for X_n after a suitable change of variables.

In 1.2 we show some sufficient condition to determine the uniqueness of the solution, which is guaranteed in our case. We proceed to show the smoothness of the solution via the invariant region lemma and a standard argument based on the contraction mapping theorem.

In chapter 2, we consider the tree dyadic model and we aim to prove the same regularity results. If N^* is the cardinality of O_j , we assume N^* is the same for each $j \in J$ and we prove the invariant region lemma under different hypothesis about N^* ; we also show why we cannot apply the same lemma if the cardinality of O_j is no more constant through the eddies.

In 2.2 we consider a slightly different version of 0.1, making the model more complete and at the same time more difficult to study. We introduce some coefficients $d_k \in [0, 1]$ which are weights to denote which percentage of the possible energy flux between j and $k \in O_j$ is actually going through. The main result of this thesis is the generalization of the invariant region lemma: since we could not deal with the general case, we assumed the weights $d_k \neq 1$ to be all the same value d and we prove the lemma for $d \in [\frac{5}{6}, 1]$.

In 2.3 we proceed to show the smoothness of the solution via the new invariant region lemma and a standard argument based on the contraction mapping theorem, which is adapted to the new formulation of the problem.

Capitolo 1

Smooth solutions for the dyadic model

We consider the following problem. which is called the viscous problem.
Let $\nu > 0$ and $\beta > 0$:

$$\begin{cases} \dot{X}_n = -\nu\lambda_n^2 X_n + \lambda_{n-1}^\beta X_{n-1}^2 - \lambda_n^\beta X_n X_{n+1} \\ X_n(0) = x_n \end{cases} \quad \forall n \geq 1, \forall t \geq 0 \quad (1.1)$$

where $X_0 = 0$, $\lambda_0 = 0$, $\lambda_n = \lambda^n$ and in particular $\lambda = 2$. It may look like a strange notation but later the benefits will be clear. We assume that $x_n \geq 0$: by the method of variation of parameters, this implies that the solution will be positive at all times t .

The parameter β measures the strength of the nonlinearity compared to the dissipative term: we are interested in the range of values $[2, \frac{5}{2}]$.

1.1 The invariant region lemma

Let $\{X_n\}_{n \geq 1}$ be a solution to problem 1.1 on a time interval $[0, T]$. We apply the following change of variables:

$$Y_n = \lambda_n^{\beta-2+\varepsilon} X_n \quad (1.2)$$

where $\varepsilon > 0$ is a positive number whose choice will be made later.

By substitution and some simple calculations we can find the dynamics of $\{Y_n\}$:

$$\begin{cases} \dot{Y}_n = -\nu\lambda_n^2 Y_n + \lambda_n^{2-\varepsilon} \lambda^{\beta-4+2\varepsilon} Y_{n-1}^2 - \lambda_n^{2-\varepsilon} \lambda^{2-\beta-\varepsilon} Y_n Y_{n+1} \\ Y_n(0) = y_n \end{cases} \quad (1.3)$$

for $n \geq 1$ and $t \in [0, T]$. Clearly y_n is the image of x_n after changing variables. Consider now a finite dimensional version of the equations for Y : for every $N \geq 1$ we will denote with $\{Y_n^{(N)}\}_{n \leq N}$ the solution to

$$\begin{cases} \dot{Y}_n^{(N)} = -\nu \lambda_n^2 Y_n^{(N)} + \lambda_n^{2-\varepsilon} \lambda^{\beta-4+2\varepsilon} \left(Y_{n-1}^{(N)}\right)^2 - \lambda_n^{2-\varepsilon} \lambda^{2-\beta-\varepsilon} Y_n^{(N)} Y_{n+1}^{(N)} \\ Y_n(0) = y_n \end{cases}$$

for $1 \leq n \leq N$, i.e. we are taking into consideration just the first N equations. For simplicity, we set $Y_0^{(N)} = 0$ and $Y_{N+1}^{(N)} = Y_N^{(N)}$, so the border equations still have that form.

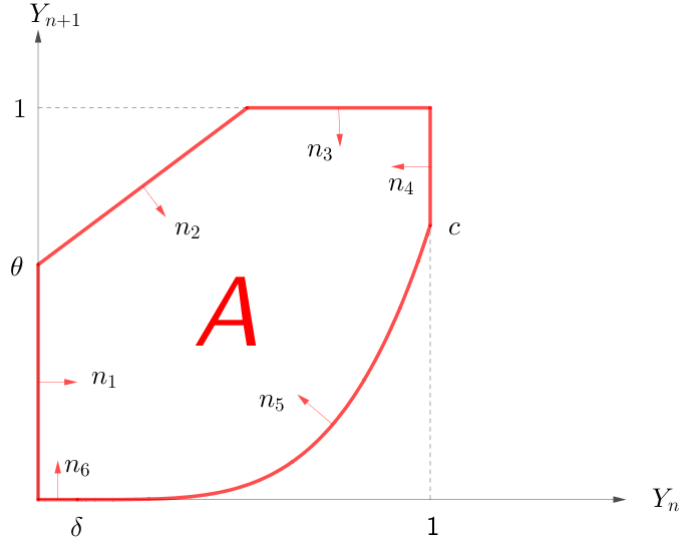
Now we introduce the following region contained in $\mathbb{R}^2$:

$$A := \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, h(x) < y < g(x)\}$$

where

$$g(x) = \min\{mx + \theta, 1\}, \quad h(x) = \begin{cases} 0 & x \leq \delta \\ c \left(\frac{x - \delta}{1 - \delta}\right)^{\lambda^2} & x > \delta \end{cases}$$

are the functions providing the lower and upper bound of A . The only conditions on parameters δ, θ is that $\delta \leq \theta < 1$ so the little square $[0, \delta]^2$ is all inside A .



Lemma 1.4. (*Invariant region*)

There exists $\delta \in (0, 1)$, $c \in (0, 1)$, $\theta \in (\delta, 1)$, $m > 1 - \theta$ and $\varepsilon_0 > 0$ such that: for every $\beta \in (2, \frac{5}{2})$, $\nu \geq 0$ and $\varepsilon \in (0, \varepsilon_0)$ the following statement is true. If $N \geq 1$ and if $(y_n, y_{n+1}) \in A$ for all $n \leq N$,

$$(Y_n^{(N)}(t), Y_{n+1}^{(N)}(t)) \in A \quad \forall n \leq N, \forall t \geq 0$$

where $\{Y_n^{(N)}\}$ is the same as before.

Dim.: for the sake of elegance, we drop the superscript (N) along the proof. It is sufficient to show that the time derivative of (Y_n, Y_{n+1}) points inward on the border of A when $(Y_n, Y_{n+1}) \in A$. We check by computing the scalar product of the inward normal on the border of A with the following vector field $\mathcal{B}$:

$$\begin{pmatrix} \dot{Y}_n \\ \dot{Y}_{n+1} \end{pmatrix} = \nu \lambda_n^2 \begin{pmatrix} -Y_n \\ -\lambda^2 Y_{n+1} \end{pmatrix} + \lambda_n^{2-\varepsilon} \lambda^{\beta-4+2\varepsilon} \begin{pmatrix} Y_{n-1}^2 - \lambda^{6-2\beta-3\varepsilon} Y_n Y_{n+1} \\ \lambda^{2-\varepsilon} (Y_n^2 - \lambda^{6-2\beta-3\varepsilon} Y_{n+1} Y_{n+2}) \end{pmatrix}$$

and verifying it is positive when $(Y_n, Y_{n+1}) \in A$ for all $n = 1, \dots, N$. Since A is a convex set, we can compute separately the scalar product with just the viscous term and just the inviscid one.

We denote the inward normals as in figure, so their expressions are the following:

$$\begin{aligned} \vec{n}_1 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{n}_3 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \vec{n}_4 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \vec{n}_6 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \vec{n}_2 &= \begin{pmatrix} g'(x) \\ -1 \end{pmatrix}, \quad \vec{n}_5 = \begin{pmatrix} -h'(x) \\ 1 \end{pmatrix} \end{aligned}$$

We start with the viscous term, which will be denoted as $\mathcal{B}_v$. Note that there is no need to consider the positive constant $\nu \lambda_n^2$.

The scalar products are evaluated on the respective pieces of the border of A :

$$\begin{aligned} \mathcal{B}_v \cdot \vec{n}_1 &= -Y_n = 0, & \mathcal{B}_v \cdot \vec{n}_3 &= \lambda^2 Y_{n+1} = \lambda^2 \\ \mathcal{B}_v \cdot \vec{n}_4 &= Y_n = 1, & \mathcal{B}_v \cdot \vec{n}_6 &= -\lambda^2 Y_{n+1} = 0 \end{aligned}$$

These are positive. To compute the remaining scalar products, we set $x = Y_n$. Along g , $Y_{n+1} = g(x)$:

$$\mathcal{B}_v \cdot \vec{n}_2 = -xg'(x) + \lambda^2 Y_{n+1} = -xg'(x) + \lambda^2 g(x) = mx(-1 + \lambda^2) + \lambda^2 \theta > 0$$

Along h , $Y_{n+1} = h(x)$:

$$\begin{aligned} \mathcal{B}_v \cdot \vec{n}_5 &= xh'(x) - \lambda^2 Y_{n+1} = xh'(x) - \lambda^2 h(x) = x \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} - \lambda^2 h(x) \\ &\geq \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} - \lambda^2 h(x) = \frac{\lambda^2}{1-\delta} h(x) - \lambda^2 h(x) = \lambda^2 h \frac{\delta}{1-\delta} > 0 \end{aligned}$$

We proceed with the inviscid term, which will be denoted as $\mathcal{B}_i$. Just as before, there is no need to consider the positive constant.

We set again $x = Y_n$ and for simplicity we also set $\gamma = 6v - 2\beta - 3\varepsilon$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_1 &= Y_{n-1}^2 - \lambda^\gamma x Y_{n+1} = Y_{n-1}^2 \geq 0 \\ \mathcal{B}_i \cdot \vec{n}_6 &= \lambda^{2-\varepsilon} (x^2 - \lambda^\gamma Y_{n+1} Y_{n+2}) = \lambda^{2-\varepsilon} x^2 \geq 0 \end{aligned}$$

Consider the piece of the border of A corresponding to $\vec{n}_3$. Here $x \leq 1$, $Y_{n+1} = 1$ and since $(Y_{n+1}, Y_{n+2}) \in A$ we have $Y_{n+2} \geq c$. Let us exploit these inequalities:

$$\mathcal{B}_i \cdot \vec{n}_3 = \lambda^{2-\varepsilon} (\lambda^\gamma Y_{n+1} Y_{n+2} - x^2) \geq \lambda^{2-\varepsilon} (\lambda^\gamma c - 1)$$

Consider the piece of the border of A corresponding to $\vec{n}_4$. Here $x = 1$, $Y_{n-1} \leq 1$ and since $(Y_n, Y_{n+1}) \in A$ we have $Y_{n+1} \geq c$. Let us exploit these inequalities:

$$\mathcal{B}_i \cdot \vec{n}_4 = \lambda^\gamma x Y_{n+1} - Y_{n-1}^2 \geq (\lambda^\gamma c - 1)$$

We choose c in order to have $\lambda^\gamma c = 1$, so the previous are positive. Note that

$$\begin{aligned} \beta \geq 2 &\Rightarrow \gamma \leq 2 - 3\varepsilon \\ \beta \leq \frac{5}{2} &\Rightarrow \gamma \geq 1 - 3\varepsilon \end{aligned}$$

Consider the piece of the border of A corresponding to $\vec{n}_2$. Here $Y_{n+1} = g(x)$ and since $(Y_{n+1}, Y_{n+2}) \in A$ we have $Y_{n+2} \geq h(Y_{n+1}) = h(g(x))$. Let us exploit this and the upper bound for γ :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_2 &= g'(x) (Y_{n-1}^2 - \lambda^\gamma x Y_{n+1}) - \lambda^{2-\varepsilon} (x^2 - \lambda^\gamma Y_{n+1} Y_{n+2}) \\ &\geq m(-\lambda^\gamma x Y_{n+1}) - \lambda^{2-\varepsilon} [x^2 - \lambda^\gamma g(x) h(g(x))] \\ &= -\lambda^\gamma m x (mx + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (mx + \theta) \lambda^\gamma c \left(\frac{mx + \theta - \delta}{1 - \delta} \right)^{\lambda^2} \\ &\geq -\lambda^{2-3\varepsilon} m x (mx + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (mx + \theta) \left(\frac{mx + \theta - \delta}{1 - \delta} \right)^{\lambda^2} \end{aligned}$$

Consider the piece of the border of A corresponding to $\vec{n}_5$. Here $Y_{n+1} = h(x)$ and since $(Y_{n+1}, Y_{n+2}) \in A$ we have $Y_{n+2} \leq g(Y_{n+1}) = g(h(x))$. Let us exploit this and the lower bound for γ :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &= \lambda^{2-\varepsilon} (x^2 - \lambda^\gamma Y_{n+1} Y_{n+2}) - h'(x) (Y_{n-1}^2 - \lambda^\gamma x Y_{n+1}) \\ &\geq \lambda^{2-\varepsilon} [x^2 - \lambda^\gamma h(x) g(h(x))] - h'(x) \left[1 - x (\lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \\ &= \lambda^{2-\varepsilon} [x^2 - \lambda^\gamma h(x) (mh(x) + \theta)] - h'(x) \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \\ &= \lambda^{2-\varepsilon} \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - m \lambda^{2-\varepsilon} \lambda^{-\gamma} \lambda^{2\gamma} h^2 - h' \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \\ &= \lambda^{2-\varepsilon} \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - m \lambda^{2-\varepsilon} \lambda^{-\gamma} \left(\frac{x - \delta}{1 - \delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^2 c}{1 - \delta} \lambda^\gamma \lambda^{-\gamma} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \\ &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - m \lambda^{2-\varepsilon} \lambda^{3\varepsilon - 1} \left(\frac{x - \delta}{1 - \delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^2}{1 - \delta} \lambda^{3\varepsilon - 1} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \end{aligned}$$

$$\begin{aligned} &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] - m\lambda^{1+2\varepsilon} \left(\frac{x-\delta}{1-\delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^{1+3\varepsilon}}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} \left[1 - x \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] \end{aligned}$$

Let $\psi_2(x)$ and $\psi_5(x)$ be the lower bounds we just obtained for $\mathcal{B}_i \cdot \vec{n}_2$ and $\mathcal{B}_i \cdot \vec{n}_5$, respectively, when $\varepsilon = 0$:

$$\begin{aligned} \psi_2(x) &= -\lambda^2 m x (mx + \theta) - \lambda^2 x^2 + \lambda^2 (mx + \theta) \left(\frac{mx + \theta - \delta}{1-\delta} \right)^{\lambda^2} \\ \psi_5(x) &= \lambda^2 \left[x^2 - \theta \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] - m\lambda \left(\frac{x-\delta}{1-\delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} \left[1 - x \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] \end{aligned}$$

It is sufficient to show that both ψ_2 and ψ_5 have positive minimal values: indeed, continuity implies that it will be true also for small values of ε , i.e. $\varepsilon \in [0, \varepsilon_0]$. The minimal values can be computed explicitly once we fix all parameters: by choosing $\delta = \frac{1}{10}$, $\theta = \frac{3}{5}$ and $m = \frac{3}{4}$, they are both positive. $\square$

1.2 Uniqueness and regularity in viscous case

We will consider the space of square summable, real sequences:

$$H = \left\{ x = \{x_n\}_{n \geq 1} \subseteq \mathbb{R} : \|x\|_H^2 := \sum_{n=1}^{\infty} x_n^2 < \infty \right\}$$

A weak solution to problem 1.1 on $[0, T]$ is a sequence of functions $X = \{X_n\}_{n \geq 1}$ such that $X_n \in C^1([0, T]; \mathbb{R})$ for every $n \geq 1$ and the equations in 1.1 hold.

A positive solution X is a weak solution having all components X_n positive.

By the method of variation of parameters:

$$\begin{aligned} X_n(t) &= X_n(0) \exp \left[- \int_0^t (\nu \lambda_n^2 + \lambda_n^\beta X_{n+1}(r)) dr \right] \\ &\quad + \int_0^t \lambda_{n-1}^\beta X_{n-1}^2(s) \exp \left[- \int_s^t (\nu \lambda_n^2 + \lambda_n^\beta X_{n+1}(r)) dr \right] ds \end{aligned}$$

This implies that if the initial condition $X(0)$ is positive, i.e. $X_n(0) \geq 0$ for all $n \geq 1$, then X_n stays positive for all t which means X is a positive solution.

A Leray-Hopf solution is a weak solution X with values in H and satisfying the following, called energy inequality,

$$\|X(t)\|_H^2 + 2\nu \int_s^t \sum_{n=1}^{\infty} (\lambda_n X_n(r))^2 dr \leq \|X(s)\|_H^2$$

for a.e. $s \geq 0$ and $t > s$.

In [5] it has been proved that if the initial condition $X(0)$ is positive, weak solutions are Leray-Hopf solutions and the energy inequality holds for all $t > s$.

Proposition 1.5. (*Uniqueness*)

Let $X = \{X_n\}_{n \geq 1}$ be a positive solution to 1.1 on $[0, T]$ with initial condition $X(0) \in H$ and such that

$$\sup_{t \in [0, T], n \geq 1} [\lambda_n^{\beta-3} X_n(t)] \leq C$$

for some $C \in \mathbb{R}$. Then X is the unique solution with initial condition $X(0)$.

Dim.: let $Y = \{Y_n\}_{n \geq 1}$ be another solution with the same initial condition. We set $Z_n = Y_n - X_n$, $\bar{W}_n = Y_n + X_n$ and let us compute the dynamics for Z :

$$\begin{aligned} \dot{Z}_n &= -\nu \lambda_n^2 Z_n + \lambda_{n-1}^\beta Z_{n-1} W_{n-1} - \frac{1}{2} \lambda_n^\beta (Z_n W_{n+1} + Z_{n+1} W_n) \\ \dot{Z}_n^2 &= -2\nu \lambda_n^2 Z_n^2 + 2\lambda_{n-1}^\beta Z_n Z_{n-1} W_{n-1} - \lambda_n^\beta Z_n^2 W_{n+1} - \lambda_n^\beta Z_{n+1} Z_n W_n \\ \frac{1}{2^n} \dot{Z}_n^2 &= -2\nu \frac{\lambda_n^2}{2^n} Z_n^2 + \frac{\lambda_{n-1}^\beta}{2^{n-1}} Z_n Z_{n-1} W_{n-1} - \frac{\lambda_n^\beta}{2^n} Z_n^2 W_{n+1} - \frac{\lambda_n^\beta}{2^n} Z_{n+1} Z_n W_n \end{aligned}$$

We used $\dot{Z}^2 = 2Z\dot{Z}$. Note that $Z_0 = 0$ since X and Y share the initial condition. Now fix $N \geq 1$ and set the following:

$$\psi_N(t) = \sum_{n=1}^N \frac{1}{2^n} Z_n^2$$

We can compute the time derivative of ψ_N by a telescopicity argument in the previous equation:

$$\begin{aligned} \frac{d}{dt} \psi_N(t) &= -2\nu \sum_{n=1}^N \frac{\lambda_n^2}{2^n} Z_n^2 - \sum_{n=1}^N \frac{\lambda_n^\beta}{2^n} Z_n^2 W_{n+1} \\ &\quad + \sum_{n=1}^N \frac{\lambda_{n-1}^\beta}{2^{n-1}} Z_n Z_{n-1} W_{n-1} - \sum_{N=1}^N \frac{\lambda_n^\beta}{2^n} Z_{n+1} Z_n W_n \end{aligned}$$

It is clear that the terms in the second row cancel each other. Since we want an upper bound for $\frac{d}{dt} \psi_N(t)$, we do not consider the first two sums. It becomes:

$$\begin{aligned} \frac{d}{dt} \psi_N(t) &\leq \lambda_0^\beta Z_1 Z_0 W_0 - \frac{\lambda_N^\beta}{2^N} Z_{N+1} Z_N W_N = -\frac{\lambda_N^\beta}{2^N} Z_{N+1} Z_N W_N \\ &\leq -\lambda_N^{\beta-1} (Y_N^2 Y_{N+1} + X_N^2 X_{N+1} - X_{N+1} Y_N^2 - Y_{N+1} X_N^2) \end{aligned}$$

We will just throw away negative terms and use the constant C :

$$\begin{aligned} \frac{d}{dt} \psi_N(t) &\leq \lambda_N^{\beta-1} (X_{N+1} Y_N^2 + Y_{N+1} X_N^2) = \lambda_N^2 \lambda_N^{\beta-3} (X_{N+1} Y_N^2 + Y_{N+1} X_N^2) \\ &\leq \lambda_N^2 (\lambda_N^{\beta-3} Y_N) X_{N+1} Y_N + \lambda_N^2 (\lambda_N^{\beta-3} X_N) Y_{N+1} X_N \\ &\leq \lambda_N^2 C (X_{N+1} Y_N + Y_{N+1} X_N) \leq \lambda_N^2 C \left(\frac{X_{N+1}^2 + Y_N^2}{2} + \frac{Y_{N+1}^2 + X_N^2}{2} \right) \end{aligned}$$

where the last inequality comes from a square of a binomial being positive. By applying the comparison theorem for differential equations, since $\psi_N(0) = 0$:

$$\psi_N(t) \leq \int_0^t \frac{\lambda_N^2 C}{2} (X_{N+1}^2 + Y_N^2 + Y_{N+1}^2 + X_N^2) ds$$

Since X and Y are Leray-Hopf solutions, the right hand side tends to 0 as $N \rightarrow \infty$. Hence $\psi_N(t) \rightarrow 0$ for any $t \geq 0$. Since ψ_n is an increasing function of n , this implies $\psi_n(t) = 0$ for all $n \geq 1$, thus we have $X_n(t) = Y_n(t)$ as desired.

Corollary 1.6. *If $\beta \leq 3$ in the formulation of problem 1.1, there is a unique solution for any positive initial condition in H .*

Dim.: we check the boundness condition to apply the previous proposition. Note that $\lambda_n^{\beta-3} \leq 1$ since $\beta \leq 3$. Hence the following inequalities

$$\lambda_n^{\beta-3} X_n(t) \leq X_n(t) \leq |X_n(t)| \leq \|X(t)\|_H < \infty$$

hold for all $n \geq 1$ and $t \geq 0$. The boundness condition on the sup is verified. $\square$

We will now introduce the main theorem regarding uniqueness and regularity in the viscous case. In the proof we will make use of Lemma 1.4.

Theorem 1.7. *Let $\beta \in [2, \frac{5}{2}]$. Set*

$$\mathcal{D}^\infty = \left\{ x = \{x_n\}_{n \geq 1} : \sup_{n \geq 1} \lambda_n^\gamma |x_n| < \infty \text{ for all } \gamma > 0 \right\}$$

For any positive initial condition $X(0) \in H$, there exists a unique weak solution to problem 1.1. Moreover it is smooth, i.e. $X(t) \in \mathcal{D}^\infty$ for every $t > 0$.

Dim.: let X be the weak solution starting at $X(0) = x$. Since from the hypothesis $\beta \in [2, \frac{5}{2}]$, we know X is unique from Corollary 1.6. It is not restrictive to assume that $\sup_n \lambda_n x_n < \infty$, a smooth initial condition. Indeed, if that is not the case, we can consider a sequence $\{t_k\}$ such that

$$t_k \rightarrow 0 \quad \text{and} \quad \sup_n \lambda_n X_n(t_k) < \infty \quad \text{for all } k$$

The existence of such a sequence follows from the energy inequality:

$$\sum_{n=1}^{\infty} \int_s^t (\lambda_n X_n(r))^2 dr < \infty \quad \text{for all } t > s$$

Indeed if $\sup_n \lambda_n X_n(r) = \infty$ on a nonzero measure set, we have a contradiction. So $\sup_n \lambda_n X_n(t) < \infty$ for a.e. $t > 0$ and it allows us to choose such a sequence. Proving that $X(t) \in \mathcal{D}^\infty$ for all $t > 0$ is equivalent to proving that, for each k , $X(t) \in \mathcal{D}^\infty$ for all $t \geq t_k$. Since the latter problem has smooth initial conditions for each k , the former may have this hypothesis as well.

Let ε_0 be as in the statement of Lemma 1.4. We choose $\varepsilon = \min \{3 - \beta, \varepsilon_0\}$:

since $\beta + \varepsilon - 2 \leq 1$, we have $\lambda_n^{\beta+\varepsilon-2} x_n \leq \lambda_n x_n$.

Since $\sup_n \lambda_n x_n$ is finite, same holds for the left side. Set $\sup_n \lambda_n^{\beta+\varepsilon-2} x_n = K_0$, we define for $n \in \mathbb{N}$:

$$\bar{Y}_n(t) = \frac{\delta}{K_0} \lambda_n^{\beta+\varepsilon-2} X_n \left(\frac{\delta}{K_0} t \right)$$

with δ as in Lemma 1.4.

Since (Y_n) defined in 1.2 was a solution to problem 1.3, then $(\bar{Y}_n)$ is a solution to the same problem but with a different viscosity, namely $\bar{\nu} = \frac{\delta}{K_0} \nu$.

Now let $\bar{Y}_n^{(N)}$ be a solution to the N -dimensional version of problem 1.3. We can prove by a standard argument that it converges to $(\bar{Y}_n)$ as $N \rightarrow \infty$.

Our idea is to apply the invariant region lemma to $(\bar{Y}_n)$. We consider the finite dimensional approximations, which all share the initial conditions with $(\bar{Y}_n)$.

The following holds for all $N \geq 1$:

$$\sup_{n \leq N} \bar{Y}_n^{(N)}(0) \leq \frac{\delta}{K_0} \cdot \sup_{n \geq 1} \lambda_n^{\beta+\varepsilon-2} x_n = \frac{\delta}{K_0} \cdot K_0 = \delta$$

$$\Rightarrow (\bar{Y}_n^{(N)}(0), \bar{Y}_{n+1}^{(N)}(0)) \in [0, \delta]^2 \subset A, \text{ for } n \leq N.$$

$$\Rightarrow (\bar{Y}_n^{(N)}(t), \bar{Y}_{n+1}^{(N)}(t)) \in A, \text{ for } n \leq N, t \geq 0 \text{ by the Lemma 1.4.}$$

$$\Rightarrow \bar{Y}_n^{(N)}(t) \leq 1, \text{ for } n \leq N, t \geq 0.$$

Note the same estimate holds for $\bar{Y}_n(t)$ for $n \geq 1, t \geq 0$ by the convergence of finite dimensional approximations. So:

$$\begin{aligned} \sup_{n \geq 1} \sup_{t \geq 0} \frac{\delta}{K_0} \lambda_n^{\beta+\varepsilon-2} X_n(t) &= \sup_{n \geq 1} \sup_{t \geq 0} \bar{Y}_n \left(\frac{K_0}{\delta} t \right) \leq 1 \\ \Rightarrow \sup_{n \geq 1} \sup_{t \geq 0} \lambda_n^{\beta+\varepsilon-2} X_n(t) &\leq \frac{K_0}{\delta} \end{aligned} \tag{1.8}$$

We will prove that there is $\eta > 0$ such that $X(t) \in \mathcal{D}^\infty$, for $t \in [0, \eta]$. A similar argument will work on any interval $[t_0, t_0 + \eta]$, allowing to cover the whole set of times $t > 0$.

Let $V_n(t) = X_n(t) \exp[\nu \lambda_n t]$. We compute the dynamics for V_n :

$$\begin{cases} \dot{V}_n = -\nu (\lambda_n^2 - \lambda_n) V_n + \lambda_{n-1}^\beta V_{n-1}^2 - \lambda_n^\beta \exp[-\nu \lambda_{n+1} t] V_n V_{n+1} \\ V_n(0) = X_n(0) \end{cases}$$

Our idea is to prove smoothness of X by showing that V is uniformly bounded in n ; we will show later why the latter implies the former.

For now, let us define

$$\mathcal{V}_\varepsilon = \left\{ x = \{x_n\}_{n \geq 1} : \|x\|_\varepsilon := \sup_{n \geq 1} (\lambda_n^{\beta-2+\varepsilon} |x_n|) < \infty \right\}$$

We also define a map $\mathcal{F}$ from the space $L^\infty([0, \eta]; \mathcal{V}_\varepsilon)$ into itself, $Z \mapsto \mathcal{F}(Z)$. The components $\mathcal{F}_n(Z)$ follow:

$$\begin{aligned} \mathcal{F}_n(Z)(t) = & \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \left(\lambda_{n-1}^\beta Z_{n-1}^2 - \lambda_n^\beta e^{-\nu\lambda_{n+1}s} Z_n Z_{n+1} \right) ds \\ & + e^{-\nu(\lambda_n^2 - \lambda_n)t} X_n(0) \end{aligned}$$

for $t \geq 0$. The reason of this definition is simple: one can prove via method of variation of parameters that the integral formula for $V_n(t)$ is exactly $\mathcal{F}_n(V)(t)$ i.e. V is a fixed point for the map $\mathcal{F}$. Still it is unclear whether or not $V(t) \in \mathcal{V}_\varepsilon$ for all $t \in [0, \eta]$.

We will use Banach's fixed point theorem, also known as contraction mapping theorem. In order to do that, we need to show that for some $\eta > 0$, $R > 0$:

- $\mathcal{F}$ maps a ball of $L^\infty([0, \eta]; \mathcal{V}_\varepsilon)$ centered in $X(t) \equiv X(0)$ (initial condition) having radius R into itself.
- $\mathcal{F}$ is a contraction on this ball.

We will need the following, which come from the argument being in $[0, 1]$:

$$\begin{aligned} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} ds & \leq \int_0^t 1 ds = t \leq \eta \\ \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} ds & \leq \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)r} dr \leq \int_0^\infty e^{-\nu(\lambda_n^2 - \lambda_n)r} dr \\ & \leq \frac{1}{\nu(\lambda_n^2 - \lambda_n)} \end{aligned}$$

Hence, it is obvious that for any $0 \leq \alpha \leq 1$ the next inequalities hold:

$$\begin{aligned} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} ds & \leq \eta^\alpha \cdot \left(\frac{1}{\nu(\lambda_n^2 - \lambda_n)} \right)^{1-\alpha} \\ \Rightarrow \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} ds & \leq \eta^\alpha \cdot \left(\frac{2}{\nu\lambda_n^2} \right)^{1-\alpha} \end{aligned} \quad (1.9)$$

since $\lambda_n^2 - \lambda_n \geq \frac{1}{2}\lambda_n^2$.

For Z a.e. bounded on $[0, \eta]$, let $P_n(t) = \lambda_n^{\beta-2+\varepsilon} |\mathcal{F}_n(Z)(t) - X_n(t)|$. By the triangular inequality, and by multiplying and dividing for suitable powers of λ :

$$\begin{aligned} P_n(t) & \leq \lambda_n^{\beta-2+\varepsilon} |X_n(0)| \\ & + \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \left(\lambda_{n-1}^\beta |Z_{n-1}^2| + \lambda_n^\beta |Z_n Z_{n+1}| \right) ds \\ & \leq \|X(0)\|_\varepsilon \\ & + \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \lambda_{n-1}^{\beta-2+\varepsilon} |Z_{n-1}| \cdot \lambda_{n-1}^{2-\varepsilon} \cdot \lambda_{n-1}^{\beta-2+\varepsilon} |Z_{n-1}| ds \\ & + \lambda_n^{-(\beta-2+\varepsilon)} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \lambda_{n+1}^{\beta-2+\varepsilon} |Z_{n+1}| \cdot \lambda_n^{2-\varepsilon} \cdot \lambda_n^{\beta-2+\varepsilon} |Z_n| ds \end{aligned}$$

Let $\|\cdot\|_{\infty,\varepsilon}$ be the sup norm in t of $\|\cdot\|_\varepsilon$. By taking out of the integrals the $\|\cdot\|_{\infty,\varepsilon}$ factors in the last two rows, we can find a constant $c > 0$ such that

$$P_n(t) \leq \|X(0)\|_\varepsilon + c \|Z\|_{\infty,\varepsilon}^2 \cdot \lambda_n^{2-\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} ds$$

We use 1.9 to estimate the last integral: by choosing $\alpha = \varepsilon/2$ we get

$$P_n(t) \leq \|X(0)\|_\varepsilon + \bar{c} \|Z\|_{\infty,\varepsilon}^2 \eta^{\frac{\varepsilon}{2}} \quad (1.10)$$

where $\bar{c} > 0$ depends only on λ, β, ν and ε .

For $Z, \bar{Z}$ a.e. bounded on $[0, \eta]$, let $Q_n(t) = \lambda_n^{\beta-2+\varepsilon} |\mathcal{F}_n(Z)(t) - \mathcal{F}_n(\bar{Z})(t)|$:

$$\begin{aligned} Q_n(t) &\leq \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \lambda_{n-1}^\beta |Z_{n-1}^2 - \bar{Z}_{n-1}^2| ds \\ &\quad + \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \lambda_n^\beta |Z_n Z_{n+1} - \bar{Z}_n \bar{Z}_{n+1}| ds \\ &\leq \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \lambda_{n-1}^\beta |Z_{n-1} - \bar{Z}_{n-1}| |Z_{n-1} + \bar{Z}_{n-1}| ds \\ &\quad + \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \lambda_n^\beta |Z_n Z_{n+1} - \bar{Z}_n \bar{Z}_{n+1}| ds \end{aligned}$$

We will make use of the following equality:

$$Z_n Z_{n+1} - \bar{Z}_n \bar{Z}_{n+1} = \frac{(Z_n + \bar{Z}_n)(Z_{n+1} - \bar{Z}_{n+1}) + (Z_n - \bar{Z}_n)(Z_{n+1} + \bar{Z}_{n+1})}{2}$$

Apply this the second part of the sum:

$$\begin{aligned} Q_n(t) &\leq \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \lambda_{n-1}^\beta |Z_{n-1} - \bar{Z}_{n-1}| |Z_{n-1} + \bar{Z}_{n-1}| ds \\ &\quad + \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \frac{\lambda_n^\beta}{2} |Z_n + \bar{Z}_n| |Z_{n+1} - \bar{Z}_{n+1}| ds \\ &\quad + \lambda_n^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} \frac{\lambda_n^\beta}{2} |Z_{n+1} + \bar{Z}_{n+1}| |Z_n - \bar{Z}_n| ds \end{aligned}$$

By multiplying and dividing for suitable powers of λ , in a similar way to what we have done before, then by taking out of the integrals the $\|\cdot\|_{\infty,\varepsilon}$ factors in the last three rows, we can find a constant $c > 0$ such that

$$Q_n(t) \leq c \|Z - \bar{Z}\|_{\infty,\varepsilon} \|Z + \bar{Z}\|_{\infty,\varepsilon} \cdot \lambda_n^{2-\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} ds$$

We use 1.9 to estimate the last integral: by choosing $\alpha = \varepsilon/2$ we get

$$Q_n(t) \leq \bar{c} \|Z - \bar{Z}\|_{\infty,\varepsilon} \|Z + \bar{Z}\|_{\infty,\varepsilon} \eta^{\frac{\varepsilon}{2}} \quad (1.11)$$

where $\bar{c} > 0$ is a constant depending only on λ, β, ν and ε .

Let us prove that $\mathcal{F}$ maps a ball of $L^\infty([0, \eta]; \mathcal{V}_\varepsilon)$ centered in $X(t) \equiv X(0)$, having radius R into itself. Suppose Z belongs to this ball, i.e.

$$\lambda_n^{\beta-2+\varepsilon} |Z_n(t) - X_n(t)| \leq R$$

for all $n \geq 1, t \in [0, \eta]$.

By the triangular inequality and a suitable choice of R , namely $R \geq 2\|X(0)\|_\varepsilon$:

$$\lambda_n^{\beta-2+\varepsilon} |Z_n(t)| \leq R + \lambda_n^{\beta-2+\varepsilon} |X_n(t)| \leq 2R \quad (1.12)$$

so $2R$ is an upper bound for $\|Z\|_{\infty, \varepsilon}$. We want to show that

$$\lambda_n^{\beta-2+\varepsilon} |\mathcal{F}_n(Z)(t) - X_n(t)| \leq R$$

for all $n \geq 1, t \in [0, \eta]$. The left hand side is exactly what we called $P_n(t)$ and we already have an upper bound for it. We substitute 1.12 in the estimate 1.10:

$$P_n(t) \leq \frac{1}{2}R + 4R^2c \cdot \eta^{\frac{\varepsilon}{2}} \leq R$$

and the last inequality is true if we choose η such that $\eta^{\frac{\varepsilon}{2}} \leq (8Rc)^{-1}$.

Let us prove that $\mathcal{F}$ is a contraction. We want to show that, for some $\Lambda < 1$,

$$\|\mathcal{F}(Z) - \mathcal{F}(\bar{Z})\|_{\infty, \varepsilon} \leq \Lambda \|Z - \bar{Z}\|_{\infty, \varepsilon}$$

which means

$$\lambda_n^{\beta-2+\varepsilon} |\mathcal{F}_n(Z)(t) - \mathcal{F}_n(\bar{Z})(t)| \leq \Lambda \|Z - \bar{Z}\|_{\infty, \varepsilon}$$

for all $n \geq 1, t \in [0, \eta]$. The left hand side is exactly what we called $Q_n(t)$ and we already have an upper bound for it. We substitute 1.12 in the estimate 1.11:

$$\begin{aligned} Q_n(t) &\leq c \|Z - \bar{Z}\|_{\infty, \varepsilon} \|Z + \bar{Z}\|_{\infty, \varepsilon} \eta^{\frac{\varepsilon}{2}} \\ &\leq c \|Z - \bar{Z}\|_{\infty, \varepsilon} \left(\|Z\|_{\infty, \varepsilon} + \|\bar{Z}\|_{\infty, \varepsilon} \right) \eta^{\frac{\varepsilon}{2}} \\ &\leq (4Rc \cdot \eta^{\frac{\varepsilon}{2}}) \|Z - \bar{Z}\|_{\infty, \varepsilon} \end{aligned}$$

We need to choose η such that $\eta^{\frac{\varepsilon}{2}} \leq (4Rc)^{-1}$ to have $\Lambda < 1$. This is easily implied by the previous condition, $\eta^{\frac{\varepsilon}{2}} \leq (8Rc)^{-1}$.

We conclude by Banach that there is a unique fixed point for the map $\mathcal{F}$ and this is necessarily V defined before. This allows us to conclude that $V(t) \in \mathcal{V}_\varepsilon$ for $t \in [0, \eta]$. It remains to show why $V(t) \in \mathcal{V}_\varepsilon$ implies $X(t) \in \mathcal{D}^\infty$. Since

$$\sup_{n \geq 1} |V_n(t)| \leq \sup_{n \geq 1} \lambda_n^{\beta-2+\varepsilon} |V_n(t)| < \infty$$

Now note that for sufficiently large n , say $n \geq N$, the following estimate holds since a exponentially-growing exponent beats a linearly-growing exponent:

$$\lambda_n^\gamma |X_n(t)| \leq e^{\nu \lambda_n t} |X_n(t)| \implies \sup_{n \geq N} \lambda_n^\gamma |X_n(t)| \leq \sup_{n \geq N} |V_n(t)| < \infty$$

for all $\gamma > 0$. Obviously the same holds for $\sup_{|j| \leq N} \lambda_{|j|}^\gamma |X_j(t)|$ since we take sup on a finite set. Thus $X(t) \in \mathcal{D}^\infty$ as desired. $\square$

Capitolo 2

Dyadic model on a tree

2.1 The tree formulation

We consider a more general version of the dyadic model: let J be the set of nodes of a tree, where each node "father" has a certain number of "sons".

Let O_j denote the set of nodes which descend from node j . For now, we will suppose this quantity does not depend on j and we pose $|O_j| = N^*$ for all j . The dynamic of the system is described by the following equations:

$$\begin{cases} \dot{X}_j = -\nu\lambda_{|j|}^2 X_j + \lambda_{|j|}^\beta X_j^2 - \sum_{k \in O_j} \lambda_{|k|}^\beta X_j X_k \\ X_j(0) = x_j \end{cases} \quad \forall j \in J, \forall t \geq 0 \quad (2.1)$$

Comparing systems 1.1 and 2.1, there is one difference that meets the eye. Let us say $|j| = n$. While the viscous terms match each other perfectly:

- the equivalent of $\lambda_{|j|}^\beta X_j^2$ should be $\lambda_n^\beta X_{n-1}^2$ but in 1.1 there is $\lambda_{n-1}^\beta X_{n-1}^2$;
- the equivalent of $\lambda_{|k|}^\beta X_j X_k$ should be $\lambda_{n+1}^\beta X_n X_{n+1}$ but in 1.1 there is $\lambda_n^\beta X_n X_{n+1}$.

We just need to multiply for λ^β to have the nonlinear terms match other; for what regards the viscous term, multiplying for λ^β can be considered as changing the viscosity coefficient. However this does not make any difference when we are interested in studying the regularity of the solution.. Thus, the argument we presented in chapter 1 is also valid for the following problem:

$$\begin{cases} \dot{X}_n = -\nu\lambda_n^2 X_n + \lambda_n^\beta X_{n-1}^2 - \lambda_{n+1}^\beta X_n X_{n+1} \\ X_n(0) = x_n \end{cases} \quad \forall n \geq 1, \forall t \geq 0 \quad (2.2)$$

We apply the following change of variables:

$$Y_j = \lambda_{|j|}^{\beta-2+\varepsilon} X_j \quad (2.3)$$

where $\varepsilon > 0$ is a positive number whose choice will be made later.

By substitution and some simple calculations we can find the dynamics of $\{Y_j\}$:

$$\begin{cases} \dot{Y}_j = -\nu\lambda_{|j|}^2 Y_j + \lambda^{2\beta-4+2\varepsilon} \cdot \lambda_{|j|}^{2-\varepsilon} Y_j^2 - \lambda^{2-\varepsilon} \cdot \lambda_{|j|}^{2-\varepsilon} \sum_{k \in O_j} Y_j Y_k \\ Y_j(0) = y_j \end{cases} \quad (2.4)$$

for $j \in J$ and $t \in [0, T]$. Clearly y_j is the image of x_j after changing variables. Consider now a finite dimensional version of the equations for Y : for every $N \geq 1$ we will denote with $\left\{Y_j^{(N)}\right\}_{|j| \leq N}$ the solution to

$$\begin{cases} \dot{Y}_j^{(N)} = -\nu\lambda_{|j|}^2 Y_j^{(N)} + \lambda^{2\beta-4+2\varepsilon} \cdot \lambda_{|j|}^{2-\varepsilon} \left(Y_j^{(N)}\right)^2 - \lambda^{2-\varepsilon} \cdot \lambda_{|j|}^{2-\varepsilon} \sum_{k \in O_j} Y_j^{(N)} Y_k^{(N)} \\ Y_j(0) = y_j \end{cases}$$

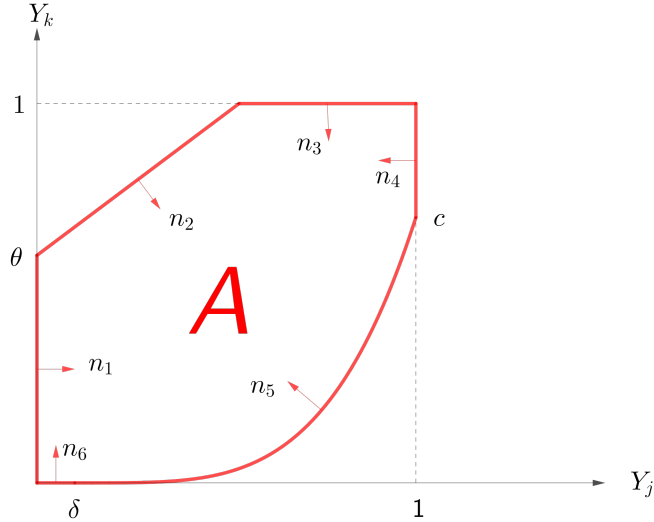
for $1 \leq |j| \leq N$, i.e. we are taking into consideration just the first N generations. Now we introduce the following region contained in $\mathbb{R}^2$:

$$A := \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, h(x) < y < g(x)\}$$

where

$$g(x) = \min\{mx + \theta, 1\}, \quad h(x) = \begin{cases} 0 & x \leq \delta \\ c \left(\frac{x - \delta}{1 - \delta}\right)^{\lambda^2} & x > \delta \end{cases}$$

are the functions providing the lower and upper bound of A . The only conditions on parameters δ, θ is that $\delta \leq \theta < 1$ so the little square $[0, \delta]^2$ is all inside A .



2.1.1 Case $N^* = 8$

In chapter we dealt with the case $N^* = 1$, indeed every node X_n had a unique son X_{n+1} . Now we will consider the case when $N^* = 8$ for all j .

Lemma 2.5. (*Invariant region, $N^* = 8$*)

There exists $\delta \in (0, 1)$, $c \in (0, 1)$, $\theta \in (\delta, 1)$, $m > 1 - \theta$ and $\varepsilon_0 > 0$ such that: for every $\beta \in (2, \frac{5}{2})$, $\nu \geq 0$ and $\varepsilon \in (0, \varepsilon_0)$ the following statement is true. If $(y_j, y_k) \in A$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$,

$$(Y_j^{(N)}(t), Y_k^{(N)}(t)) \in A \quad \forall |j| \leq N, \forall k \in O_j, \forall t \geq 0$$

where $\{Y_j^{(N)}\}$ is the same as before.

Dim: for the sake of elegance, we drop the superscript $^{(N)}$ along the proof. It is sufficient to show that the time derivative of (Y_j, Y_k) points inward on the border of A when $(Y_j, Y_k) \in A$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$. We check by computing the scalar product of the inward normal on the border of A with the following vector field $\mathcal{B}$:

$$\begin{pmatrix} \dot{Y}_j \\ \dot{Y}_k \end{pmatrix} = \nu \lambda_{|j|}^2 \begin{pmatrix} -Y_j \\ -\lambda^2 Y_k \end{pmatrix} + \lambda^{2\beta-4+2\varepsilon} \lambda_{|j|}^{2-\varepsilon} \begin{pmatrix} Y_j^2 - \lambda^{6-2\beta-3\varepsilon} \sum_{l \in O_j} Y_j Y_l \\ \lambda^{2-\varepsilon} \left(Y_j^2 - \lambda^{6-2\beta-3\varepsilon} \sum_{m \in O_k} Y_k Y_m \right) \end{pmatrix}$$

and verifying it is positive when $(Y_j, Y_k) \in A$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$. Since A is a convex set, we can compute separately the scalar product with just the viscous term and just the inviscid one.

We denote the inward normals as in figure, so their expressions are the following:

$$\begin{aligned} \vec{n}_1 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{n}_3 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \vec{n}_4 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \vec{n}_6 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \vec{n}_2 &= \begin{pmatrix} g'(x) \\ -1 \end{pmatrix}, \quad \vec{n}_5 = \begin{pmatrix} -h'(x) \\ 1 \end{pmatrix} \end{aligned}$$

We start with the viscous term, which will be denoted as $\mathcal{B}_v$. Note that there is no need to consider the positive constant $\nu \lambda_n^2$.

The scalar products are evaluated on the respective pieces of the border of A :

$$\begin{aligned} \mathcal{B}_v \cdot \vec{n}_1 &= -Y_j = 0, & \mathcal{B}_v \cdot \vec{n}_3 &= \lambda^2 Y_k = \lambda^2 \\ \mathcal{B}_v \cdot \vec{n}_4 &= Y_j = 1, & \mathcal{B}_v \cdot \vec{n}_6 &= -\lambda^2 Y_k = 0 \end{aligned}$$

These are positive. To compute the remaining scalar products, we set $x = Y_j$. Along g , $Y_k = g(x)$:

$$\mathcal{B}_v \cdot \vec{n}_2 = -xg'(x) + \lambda^2 Y_k = -xg'(x) + \lambda^2 g(x) = mx(-1 + \lambda^2) + \lambda^2 \theta > 0$$

Along h , $Y_k = h(x)$:

$$\begin{aligned} \mathcal{B}_v \cdot \vec{n}_5 &= xh'(x) - \lambda^2 Y_k = xh'(x) - \lambda^2 h(x) = x \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} - \lambda^2 h(x) \\ &\geq \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} - \lambda^2 h(x) = \frac{\lambda^2}{1-\delta} h(x) - \lambda^2 h(x) = \lambda^2 h \frac{\delta}{1-\delta} > 0 \end{aligned}$$

We proceed with the inviscid term, which will be denoted as $\mathcal{B}_i$. Just as before, there is no need to consider the positive constant.

We will use again $x = Y_j$ and for simplicity we set $\gamma = 6 - 2\beta - 3\varepsilon$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_1 &= Y_j^2 - \lambda^\gamma \sum_{l \in O_j} Y_j Y_l = Y_j^2 \geq 0 \\ \mathcal{B}_i \cdot \vec{n}_6 &= \lambda^{2-\varepsilon} \left(Y_j^2 - \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) = \lambda^{2-\varepsilon} Y_j^2 \geq 0 \end{aligned}$$

Consider the piece of the border of A corresponding to $\vec{n}_3$. Here $Y_j \leq 1$, $Y_k = 1$ and $(Y_k, Y_m) \in A$ for all $m \in O_k$ so that we have $Y_m \geq c$ for all $m \in O_k$. Let us exploit these inequalities:

$$\mathcal{B}_i \cdot \vec{n}_3 = \lambda^{2-\varepsilon} \left(-Y_j^2 + \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) \geq \lambda^{2-\varepsilon} (8\lambda^\gamma c - 1)$$

Consider the piece of the border of A corresponding to $\vec{n}_4$. Here $Y_j = 1$, $Y_j \leq 1$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \geq c$ for all $l \in O_j$. Let us exploit these inequalities:

$$\mathcal{B}_i \cdot \vec{n}_4 = -Y_j^2 + \lambda^\gamma \sum_{l \in O_j} Y_j Y_l \geq (8\lambda^\gamma c - 1)$$

We choose c in order to have $8\lambda^\gamma c = 1$, so the previous are positive. Note that

$$\begin{aligned} \beta \geq 2 &\Rightarrow \gamma \leq 2 - 3\varepsilon \\ \beta \leq \frac{5}{2} &\Rightarrow \gamma \geq 1 - 3\varepsilon \end{aligned}$$

Let $Y_j = x$. Consider the piece of the border of A corresponding to $\vec{n}_2$. Here $Y_k = g(x)$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \leq g(x)$. Also $(Y_m, Y_k) \in A$ for all $m \in O_k$ so that we have $Y_m \geq h(Y_k) = h(g(x))$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_2 &= g'(x) \left(Y_j^2 - \lambda^\gamma \sum_{l \in O_j} x Y_l \right) - \lambda^{2-\varepsilon} \left(x^2 - \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) \\ &\geq g'(x) (-8\lambda^\gamma x \cdot g(x)) - \lambda^{2-\varepsilon} [x^2 - 8\lambda^\gamma g(x) h(g(x))] \end{aligned}$$

Substitute for g, h :

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -8\lambda^\gamma m x (mx + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (mx + \theta) (8\lambda^\gamma c) \left(\frac{mx + \theta - \delta}{1-\delta} \right)^{\lambda^2}$$

Remember $8\lambda^\gamma c = 1$ and the upper bound for γ :

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -8\lambda^{2-3\varepsilon} mx (mx + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (mx + \theta) \left(\frac{mx + \theta - \delta}{1 - \delta} \right)^{\lambda^2}$$

Consider the piece of the border of A corresponding to $\vec{n}_5$. Here $Y_k = h(x)$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \geq h(x)$. Also $(Y_k, Y_m) \in A$ for all $m \in O_k$ so that we have $Y_m \leq g(Y_k) = g(h(x))$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &= \lambda^{2-\varepsilon} \left(x^2 - \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) - h'(x) \left(Y_j^2 - \lambda^\gamma \sum_{l \in O_j} x Y_l \right) \\ &\geq \lambda^{2-\varepsilon} \left[x^2 - 8\lambda^\gamma h(x) g(h(x)) \right] - h'(x) \left[1 - 8\lambda^\gamma x \cdot h(x) \right] \\ &\geq \lambda^{2-\varepsilon} \left[x^2 - 8\lambda^\gamma h(x) (mh + \theta) \right] - h'(x) \left[1 - 8\lambda^\gamma x \cdot h(x) \right] \end{aligned}$$

Substitute for h :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta (8\lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - m\lambda^{2-\varepsilon} \cdot 8\lambda^\gamma h(x)^2 \\ &\quad - h'(x) \left[1 - (8\lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} x \right] \end{aligned}$$

Substitute for h' :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta (8\lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - m\lambda^{2-\varepsilon} \cdot 8\lambda^\gamma h(x)^2 \\ &\quad - \frac{\lambda^2 c}{1 - \delta} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - (8\lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} x \right] \end{aligned}$$

Multiply and divide for $8\lambda^\gamma$ both the second and the third term. Since $8\lambda^\gamma c = 1$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - \frac{m}{8\lambda^\gamma} \lambda^{2-\varepsilon} \left(\frac{x - \delta}{1 - \delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^2}{8\lambda^\gamma (1 - \delta)} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \end{aligned}$$

Now we make use of the lower bound for γ :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - \frac{m}{8} \lambda^{2-\varepsilon} \lambda^{3\varepsilon - 1} \left(\frac{x - \delta}{1 - \delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^2}{8(1 - \delta)} \lambda^{3\varepsilon - 1} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \end{aligned}$$

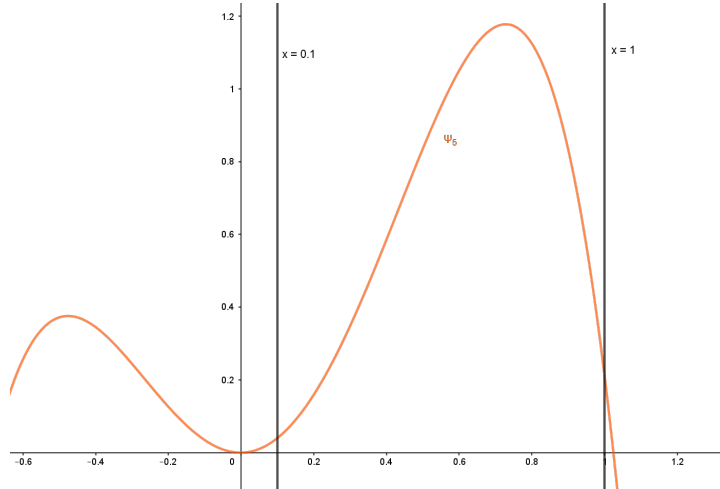
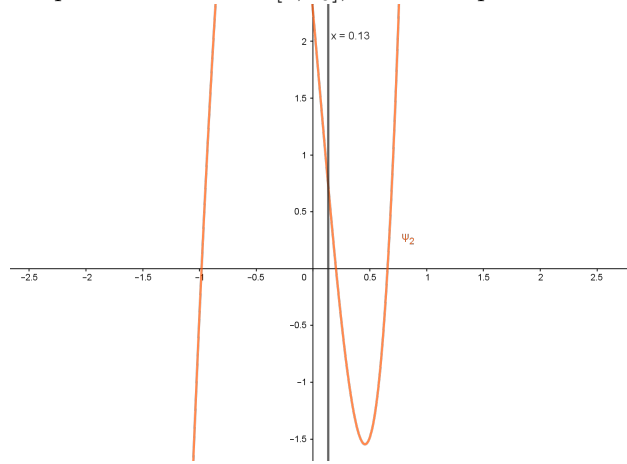
Let $\psi_2(x)$, $\psi_5(x)$ be the lower bounds we obtained for $\mathcal{B}_i \cdot \vec{n}_2$, $\vec{n}_5$ when $\varepsilon = 0$:

$$\psi_2(x) = -8\lambda^2 mx (mx + \theta) - \lambda^2 x^2 + \lambda^2 (mx + \theta) \left(\frac{mx + \theta - \delta}{1 - \delta} \right)^{\lambda^2}$$

$$\begin{aligned} \psi_5(x) = & \lambda^2 \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - \frac{m}{8} \lambda \left(\frac{x - \delta}{1 - \delta} \right)^{2\lambda^2} \\ & - \frac{\lambda}{8(1 - \delta)} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \end{aligned}$$

Now, ψ_2 and ψ_5 must be positive on the right interval, which is $[0, \frac{1-\theta}{m}]$ for ψ_2 , and $[\delta, 1]$ for ψ_5 . Below you can see the plot of both functions for $m = \frac{3}{4}$, $\theta = \frac{9}{10}$ e $\delta = \frac{1}{10}$ and they are both positive with this choice.

Since the estimate for $\mathcal{B}_i \cdot \vec{n}_2$ and $\mathcal{B}_i \cdot \vec{n}_5$ are continuous functions in ε and they are positive for $\varepsilon = 0$, we conclude that there exists ε_0 such that scalar products are positive for all $\varepsilon \in [0, \varepsilon_0]$, which is equivalent to the lemma. $\square$



2.1.2 Case when N^* is fixed

In this section we are going to extend the invariant region lemma found in the previous one. We know that it holds when $N^* = 8$, now we show that it also holds for any fixed $N^* \leq 8$. Again, each node has the same number of sons.

Lemma 2.6. (*Invariant region, N^* is fixed, $1 \leq N^* \leq 8$)*

There exists $\delta \in (0, 1)$, $c \in (0, 1)$, $\theta \in (\delta, 1)$, $m > 1 - \theta$ and $\varepsilon_0 > 0$ such that: for every $\beta \in (2, \frac{5}{2})$, $\nu \geq 0$ and $\varepsilon \in (0, \varepsilon_0)$ the following statement is true. If $(y_j, y_k) \in A$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$,

$$(Y_j^{(N)}(t), Y_k^{(N)}(t)) \in A \quad \forall |j| \leq N, \forall k \in O_j, \forall t \geq 0$$

where $\{Y_j^{(N)}\}$ is the same as before.

Dim: for the sake of elegance, we drop the superscript (N) along the proof. It is sufficient to show that the time derivative of (Y_j, Y_k) points inward on the border of A when $(Y_j, Y_k) \in A$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$. We check by computing the scalar product of the inward normal on the border of A with the following vector field $\mathcal{B}$:

$$\begin{pmatrix} \dot{Y}_j \\ \dot{Y}_k \end{pmatrix} = \nu \lambda_{|j|}^2 \begin{pmatrix} -Y_j \\ -\lambda^2 Y_k \end{pmatrix} + \lambda^{2\beta-4+2\varepsilon} \lambda_{|j|}^{2-\varepsilon} \begin{pmatrix} Y_j^2 - \lambda^{6-2\beta-3\varepsilon} \sum_{l \in O_j} Y_j Y_l \\ \lambda^{2-\varepsilon} \left(Y_j^2 - \lambda^{6-2\beta-3\varepsilon} \sum_{m \in O_k} Y_k Y_m \right) \end{pmatrix}$$

and verifying it is positive when $(Y_j, Y_k) \in A$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$. Since A is a convex set, we can compute separately the scalar product with just the viscous term and just the inviscid one.

We denote the inward normals as in figure, so their expressions are the following:

$$\begin{aligned} \vec{n}_1 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{n}_3 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \vec{n}_4 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \vec{n}_6 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \vec{n}_2 &= \begin{pmatrix} g'(x) \\ -1 \end{pmatrix}, \quad \vec{n}_5 = \begin{pmatrix} -h'(x) \\ 1 \end{pmatrix} \end{aligned}$$

We start with the viscous term, which will be denoted as $\mathcal{B}_v$. Note that there is no need to consider the positive constant $\nu \lambda_n^2$.

The scalar products are evaluated on the respective pieces of the border of A :

$$\begin{aligned} \mathcal{B}_v \cdot \vec{n}_1 &= -Y_j = 0, & \mathcal{B}_v \cdot \vec{n}_3 &= \lambda^2 Y_k = \lambda^2 \\ \mathcal{B}_v \cdot \vec{n}_4 &= Y_j = 1, & \mathcal{B}_v \cdot \vec{n}_6 &= -\lambda^2 Y_k = 0 \end{aligned}$$

These are positive. To compute the remaining scalar products, we set $x = Y_j$. Along g , $Y_k = g(x)$:

$$\mathcal{B}_v \cdot \vec{n}_2 = -xg'(x) + \lambda^2 Y_k = -xg'(x) + \lambda^2 g(x) = mx(-1 + \lambda^2) + \lambda^2 \theta > 0$$

Along h , $Y_k = h(x)$:

$$\begin{aligned}\mathcal{B}_v \cdot \vec{n}_5 &= xh'(x) - \lambda^2 Y_k = xh'(x) - \lambda^2 h(x) = x \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} - \lambda^2 h(x) \\ &\geq \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} - \lambda^2 h(x) = \frac{\lambda^2}{1-\delta} h(x) - \lambda^2 h(x) = \lambda^2 h \frac{\delta}{1-\delta} > 0\end{aligned}$$

We proceed with the inviscid term, which will be denoted as $\mathcal{B}_i$. Just as before, there is no need to consider the positive constant.

We will use again $x = Y_j$ and for simplicity we set $\gamma = 6 - 2\beta - 3\varepsilon$:

$$\begin{aligned}\mathcal{B}_i \cdot \vec{n}_1 &= Y_j^2 - \lambda^\gamma \sum_{l \in O_j} Y_j Y_l = Y_j^2 \geq 0 \\ \mathcal{B}_i \cdot \vec{n}_6 &= \lambda^{2-\varepsilon} \left(Y_j^2 - \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) = \lambda^{2-\varepsilon} Y_j^2 \geq 0\end{aligned}$$

Consider the piece of the border of A corresponding to $\vec{n}_3$. Here $Y_j \leq 1$, $Y_k = 1$ and $(Y_k, Y_m) \in A$ for all $m \in O_k$ so that we have $Y_m \geq c$ for all $m \in O_k$. Let us exploit these inequalities:

$$\mathcal{B}_i \cdot \vec{n}_3 = \lambda^{2-\varepsilon} \left(-Y_j^2 + \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) \geq \lambda^{2-\varepsilon} (N^* \lambda^\gamma c - 1)$$

Consider the piece of the border of A corresponding to $\vec{n}_4$. Here $Y_j = 1$, $Y_j \leq 1$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \geq c$ for all $l \in O_j$. Let us exploit these inequalities:

$$\mathcal{B}_i \cdot \vec{n}_4 = -Y_j^2 + \lambda^\gamma \sum_{l \in O_j} Y_j Y_l \geq (N^* \lambda^\gamma c - 1)$$

We choose c so that $N^* \lambda^\gamma c = 1$, so the previous are positive. Note that

$$\begin{aligned}\beta \geq 2 &\Rightarrow \gamma \leq 2 - 3\varepsilon \\ \beta \leq \frac{5}{2} &\Rightarrow \gamma \geq 1 - 3\varepsilon\end{aligned}$$

Let $Y_j = x$. Consider the piece of the border of A corresponding to $\vec{n}_2$. Here $Y_k = g(x)$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \leq g(x)$. Also $(Y_m, Y_k) \in A$ for all $m \in O_k$ so that we have $Y_m \geq h(Y_k) = h(g(x))$:

$$\begin{aligned}\mathcal{B}_i \cdot \vec{n}_2 &= g'(x) \left(Y_j^2 - \lambda^\gamma \sum_{l \in O_j} x Y_l \right) - \lambda^{2-\varepsilon} \left(x^2 - \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) \\ &\geq g'(x) (-N^* \lambda^\gamma x \cdot g(x)) - \lambda^{2-\varepsilon} [x^2 - N^* \lambda^\gamma g(x) h(g(x))]\end{aligned}$$

Substitute for g, h :

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -N^* \lambda^\gamma m x (m x + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (m x + \theta) (N^* \lambda^\gamma c) \left(\frac{m x + \theta - \delta}{1-\delta} \right)^{\lambda^2}$$

Remember $N^* \lambda^\gamma c = 1$ and the upper bound for γ :

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -N^* \lambda^{2-3\varepsilon} m x (m x + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (m x + \theta) \left(\frac{m x + \theta - \delta}{1 - \delta} \right)^{\lambda^2}$$

Consider the piece of the border of A corresponding to $\vec{n}_5$. Here $Y_k = h(x)$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \geq h(x)$. Also $(Y_k, Y_m) \in A$ for all $m \in O_k$ so that we have $Y_m \leq g(Y_k) = g(h(x))$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &= \lambda^{2-\varepsilon} \left(x^2 - \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) - h'(x) \left(Y_j^2 - \lambda^\gamma \sum_{l \in O_j} x Y_l \right) \\ &\geq \lambda^{2-\varepsilon} \left[x^2 - N^* \lambda^\gamma h(x) g(h(x)) \right] - h'(x) \left[1 - N^* \lambda^\gamma x \cdot h(x) \right] \\ &\geq \lambda^{2-\varepsilon} \left[x^2 - N^* \lambda^\gamma h(x) (m h + \theta) \right] - h'(x) \left[1 - N^* \lambda^\gamma x \cdot h(x) \right] \end{aligned}$$

Substitute for h :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta (N^* \lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - m \lambda^{2-\varepsilon} \cdot N^* \lambda^\gamma h(x)^2 \\ &\quad - h'(x) \left[1 - (N^* \lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} x \right] \end{aligned}$$

Substitute for h' :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta (N^* \lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - m \lambda^{2-\varepsilon} \cdot N^* \lambda^\gamma h(x)^2 \\ &\quad - \frac{\lambda^2 c}{1 - \delta} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - (N^* \lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} x \right] \end{aligned}$$

Multiply and divide for $N^* \lambda^\gamma$ the second and the third term. Since $N^* \lambda^\gamma c = 1$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - \frac{m}{N^* \lambda^\gamma} \lambda^{2-\varepsilon} \left(\frac{x - \delta}{1 - \delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^2}{N^* \lambda^\gamma (1 - \delta)} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \end{aligned}$$

Now we make use of the lower bound for γ :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - \frac{m}{N^*} \lambda^{2-\varepsilon} \lambda^{3\varepsilon - 1} \left(\frac{x - \delta}{1 - \delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^2}{N^* (1 - \delta)} \lambda^{3\varepsilon - 1} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \end{aligned}$$

Let $\psi_{2,N^*}(x)$, $\psi_{5,N^*}(x)$ be the lower bounds we obtained for $\mathcal{B}_i \cdot \vec{n}_2$ and $\mathcal{B}_i \cdot \vec{n}_5$ when $\varepsilon = 0$:

$$\psi_{2,N^*}(x) = -N^* \lambda^2 m x (mx + \theta) - \lambda^2 x^2 + \lambda^2 (mx + \theta) \left(\frac{mx + \theta - \delta}{1 - \delta} \right)^{\lambda^2}$$

$$\begin{aligned} \psi_{5,N^*}(x) = & \lambda^2 \left[x^2 - \theta \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] - \frac{m}{N^*} \lambda \left(\frac{x - \delta}{1 - \delta} \right)^{2\lambda^2} \\ & - \frac{\lambda}{N^* (1 - \delta)} \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2 - 1} \left[1 - x \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} \right] \end{aligned}$$

Note that ψ_{2,N^*} is evaluated on $[0, \frac{1-\theta}{m}]$, while ψ_{5,N^*} is evaluated on $[\delta, 1]$. It is clear that, on the mutual intervals, $\psi_{2,8}(x) \leq \psi_{2,N^*}(x)$ while $\psi_{5,1}(x) \leq \psi_{5,N^*}(x)$. In the next page we plotted $\psi_{2,8}$ and $\psi_{5,1}$ for

$$m = \frac{150}{77} \frac{1 - \left(\frac{77}{150}\right)^2}{8} = 0,179... \quad \theta = 1 - \frac{1 - \left(\frac{77}{150}\right)^2}{8} = 0,908... \quad \delta = \frac{1}{10}$$

We show that $\psi_{2,8}(x)$ is positive:

$$\psi_{2,8}(x) = -8\lambda^2 m x (mx + \theta) - \lambda^2 x^2 + \lambda^2 (mx + \theta) \left(\frac{mx + \theta - \delta}{1 - \delta} \right)^4$$

We show that it is a decreasing function of x on $[0, \frac{1-\theta}{m}]$. Since $mx + \theta \leq 1$:

$$\begin{aligned} \psi'_{2,8}(x) & \leq -8\lambda^2 m \theta + \frac{4\lambda^2 m}{1 - \delta} (mx + \theta) \left(\frac{mx + \theta - \delta}{1 - \delta} \right)^3 + \lambda^2 m \left(\frac{mx + \theta - \delta}{1 - \delta} \right)^4 \\ & \leq -8\lambda^2 m \theta + \frac{4\lambda^2 m}{1 - \delta} + \lambda^2 m \leq \lambda^2 m \left(-8\theta + \frac{4}{1 - \delta} + 1 \right) < 0 \end{aligned}$$

Let us compute its value in $x = \frac{1-\theta}{m}$:

$$\psi_{2,8} \left(\frac{1-\theta}{m} \right) = -8\lambda^2 (1 - \theta) - \lambda^2 \left(\frac{1-\theta}{m} \right)^2 + \lambda^2 = \lambda^2 \left[1 - 8(1 - \theta) - \left(\frac{1-\theta}{m} \right)^2 \right]$$

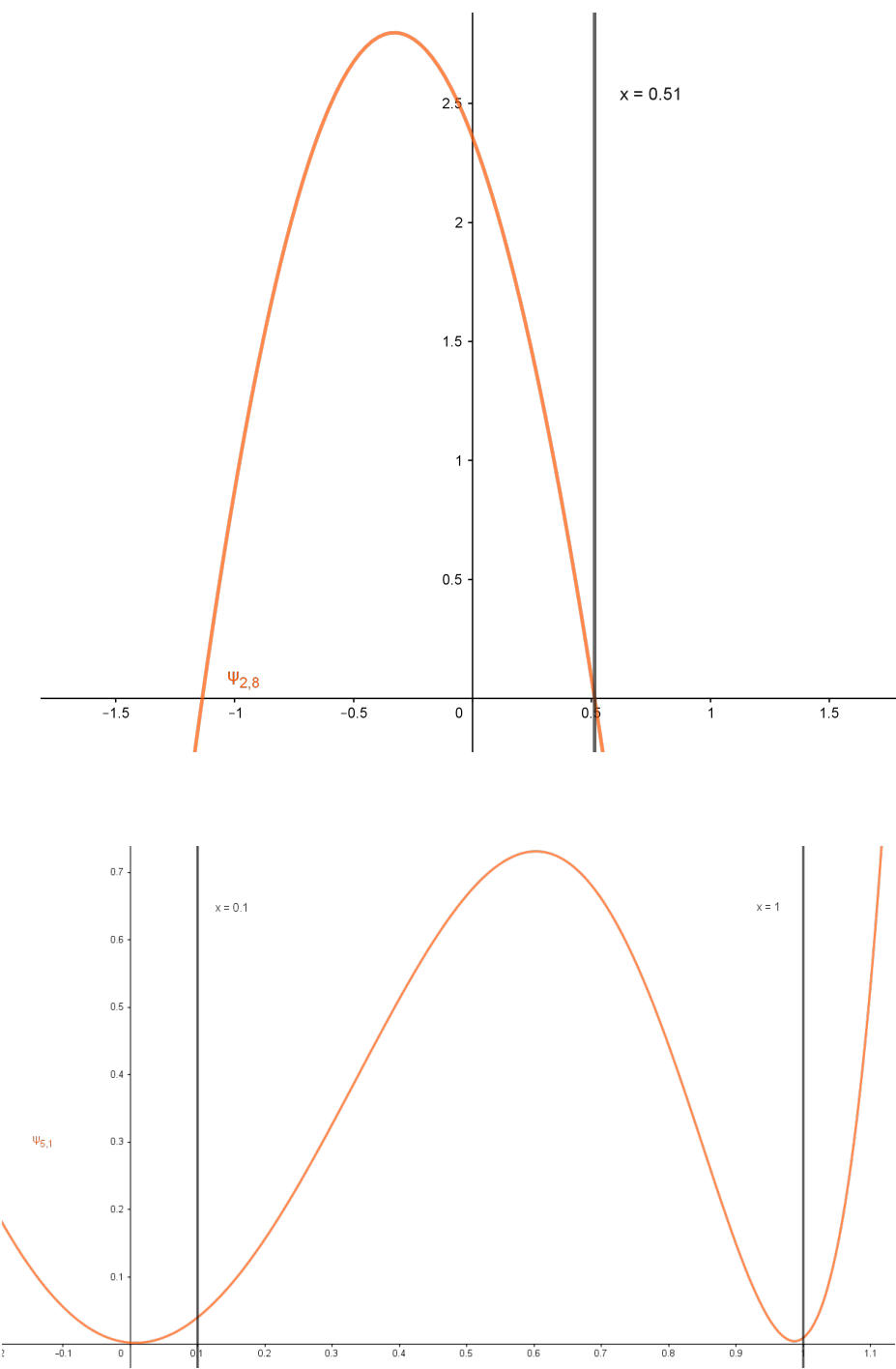
and the second factor is 0 for our choice of m, θ .

Let us consider $\psi_{5,1}$. It is not sufficient to ask it cancels for $x = 1$ because $\psi_{5,1}$ would present a local minimum $M < 0$ for some $x_M < 1$. Indeed, we have chosen those m, θ to shift $\psi_{5,1}$'s plot of a quantity greater than $|M|$. So

$$\psi_{2,N^*}(x) \geq \psi_{2,8}(x) \geq 0 \quad \text{for } x \in \left[0, \frac{1-\theta}{m} \right]$$

$$\psi_{5,N^*}(x) \geq \psi_{5,1}(x) \geq 0 \quad \text{for } x \in [\delta, 1]$$

for any fixed N^* . We conclude using the same argument based on the continuity in ε that worked in the previous section. $\square$



2.1.3 Case when N^* is variable

First of all we must clarify what we mean saying that N^* is variable. Consider:

$$\{n \text{ such that } n = |O_j| \text{ for some } j \in J\}$$

So far we assumed this set to contain one distinct element, which was N^* . Now we examine the case when there are more than 1: let a be the minimum value and b the maximum, with $a \neq b$ and $1 \leq a, b \leq 8$. From now on N^* will always denote $|O_j|$ for some j and we will use $a \leq N^* \leq b$.

In this section we are going to show why the proof used in the previous section does not work. Our focus lies entirely on the scalar products involving $\mathcal{B}_i$. Just like before:

$$\mathcal{B}_i \cdot \vec{n}_3 = \lambda^{2-\varepsilon} \left(-Y_j^2 + \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) \geq \lambda^{2-\varepsilon} (N^* \lambda^\gamma c - 1)$$

$$\mathcal{B}_i \cdot \vec{n}_4 = -Y_j^2 + \lambda^\gamma \sum_{l \in O_j} Y_j Y_l \geq N^* \lambda^\gamma c - 1 \geq a \lambda^\gamma c - 1$$

We choose c so that $a \lambda^\gamma c = 1$, so the previous are positive. Note that

$$\beta \geq 2 \Rightarrow \gamma \leq 2 - 3\varepsilon$$

$$\beta \leq \frac{5}{2} \Rightarrow \gamma \geq 1 - 3\varepsilon$$

Let $Y_j = x$. Consider the piece of the border of A corresponding to $\vec{n}_2$. Here $Y_k = g(x)$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \leq g(x)$. Also $(Y_m, Y_k) \in A$ for all $m \in O_k$ so that we have $Y_m \geq h(Y_k) = h(g(x))$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_2 &= g'(x) \left(Y_j^2 - \lambda^\gamma \sum_{l \in O_j} x Y_l \right) - \lambda^{2-\varepsilon} \left(x^2 - \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) \\ &\geq g'(x) (-N^* \lambda^\gamma x \cdot g(x)) - \lambda^{2-\varepsilon} [x^2 - N^* \lambda^\gamma g(x) h(g(x))] \end{aligned}$$

Substitute for g, h :

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -N^* \lambda^\gamma m x (m x + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (m x + \theta) (N^* \lambda^\gamma c) \left(\frac{m x + \theta - \delta}{1 - \delta} \right)^{\lambda^2}$$

Remember $N^* \lambda^\gamma c \geq 1$ and the upper bound for γ :

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -N^* \lambda^{2-3\varepsilon} m x (m x + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (m x + \theta) \left(\frac{m x + \theta - \delta}{1 - \delta} \right)^{\lambda^2}$$

Since $N^* \leq b$:

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -b \lambda^{2-3\varepsilon} m x (m x + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (m x + \theta) \left(\frac{m x + \theta - \delta}{1 - \delta} \right)^{\lambda^2}$$

Consider the piece of the border of A corresponding to $\vec{n}_5$. Here $Y_k = h(x)$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \geq h(x)$. Also $(Y_k, Y_m) \in A$ for all $m \in O_k$ so that we have $Y_m \leq g(Y_k) = g(h(x))$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &= \lambda^{2-\varepsilon} \left(x^2 - \lambda^\gamma \sum_{m \in O_k} Y_k Y_m \right) - h'(x) \left(Y_j^2 - \lambda^\gamma \sum_{l \in O_j} x Y_l \right) \\ &\geq \lambda^{2-\varepsilon} \left[x^2 - N^* \lambda^\gamma h(x) g(h(x)) \right] - h'(x) \left[1 - N^* \lambda^\gamma x \cdot h(x) \right] \\ &\geq \lambda^{2-\varepsilon} \left[x^2 - N^* \lambda^\gamma h(x) (mh + \theta) \right] - h'(x) \left[1 - N^* \lambda^\gamma x \cdot h(x) \right] \end{aligned}$$

Substitute for h :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta (N^* \lambda^\gamma c) \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] - m \lambda^{2-\varepsilon} \cdot N^* \lambda^\gamma h(x)^2 \\ &\quad - h'(x) \left[1 - (N^* \lambda^\gamma c) \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} x \right] \end{aligned}$$

Substitute for h' :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta (N^* \lambda^\gamma c) \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] - m \lambda^{2-\varepsilon} \cdot N^* \lambda^\gamma h(x)^2 \\ &\quad - \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} \left[1 - (N^* \lambda^\gamma c) \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} x \right] \end{aligned}$$

Note that $1 \leq N^* \lambda^\gamma c \leq b \lambda^\gamma c = \frac{b}{a}$. Multiply and divide for $a \lambda^\gamma$ both the second and the third term:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta \frac{b}{a} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] - \frac{m}{a \lambda^\gamma} \lambda^{2-\varepsilon} \cdot \frac{b}{a} \left(\frac{x-\delta}{1-\delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^2}{a \lambda^\gamma (1-\delta)} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} \left[1 - x \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] \end{aligned}$$

Now we make use of the lower bound for γ :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[x^2 - \theta \frac{b}{a} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] - \frac{m}{a} \lambda^{2-\varepsilon} \lambda^{3\varepsilon-1} \cdot \frac{b}{a} \left(\frac{x-\delta}{1-\delta} \right)^{2\lambda^2} \\ &\quad - \frac{\lambda^2}{a(1-\delta)} \lambda^{3\varepsilon-1} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} \left[1 - x \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] \end{aligned}$$

Let $\psi_{2,b}(x)$, $\psi_{5,a,b}(x)$ be the lower bounds we obtained for $\mathcal{B}_i \cdot \vec{n}_2$ and $\mathcal{B}_i \cdot \vec{n}_5$ when $\varepsilon = 0$:

$$\psi_{2,b}(x) = -b\lambda^2 m x (mx + \theta) - \lambda^2 x^2 + \lambda^2 (mx + \theta) \left(\frac{mx + \theta - \delta}{1-\delta} \right)^{\lambda^2}$$

$$\begin{aligned} \psi_{5,a,b}(x) = & \lambda^2 \left[x^2 - \theta \frac{b}{a} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] - \frac{m\lambda}{a} \cdot \frac{b}{a} \left(\frac{x-\delta}{1-\delta} \right)^{2\lambda^2} \\ & - \frac{\lambda}{a(1-\delta)} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} \left[1 - x \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] \end{aligned}$$

Now, $\psi_{2,b}$ and $\psi_{5,a,b}$ must be positive on the right interval, which is $[0, \frac{1-\theta}{m}]$ for ψ_2 , and $[\delta, 1]$ for ψ_5 . Let us evaluate both in the right end of their interval:

$$\psi_{2,b} \left(\frac{1-\theta}{m} \right) = -b\lambda^2 (1-\theta) - \lambda^2 \left(\frac{1-\theta}{m} \right)^2 + \lambda^2 \quad (2.7)$$

$$\psi_{5,a,b}(1) = \lambda^2 \left[1 - \theta \frac{b}{a} \right] - \frac{m}{a} \lambda \cdot \frac{b}{a} \quad (2.8)$$

From 2.8 we get:

$$\psi_{5,a,b}(1) \geq 0 \Rightarrow \theta \leq \frac{a}{b} \Rightarrow 1 - \theta \geq 1 - \frac{a}{b} \Rightarrow b(1 - \theta) \geq b - a \geq 1$$

Note that $b - a \geq 1$ follows from our assumption that $a \neq b$, and an immediate consequence is $\theta \neq 1$. Let us substitute in 2.7:

$$\psi_{2,b} \left(\frac{1-\theta}{m} \right) = \lambda^2 \left(1 - b(1 - \theta) - \left(\frac{1-\theta}{m} \right)^2 \right) \leq -\lambda^2 \left(\frac{1-\theta}{m} \right)^2 < 0$$

and the second factor cannot cancel since $\theta \neq 1$. Thus we need to consider a different region if we want a chance for it to be invariant. Instead, we consider in the next section a broader version of problem (2.1).

2.2 A different formulation

We consider the following version of problem 2.1:

$$\begin{cases} \dot{X}_j = -\nu\lambda_{|j|}^2 X_j + d_j \lambda_{|j|}^\beta X_j^2 - \sum_{k \in O_j} d_k \lambda_{|k|}^\beta X_j X_k \\ X_j(0) = x_j \end{cases} \quad \forall j \in J, \forall t \geq 0 \quad (2.9)$$

where $d_j \in [0, 1]$ for all $j \in J$: they are the weights associated with the energy cascade between nodes. The closer d_j is to 1, the greater is the flux from $\bar{j}$ to j . Moreover, define $D_j = \{d_k : k \in O_j\}$: we will assume that D_j is the same set for all j , thus we will call it D . We assume that $1 \in D$, which means there is always $k \in O_j$ such that $d_k = 1$. Finally, in this section we consider $\beta = \frac{5}{2}$.

Consider the change of variables $Y_j = \lambda_{|j|}^{\frac{1}{2}+\varepsilon} X_j$. The dynamics of $\{Y_j\}$ is

$$\begin{cases} \dot{Y}_j = -\nu\lambda_{|j|}^2 Y_j + \lambda^{1+2\varepsilon} \cdot d_j \lambda_{|j|}^{2-\varepsilon} Y_j^2 - \lambda^{2-\varepsilon} \cdot \lambda_{|j|}^{2-\varepsilon} \sum_{k \in O_j} d_k Y_j Y_k \\ Y_j(0) = y_j \end{cases} \quad (2.10)$$

for $j \in J$ and $t \in [0, T]$. Clearly y_j is the image of x_j after changing variables. Consider now a finite dimensional version of the equations for Y : for every $N \geq 1$ we will denote with $\left\{Y_j^{(N)}\right\}_{|j| \leq N}$ the solution to

$$\begin{cases} \dot{Y}_j^{(N)} = -\nu\lambda_{|j|}^2 Y_j^{(N)} + \lambda^{1+2\varepsilon} \cdot d_j \lambda_{|j|}^{2-\varepsilon} \left(Y_j^{(N)}\right)^2 - \lambda^{2-\varepsilon} \cdot \lambda_{|j|}^{2-\varepsilon} \sum_{k \in O_j} d_k Y_j^{(N)} Y_k^{(N)} \\ Y_j(0) = y_j \end{cases}$$

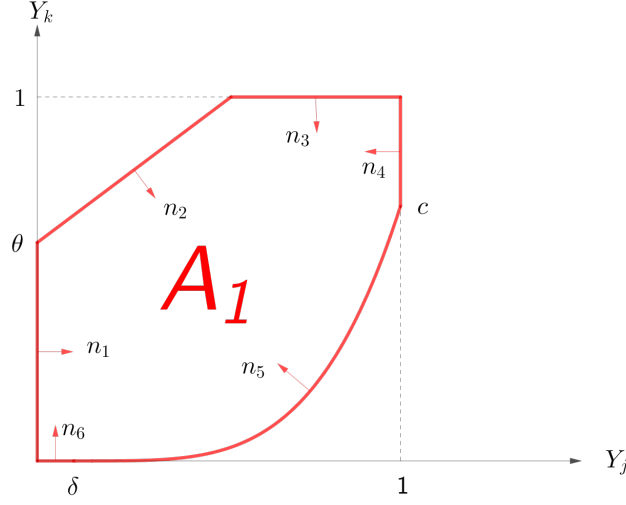
for $1 \leq |j| \leq N$, i.e. we are taking into consideration just the first N generations. Now we introduce the following family of regions contained in $\mathbb{R}^2$:

$$A_{j,k} := \left\{ (x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, \frac{d_k}{\Gamma} h(x) < y < \frac{d_k}{\Gamma} g(x) \right\}$$

where $\Gamma = \sum_{i=1}^8 d_i^2$,

$$g(x) = \min \left\{ mx + \theta, \frac{\Gamma}{d_k} \right\}, \quad h(x) = \begin{cases} 0 & x \leq \delta \\ c \left(\frac{x - \delta}{1 - \delta} \right)^{\lambda^2} & x > \delta \end{cases}$$

The expression for g is to make sure that $A_{j,k}$ does not trespass the line $y = 1$. This is the most general formulation. We are interested when D contains only two distinct elements, 1 and $d \neq 1$: hence $\Gamma = 1 + 7d^2$.



Lemma 2.11. (Invariant region, $d_1 = 1, d_2 = \dots = d_8 = d$)

Let $\frac{5}{6} \leq d \leq 1$. There exists $\delta \in (0, 1)$, $c \in (0, 1)$, θ s.t. $\frac{\theta}{1}d \in (\delta, d)$, $m > \frac{\Gamma}{d} - \theta$ and $\varepsilon_0 > 0$ such that: for every $\nu \geq 0$ and $\varepsilon \in (0, \varepsilon_0)$ the following statement is true. If $(y_j, y_k) \in A_{j,k}$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$,

$$\left(Y_j^{(N)}(t), Y_k^{(N)}(t) \right) \in A_{j,k} \quad \forall |j| \leq N, \forall k \in O_j, \forall t \geq 0$$

where $\{Y_j^{(N)}\}$ is the same as before.

Dim: for the sake of elegance, we drop the superscript (N) along the proof. It is sufficient to show that the time derivative of (Y_j, Y_k) points inward on the border of A when $(Y_j, Y_k) \in A$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$. We check by computing the scalar product of the inward normal on the border of A with the following vector field $\mathcal{B}$:

$$\begin{pmatrix} \dot{Y}_j \\ \dot{Y}_k \end{pmatrix} = \nu \lambda_{|j|}^2 \begin{pmatrix} -Y_j \\ -\lambda^2 Y_k \end{pmatrix} + \lambda^{1+2\varepsilon} \lambda_{|j|}^{2-\varepsilon} \begin{pmatrix} d_j Y_j^2 - \lambda^{6-2\beta-3\varepsilon} \sum_{l \in O_j} d_l Y_j Y_l \\ \lambda^{2-\varepsilon} \left(d_k Y_j^2 - \lambda^{6-2\beta-3\varepsilon} \sum_{m \in O_k} d_m Y_k Y_m \right) \end{pmatrix}$$

and verifying it is positive when $(Y_j, Y_k) \in A$ for all j such that $1 \leq |j| \leq N$ and for all $k \in O_j$. Since A is a convex set, we can compute separately the scalar product with just the viscous term and just the inviscid one.

We denote the inward normals as in figure, so their expressions are the following:

$$\vec{n}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{n}_3 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \vec{n}_4 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \vec{n}_6 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\vec{n}_2 = \begin{pmatrix} g'(x) \\ -1 \end{pmatrix}, \quad \vec{n}_5 = \begin{pmatrix} -h'(x) \\ 1 \end{pmatrix}$$

We start with the viscous term, which will be denoted as $\mathcal{B}_v$. Note that there is no need to consider the positive constant $\nu\lambda_n^2$.

The scalar products are evaluated on the respective pieces of the border of A :

$$\begin{aligned}\mathcal{B}_v \cdot \vec{n}_1 &= -Y_j = 0, & \mathcal{B}_v \cdot \vec{n}_3 &= \lambda^2 Y_k = \lambda^2 \\ \mathcal{B}_v \cdot \vec{n}_4 &= Y_j = 1, & \mathcal{B}_v \cdot \vec{n}_6 &= -\lambda^2 Y_k = 0\end{aligned}$$

We will abbreviate $g_k(x) = \frac{d_k}{\Gamma} g(x)$ and $h_k(x) = \frac{d_k}{\Gamma} h(x)$. Along g_k , $Y_k = g_k(x)$:

$$\mathcal{B}_v \cdot \vec{n}_2 = -xg'_k(x) + \lambda^2 Y_k = -xg'_k(x) + \lambda^2 g_k(x) = mx(-1 + \lambda^2) + \lambda^2 \theta > 0$$

Along h , $Y_k = h_k(x)$:

$$\begin{aligned}\mathcal{B}_v \cdot \vec{n}_5 &= xh'_k(x) - \lambda^2 Y_k = xh'_k(x) - \lambda^2 h_k(x) = x \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2-1} - \lambda^2 h(x) \\ &\geq \frac{\lambda^2 c}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} - \lambda^2 h(x) = \frac{\lambda^2}{1-\delta} h(x) - \lambda^2 h(x) = \lambda^2 h \frac{\delta}{1-\delta} > 0\end{aligned}$$

We proceed with the inviscid term, which will be denoted as $\mathcal{B}_i$. Just as before, there is no need to consider the positive constant.

We will use again $x = Y_j$ and for simplicity we set $\gamma = 1 - 3\varepsilon$:

$$\begin{aligned}\mathcal{B}_i \cdot \vec{n}_1 &= d_j Y_j^2 - \lambda^\gamma \sum_{l \in O_j} d_l Y_j Y_l = d_j Y_j^2 \geq 0 \\ \mathcal{B}_i \cdot \vec{n}_6 &= \lambda^{2-\varepsilon} \left(d_k Y_j^2 - \lambda^\gamma \sum_{m \in O_k} d_m Y_k Y_m \right) = \lambda^{2-\varepsilon} d_k Y_j^2 \geq 0\end{aligned}$$

Consider the piece of the border of A corresponding to $\vec{n}_3$. Here $Y_j \leq 1$, $Y_k = 1$ and $(Y_k, Y_m) \in A$ for all $m \in O_k$ so that we have $Y_m \geq c \cdot \frac{d_m}{\Gamma}$ for all $m \in O_k$. Let us exploit these inequalities:

$$\mathcal{B}_i \cdot \vec{n}_3 = \lambda^{2-\varepsilon} \left(-d_k Y_j^2 + \lambda^\gamma \sum_{m \in O_k} d_m Y_k Y_m \right) \geq \lambda^{2-\varepsilon} \left(-1 + \lambda^\gamma c \sum_{m \in O_k} \frac{d_m^2}{\Gamma} \right)$$

and the sum is equal to 1 by the definition of Γ .

Consider the piece of the border of A corresponding to $\vec{n}_4$. Here $Y_j = 1$, $Y_j \leq 1$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \geq c \cdot \frac{d_l}{\Gamma}$ for all $l \in O_j$. Let us exploit these inequalities:

$$\mathcal{B}_i \cdot \vec{n}_4 = -d_j Y_j^2 + \lambda^\gamma \sum_{l \in O_j} d_l Y_j Y_l \geq -1 + \lambda^\gamma c \sum_{l \in O_j} \frac{d_l^2}{\Gamma} \geq \lambda^\gamma c - 1$$

We choose c in order to have $\lambda^\gamma c = 1$, so the previous are positive.

Let $Y_j = x$. Consider the piece of the border of A corresponding to $\vec{n}_2$. Here $Y_k = g_k(x)$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \leq g_l(x)$. Also $(Y_m, Y_k) \in A$ for all $m \in O_k$ so that we have $Y_m \geq h_m(Y_k) = h_m(g_k(x))$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_2 &= g'_k(x) \left(d_j Y_j^2 - \lambda^\gamma \sum_{l \in O_j} d_l x Y_l \right) - \lambda^{2-\varepsilon} \left(d_k x^2 - \lambda^\gamma \sum_{m \in O_k} d_m Y_k Y_m \right) \\ &\geq g'_k(x) (-\lambda^\gamma x g(x)) - \lambda^{2-\varepsilon} \left[d_k x^2 - \lambda^\gamma g_k(x) c \left(\frac{g_k(x) - \delta}{1 - \delta} \right)^4 \right] \end{aligned}$$

We suppose here that $d_k = 1$ and we will deal later with the case $d_k = d$. Here $g_k(x) = \frac{1}{\Gamma} (mx + \theta)$:

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -\lambda^\gamma \frac{m}{\Gamma} x (mx + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (\lambda^\gamma c) g_k(x) \left(\frac{g_k(x) - \delta}{1 - \delta} \right)^4$$

Remember $\lambda^\gamma c = 1$:

$$\mathcal{B}_i \cdot \vec{n}_2 \geq -\lambda^\gamma \frac{m}{\Gamma} x (mx + \theta) - \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} g_k(x) \left(\frac{g_k(x) - \delta}{1 - \delta} \right)^4$$

Consider the piece of the border of A corresponding to $\vec{n}_5$. Here $Y_k = h_k(x)$ and $(Y_j, Y_l) \in A$ for all $l \in O_j$ so that we have $Y_l \geq h_l(x)$. Also $(Y_k, Y_m) \in A$ for all $m \in O_k$ so that we have $Y_m \leq g_m(Y_k) = g_m(h_k(x))$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &= \lambda^{2-\varepsilon} \left(d_k x^2 - \lambda^\gamma \sum_{m \in O_k} d_m Y_k Y_m \right) - h'_k(x) \left(d_j Y_j^2 - \lambda^\gamma \sum_{l \in O_j} d_l x Y_l \right) \\ &\geq \lambda^{2-\varepsilon} \left[d_k x^2 - \lambda^\gamma h_k(x) (mh_k + \theta) \right] - h'_k(x) \left[1 - \lambda^\gamma x h(x) \right] \end{aligned}$$

Substitute for h :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[d_k x^2 - \theta (\lambda^\gamma c) \frac{d_k}{\Gamma} \left(\frac{x - \delta}{1 - \delta} \right)^4 \right] - m \lambda^{2-\varepsilon} \cdot \lambda^\gamma c^2 \frac{d_k^2}{\Gamma^2} \left(\frac{x - \delta}{1 - \delta} \right)^8 \\ &\quad - h'_k(x) \left[1 - (\lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^4 x \right] \end{aligned}$$

Substitute for h' :

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[d_k x^2 - \theta (\lambda^\gamma c) \frac{d_k}{\Gamma} \left(\frac{x - \delta}{1 - \delta} \right)^4 \right] - m \lambda^{2-\varepsilon} \cdot \lambda^\gamma c^2 \frac{d_k^2}{\Gamma^2} \left(\frac{x - \delta}{1 - \delta} \right)^8 \\ &\quad - \frac{4c}{1 - \delta} \frac{d_k}{\Gamma} \left(\frac{x - \delta}{1 - \delta} \right)^3 \left[1 - (\lambda^\gamma c) \left(\frac{x - \delta}{1 - \delta} \right)^4 x \right] \end{aligned}$$

Multiply and divide for λ^γ the second and the third term. Since $\lambda^\gamma c = 1$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[d_k x^2 - \theta \frac{d_k}{\Gamma} \left(\frac{x-\delta}{1-\delta} \right)^4 \right] - \frac{m}{\lambda^\gamma} \lambda^{2-\varepsilon} \frac{d_k^2}{\Gamma^2} \left(\frac{x-\delta}{1-\delta} \right)^8 \\ &\quad - \frac{4}{\lambda^\gamma (1-\delta)} \frac{d_k}{\Gamma} \left(\frac{x-\delta}{1-\delta} \right)^3 \left[1 - \left(\frac{x-\delta}{1-\delta} \right)^4 x \right] \end{aligned}$$

Multiply both sides for $\frac{\Gamma}{d_k}$:

$$\begin{aligned} \frac{\Gamma}{d_k} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[\Gamma x^2 - \theta \left(\frac{x-\delta}{1-\delta} \right)^4 \right] - \frac{m}{\lambda^\gamma} \lambda^{2-\varepsilon} \frac{d_k}{\Gamma} \left(\frac{x-\delta}{1-\delta} \right)^8 \\ &\quad - \frac{4}{\lambda^\gamma (1-\delta)} \left(\frac{x-\delta}{1-\delta} \right)^3 \left[1 - \left(\frac{x-\delta}{1-\delta} \right)^4 x \right] \end{aligned}$$

Since $d_k \leq 1$:

$$\begin{aligned} \frac{\Gamma}{d_k} \mathcal{B}_i \cdot \vec{n}_5 &\geq \lambda^{2-\varepsilon} \left[\Gamma x^2 - \theta \left(\frac{x-\delta}{1-\delta} \right)^4 \right] - \frac{m}{\lambda^\gamma} \lambda^{2-\varepsilon} \frac{1}{\Gamma} \left(\frac{x-\delta}{1-\delta} \right)^8 \\ &\quad - \frac{4}{\lambda^\gamma (1-\delta)} \left(\frac{x-\delta}{1-\delta} \right)^3 \left[1 - \left(\frac{x-\delta}{1-\delta} \right)^4 x \right] \end{aligned}$$

Recall that Γ is a function of d , i.e. $\Gamma = 1 + 7d^2$.

Let $\psi_{2,d}(x)$, $\psi_{5,d}(x)$ be the lower bounds for $\mathcal{B}_i \cdot \vec{n}_2$ and $\frac{\Gamma}{d_k} \mathcal{B}_i \cdot \vec{n}_5$ when $\varepsilon = 0$:

$$\begin{aligned} \psi_{2,d}(x) &= -\lambda \frac{m}{\Gamma} x (mx + \theta) - \lambda^2 x^2 + \lambda^2 \frac{1}{\Gamma} (mx + \theta) \left(\frac{\frac{1}{\Gamma} (mx + \theta) - \delta}{1-\delta} \right)^4 \\ \psi_{5,d}(x) &= \lambda^2 \left[\Gamma x^2 - \theta \left(\frac{x-\delta}{1-\delta} \right)^4 \right] - \frac{m\lambda}{\Gamma} \left(\frac{x-\delta}{1-\delta} \right)^8 \\ &\quad - \frac{\lambda}{1-\delta} \left(\frac{x-\delta}{1-\delta} \right)^3 \left[1 - x \left(\frac{x-\delta}{1-\delta} \right)^{\lambda^2} \right] \end{aligned}$$

Now we are going to deal with $d_k = d$. The starting estimate was

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_2 &= g'_k(x) \left(d_j Y_j^2 - \lambda^\gamma \sum_{l \in O_j} d_l x Y_l \right) - \lambda^{2-\varepsilon} \left(d_k x^2 - \lambda^\gamma \sum_{m \in O_k} d_m Y_k Y_m \right) \\ &\geq g'_k(x) (-\lambda^\gamma x g(x)) - \lambda^{2-\varepsilon} \left[d_k x^2 - \lambda^\gamma g_k(x) c \left(\frac{g_k(x) - \delta}{1-\delta} \right)^4 \right] \end{aligned}$$

From now on it is implicate that $g_k(x) = \frac{d}{\Gamma} (mx + \theta)$:

$$\begin{aligned} \mathcal{B}_i \cdot \vec{n}_2 &\geq -\lambda^\gamma \frac{md}{\Gamma} x (mx + \theta) - d \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} (\lambda^\gamma c) g_k(x) \left(\frac{g_k(x) - \delta}{1-\delta} \right)^4 \\ &\geq -\lambda^\gamma \frac{md}{\Gamma} x (mx + \theta) - d \lambda^{2-\varepsilon} x^2 + \lambda^{2-\varepsilon} g_k(x) \left(\frac{g_k(x) - \delta}{1-\delta} \right)^4 \end{aligned}$$

For what concerns $\mathcal{B}_i \cdot \vec{n}_5$, there is no difference from the previous case. This leads us to the following definition when $\varepsilon = 0$:

$$\psi_{2,d}^*(x) = -\lambda \frac{md}{\Gamma} x (mx + \theta) - d\lambda^2 x^2 + \lambda^2 \frac{d}{\Gamma} (mx + \theta) \left(\frac{\frac{d}{\Gamma} (mx + \theta) - \delta}{1 - \delta} \right)^4$$

Now, we want to verify that $\psi_{2,d}$, $\psi_{2,d}^*$ and $\psi_{5,d}$ are positive on the right interval. These are $[0, \frac{\Gamma-\theta}{m}]$ for $\psi_{2,d}$, $[0, \frac{\frac{\Gamma}{d}-\theta}{m}]$ for $\psi_{2,d}^*$ and $[\delta, 1]$ for $\psi_{5,d}$. Let us evaluate both the second one and the third one in the interval's right end:

$$\psi_{2,d}^* \left(\frac{\frac{\Gamma}{d} - \theta}{m} \right) = -\lambda \left(\frac{\Gamma}{d} - \theta \right) - d\lambda^2 \left(\frac{\frac{\Gamma}{d} - \theta}{m} \right)^2 + \lambda^2 \geq 0 \quad (2.12)$$

$$\psi_{5,d}(1) = \lambda^2 (\Gamma - \theta) - \frac{m\lambda}{\Gamma} \geq 0 \quad (2.13)$$

From the second one get some obvious estimates like $\theta \leq \Gamma$ and $m \leq 2\Gamma^2$. This suggests to put $\theta = c_1\Gamma$, $m = c_2\Gamma^2$ to find conditions on c_1, c_2 . The (2.13) yields $2(1 - c_1) - c_2 \geq 0$. Let us assume $c_2 = 2(1 - c_1)$ and substitute:

$$\begin{aligned} \psi_{2,d}^* \left(\frac{\frac{\Gamma}{d} - \theta}{m} \right) &= -\lambda \left(\frac{1}{d} - c_1 \right) \Gamma - d\lambda^2 \left(\frac{\frac{1}{d} - c_1}{c_2} \right)^2 \frac{1}{\Gamma^2} + \lambda^2 \\ &= -\lambda \left(\frac{1}{d} - c_1 \right) \Gamma - d \frac{\lambda^2}{4} \left(\frac{\frac{1}{d} - c_1}{1 - c_1} \right)^2 \frac{1}{\Gamma^2} + \lambda^2 \\ &= -\lambda \left(\frac{1}{d} - c_1 \right) (1 + 7d^2) - \left(\frac{\frac{1}{d} - c_1}{1 - c_1} \right)^2 \frac{d}{(1 + 7d^2)^2} + \lambda^2 \end{aligned}$$

If we do not consider the mid term, which is negligible for d in a neighborhood of 1, we get the following inequality:

$$\lambda^2 - \lambda \left(\frac{1}{d} - c_1 \right) (1 + 7d^2) \geq 0$$

When $d = \frac{5}{6}$, we find $c_1 \geq 0,859\dots$ This is a necessary condition for the positivity of $\psi_{2,d}^*$ and $\psi_{5,d}$ on their intervals, but it is not a sufficient one.

We need some monotonicity results: $\psi_{5,d}(x)$ is an increasing function in Γ , hence increasing in d . Let us focus on $\psi_{2,d}^*(x)$:

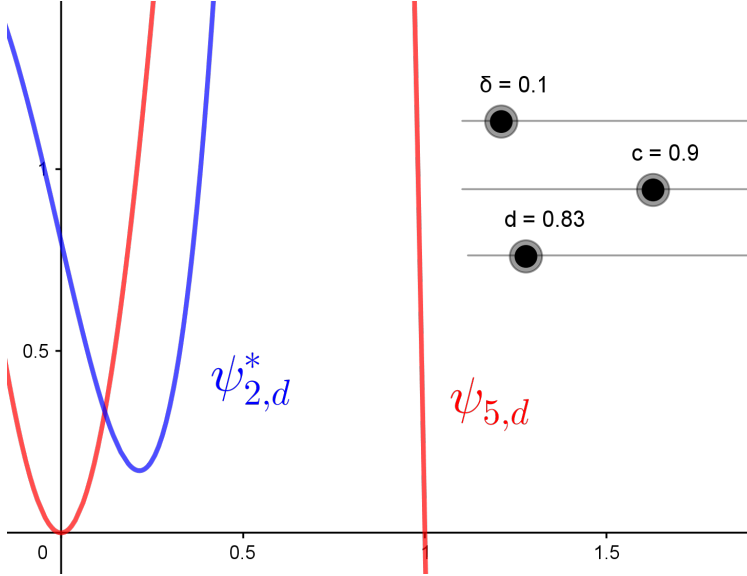
$$\psi_{2,d}^*(x) = -\lambda \frac{md}{\Gamma} x (mx + \theta) - d\lambda^2 x^2 + \lambda^2 \frac{d}{\Gamma} (mx + \theta) \left(\frac{\frac{d}{\Gamma} (mx + \theta) - \delta}{1 - \delta} \right)^4$$

Remember that $m \sim \Gamma^2$ and $\theta \sim \Gamma$. Since d is in a neighborhood of 1, Γ is in a neighborhood of 8, so we will determine the behaviour of $\psi_{2,d}^*$ by looking at the highest powers of Γ : in the meanwhile we can consider negligible δ , since we can choose it to be arbitrarily small.

The first term goes as $-\Gamma^3$ while the third term goes as $+\Gamma^5$: we conclude that $\psi_{2,d}^*(x)$ is an increasing function in Γ , hence increasing in d .

So we just need to check that both $\psi_{2,d}^*$ and $\psi_{5,d}$ are positive when $d = \frac{5}{6} \Rightarrow \Gamma = \frac{211}{36}$. We plotted for:

$$c_1 = \frac{9}{10}, \quad m = 2(1 - c_1)\Gamma^2, \quad \theta = c_1\Gamma, \quad \delta = \frac{1}{10}$$



It is not surprising that $\psi_{5,d}(x)$ cancels exactly when $x = 1$ since we assumed the equality in (2.13). We still have to discuss the positivity of $\psi_{2,d}$:

$$\psi_{2,d}(x) = -\lambda \frac{m}{\Gamma} x (mx + \theta) - \lambda^2 x^2 + \lambda^2 \frac{1}{\Gamma} (mx + \theta) \left(\frac{\frac{1}{\Gamma} (mx + \theta) - \delta}{1 - \delta} \right)^4$$

Here we do not need any plot: we just show that $\psi_{2,d} \geq \psi_{2,d}^*$ for any $\frac{5}{6} \leq d \leq 1$. Indeed, the negative terms in $\psi_{2,d}(x)$ are obtained multiplying by $\frac{1}{d}$ the negative terms in $\psi_{2,d}^*(x)$; on the other hand, considering δ negligible, the positive terms in $\psi_{2,d}(x)$ are obtained multiplying by $\frac{1}{d^5}$ the positive terms in $\psi_{2,d}^*(x)$. Since $d \leq 1$, we have $\frac{1}{d^5} \geq \frac{1}{d}$ and we conclude that $\psi_{2,d} \geq \psi_{2,d}^*$.

It is obvious to verify that θ and m , chosen as described above, satisfy the hypothesis of the lemma. Since the estimate for $\mathcal{B}_i \cdot \vec{n}_2$ and $\frac{\Gamma}{d_k} \mathcal{B}_i \cdot \vec{n}_5$ are continuous functions in ε and they are positive for $\varepsilon = 0$, we conclude that there exists ε_0 such that scalar products are positive for all $\varepsilon \in [0, \varepsilon_0]$, which is equivalent to the lemma. $\square$

2.3 Regularity in the d-tree case

In this section we make considerations about problem 2.1 but the proofs are also valid for 2.9. We consider the space of square summable, real sequences:

$$H = \left\{ x = \{x_j\}_{j \in J} \subseteq \mathbb{R} : \|x\|_H^2 := \sum_{j \in J} x_j^2 < \infty \right\}$$

A Leray-Hopf solution to 2.1 is a weak solution X with values in H and satisfying the following, called energy inequality,

$$\|X(t)\|_H^2 + 2\nu \sum_{j \in J} \int_s^t (\lambda_{|j|} X_j(r))^2 dr \leq \|X(s)\|_H^2$$

for a.e. $s \geq 0$ and $t > s$.

In [6] it has been proved that positive solutions with initial condition $X(0) \in H$ are Leray-Hopf solutions and the energy inequality holds for all $t > s$.

Proposition 2.14. (*Uniqueness*)

Let $X = \{X_j\}_{j \in J}$ be a positive solution to 2.1 on $[0, T]$ with initial condition $X(0) \in H$ and such that

$$\sup_{t \in [0, T], j \in J} [\lambda_{|j|}^{\beta-3} X_j(t)] \leq C$$

for some $C \in \mathbb{R}$. Then X is the unique solution with initial condition $X(0)$.

Dim.: let $Y = \{Y_j\}$ be another solution with the same initial condition. We set $Z_j = Y_j - X_j$, $W_j = Y_j + X_j$ and let us compute the dynamics for Z :

$$\begin{aligned} \dot{Z}_j &= -\nu \lambda_{|j|}^2 Z_j + \lambda_{|j|}^\beta Z_{\bar{j}} W_{\bar{j}} - \frac{1}{2} \lambda_{|k|}^\beta \sum_{k \in O_j} (Z_j W_k + Z_k W_j) \\ \dot{Z}_j^2 &= -2\nu \lambda_{|j|}^2 Z_j^2 + 2\lambda_{|j|}^\beta Z_j Z_{\bar{j}} W_{\bar{j}} - \lambda_{|k|}^\beta \sum_{k \in O_j} Z_j^2 W_k - \lambda_{|k|}^\beta \sum_{k \in O_j} Z_j Z_k W_j \\ \frac{1}{2^{|j|+1}} \dot{Z}_j^2 &= -\nu \frac{\lambda_{|j|}^2}{2^{|j|}} Z_j^2 + \frac{\lambda_{|j|}^\beta}{2^{|j|}} Z_j Z_{\bar{j}} W_{\bar{j}} - \sum_{k \in O_j} \frac{\lambda_{|k|}^\beta}{2^{|k|}} Z_j^2 W_k - \sum_{k \in O_j} \frac{\lambda_{|k|}^\beta}{2^{|k|}} Z_k Z_j W_j \end{aligned}$$

We used $\dot{Z}^2 = 2Z\dot{Z}$. Note that $Z_0 = 0$ since X and Y share the initial condition. Now fix $N \geq 1$ and set

$$\psi_N(t) = \sum_{j \in J, |j|=1}^N \frac{1}{2^{|j|+1}} Z_j^2$$

For the sake of elegance, we will denote as $\sum_{|j| \leq N} \frac{1}{2^{|j|+1}} Z_j^2$. We already know the time derivative of each member of the sum. In order to compute the time derivative of ψ_N we will make use of a telescopicity argument:

$$\begin{aligned} \frac{d}{dt} \psi_N(t) &= -\nu \sum_{|j| \leq N} \frac{\lambda_{|j|}^2}{2^{|j|}} Z_j^2 - \sum_{|j| \leq N} \sum_{k \in O_j} \frac{\lambda_{|k|}^\beta}{2^{|k|}} Z_j^2 W_k \\ &\quad + \sum_{|j| \leq N} \frac{\lambda_{|j|}^\beta}{2^{|j|}} Z_j Z_{\bar{j}} W_{\bar{j}} - \sum_{|j| \leq N} \sum_{k \in O_j} \frac{\lambda_{|k|}^\beta}{2^{|k|}} Z_k Z_j W_j \end{aligned}$$

It is clear that the terms in the second row cancel each other. Since we want an upper bound for $\frac{d}{dt}\psi_N(t)$, we do not consider the first two sums. It becomes:

$$\begin{aligned} \frac{d}{dt}\psi_N(t) &\leq \sum_{|j|=1} \frac{\lambda^\beta}{2} Z_j Z_0 W_0 - \sum_{|j|=N} \sum_{k \in O_j} \frac{\lambda_{N+1}^\beta}{2^{N+1}} Z_k Z_j W_j \\ &= - \sum_{|j|=N} \sum_{k \in O_j} \frac{\lambda_{N+1}^\beta}{2^{N+1}} Z_k Z_j W_j = - \sum_{|j|=N} \sum_{k \in O_j} \frac{\lambda_{N+1}^\beta}{2^{N+1}} (Y_k - X_k) (Y_j^2 - X_j^2) \end{aligned}$$

thanks to $Z_0 = 0$; observe also that the last fraction is $\lambda_{N+1}^{\beta-1}$.

We change the sign in front of the sum, then we throw away all negative terms:

$$\begin{aligned} \frac{d}{dt}\psi_N(t) &\leq \sum_{|j|=N} \sum_{k \in O_j} \lambda_{N+1}^{\beta-1} (X_k Y_j^2 + Y_k X_j^2 - Y_j^2 Y_k - X_j^2 X_k) \\ &\leq \sum_{|j|=N} \sum_{k \in O_j} \lambda_{N+1}^{\beta-1} (X_k Y_j^2 + Y_k X_j^2) = \sum_{|j|=N} \sum_{k \in O_j} \lambda_N^2 \lambda_{N+1}^{\beta-3} (X_k Y_j^2 + Y_k X_j^2) \\ &\leq \sum_{|j|=N} \sum_{k \in O_j} \left[\lambda_{N+1}^2 \left(\lambda_{N+1}^{\beta-3} Y_j \right) X_k Y_j + \lambda_{N+1}^2 \left(\lambda_{N+1}^{\beta-3} X_j \right) Y_k X_j \right] \end{aligned}$$

Now we are going to use the hypothesis on the sup. Note that it is irrelevant whether the coefficient is $\lambda_{N+1}^{\beta-3}$ or $\lambda_N^{\beta-3}$ since they differ for a constant:

$$\begin{aligned} \frac{d}{dt}\psi_N(t) &\leq \sum_{|j|=N} \sum_{k \in O_j} \lambda_{N+1}^2 C (X_k Y_j + Y_k X_j) \\ &\leq \sum_{|j|=N} \sum_{k \in O_j} \lambda_{N+1}^2 C \left(\frac{X_k^2 + Y_j^2 + Y_k^2 + X_j^2}{2} \right) \end{aligned}$$

where the last inequality comes from a square of a binomial being positive.

By applying the comparison theorem for differential equations, since $\psi_N(0) = 0$:

$$\psi_N(t) \leq \sum_{|j|=N} \sum_{k \in O_j} \int_0^t \lambda_{N+1}^2 C \left(\frac{X_k^2 + Y_j^2 + Y_k^2 + X_j^2}{2} \right) ds$$

Since X and Y are Leray-Hopf solutions, the right hand side tends to 0 as $N \rightarrow \infty$. Hence $\psi_N(t) \rightarrow 0$ for any $t \geq 0$. Since ψ_n is an increasing function of n , this implies $\psi_n(t) = 0$ for all $n \geq 1$, thus we have $X_j(t) = Y_j(t)$ as desired.

Corollary 2.15. *If $\beta \leq 3$ in the formulation of problem 2.1, there is a unique solution for any positive initial condition in H .*

Dim.: we check the boundness condition to apply the previous proposition. Note that $\lambda_{|j|}^{\beta-3} \leq 1$ since $\beta \leq 3$. Hence the following inequalities

$$\lambda_{|j|}^{\beta-3} X_j(t) \leq X_j(t) \leq \|X(t)\|_H < \infty$$

hold for all $j \in J$ and $t \geq 0$. The boundness condition on the sup is verified. $\square$

We will now introduce the main theorem regarding regularity in the d -weighted case, which means we are going to consider the formulation presented in section 2.2. In the proof we will make use of Lemma 2.11.

Theorem 2.16. *Let $\beta = \frac{5}{2}$. Set*

$$\mathcal{D}^\infty = \left\{ x = \{x_j\}_{j \in J} : \sup_{j \in J} \lambda_{|j|}^\gamma |x_j| < \infty \text{ for all } \gamma > 0 \right\}$$

For any positive initial condition $X(0) \in H$, assume there exists a weak solution to problem 2.9. Then it is smooth, i.e. $X(t) \in \mathcal{D}^\infty$ for every $t > 0$.

Dim.: let X be a weak solution starting at $X(0) = x$.

It is not restrictive to assume that $\sup_j \lambda_{|j|} x_j < \infty$, a smooth initial condition. Indeed, if that is not the case, we can consider a sequence $\{t_k\}$ such that

$$t_k \rightarrow 0 \quad \text{and} \quad \sup_j \lambda_{|j|} X_j(t_k) < \infty \quad \text{for all } k$$

The existence of such a sequence follows from the energy inequality:

$$\sum_{j \in J} \int_s^t (\lambda_{|j|} X_j(r))^2 dr < \infty \quad \text{for all } t > s$$

Indeed if $\sup_j \lambda_{|j|} X_j(r) = \infty$ on a nonzero measure set, we have contradiction. So $\sup_j \lambda_{|j|} X_j(t) < \infty$ for a.e. $t > 0$ and it allows us to choose such a sequence. Proving that $X(t) \in \mathcal{D}^\infty$ for all $t > 0$ is equivalent to proving that, for each k , $X(t) \in \mathcal{D}^\infty$ for all $t \geq t_k$. Since the latter problem has smooth initial conditions for each k , the former may have as well.

Let ε_0 be as in the statement of Lemma 2.11. We choose $\varepsilon = \min\{\frac{1}{2}, \varepsilon_0\}$: since $\frac{1}{2} + \varepsilon \leq 1$, we have $\lambda_{|j|}^{\frac{1}{2}+\varepsilon} x_j \leq \lambda_{|j|} x_j$.

Since $\sup_j \lambda_{|j|} x_j$ is finite, same holds for the left side. Set $\sup_j \lambda_{|j|}^{\frac{1}{2}+\varepsilon} x_j = K_0$, we define for $j \in J$:

$$\bar{Y}_j(t) = \frac{\delta}{K_0} \lambda_{|j|}^{\frac{1}{2}+\varepsilon} X_j \left(\frac{\delta}{K_0} t \right)$$

with δ as in Lemma 2.11.

Since (Y_j) defined in 1.2 was a solution to problem 2.10, then $(\bar{Y}_j)$ is a solution to the same problem but with a different viscosity, namely $\bar{\nu} = \frac{\delta}{K_0} \nu$.

Now let $\bar{Y}_j^{(N)}$ be a solution to the N -dimensional version of problem 2.10. We can prove by a standard argument that it converges to $(\bar{Y}_j)$ as $N \rightarrow \infty$.

Our idea is to apply the invariant region lemma to $(\bar{Y}_j)$. We consider the finite dimensional approximations, which all share the initial conditions with $(\bar{Y}_j)$.

The following holds for all $N \geq 1$:

$$\sup_{|j| \leq N} \bar{Y}_j^{(N)}(0) \leq \frac{\delta}{K_0} \cdot \sup_{j \in J} \lambda_{|j|}^{\frac{1}{2}+\varepsilon} x_j = \frac{\delta}{K_0} \cdot K_0 = \delta$$

$$\begin{aligned} &\Rightarrow (\bar{Y}_j^{(N)}(0), \bar{Y}_k^{(N)}(0)) \in [0, \delta]^2 \subset A_{j,k}, \text{ for any } j, k \text{ such that } |j| \leq N, k \in O_j \\ &\Rightarrow (\bar{Y}_j^{(N)}(t), \bar{Y}_k^{(N)}(t)) \in A_{j,k}, \text{ for any } j, k \text{ such that } |j| \leq N, k \in O_j, t \geq 0, \end{aligned}$$

using Lemma 2.11. We conclude that $\bar{Y}_j^{(N)}(t) \leq 1$, for $|j| \leq N$, $t \geq 0$.

The same holds for $\bar{Y}_j(t)$ by convergence of finite dimensional approximations.

So:

$$\begin{aligned} \sup_{j \in J} \sup_{t \geq 0} \frac{\delta}{K_0} \lambda_{|j|}^{\frac{1}{2} + \varepsilon} X_j(t) &= \sup_{j \in J} \sup_{t \geq 0} \bar{Y}_j\left(\frac{K_0}{\delta} t\right) \leq 1 \\ \Rightarrow \sup_{j \in J} \sup_{t \geq 0} \lambda_{|j|}^{\frac{1}{2} + \varepsilon} X_j(t) &\leq \frac{K_0}{\delta} \end{aligned} \quad (2.17)$$

We will prove that there is $\eta > 0$ such that $X(t) \in \mathcal{D}^\infty$, for $t \in [0, \eta]$. A similar argument will work on any interval $[t_0, t_0 + \eta]$, allowing to cover the whole set of times $t > 0$. From now we write β instead of $\frac{5}{2}$ for the sake of elegance.

Let $V_j(t) = X_j(t) \exp[\nu \lambda_{|j|} t]$. We compute the dynamics for V_j :

$$\begin{cases} \dot{V}_j = -\nu \left(\lambda_{|j|}^2 - \lambda_{|j|} \right) V_j + \lambda_{|j|}^\beta V_j^2 - \lambda_{|k|}^\beta e^{-\nu \lambda_{|k|} t} \sum_{k \in O_j} V_j V_k \\ V_j(0) = X_j(0) \end{cases}$$

Our idea is to prove smoothness of X by showing that V is uniformly bounded in $n = |j|$; we will show later why the latter implies the former. Define

$$\mathcal{V}_\varepsilon = \left\{ x = \{x_j\}_{j \in J} : \|x\|_\varepsilon := \sup_{j \in J} \left(\lambda_{|j|}^{\beta-2+\varepsilon} |x_j| \right) < \infty \right\}$$

We also define a map $\mathcal{F}$ from the space $L^\infty([0, \eta]; \mathcal{V}_\varepsilon)$ into itself, $Z \mapsto \mathcal{F}(Z)$. The components $\mathcal{F}_j(Z)$ follow:

$$\begin{aligned} \mathcal{F}_j(Z)(t) &= \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \left(\lambda_{|j|}^\beta Z_j^2 - \lambda_{|k|}^\beta e^{-\nu \lambda_{|k|} s} \sum_{k \in O_j} Z_j Z_k \right) ds \\ &\quad + e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})t} X_j(0) \end{aligned}$$

for $t \geq 0$. The reason of this definition is simple: one can prove via method of variation of parameters that the integral formula for $V_j(t)$ is exactly $\mathcal{F}_j(V)(t)$ i.e. V is a fixed point for the map $\mathcal{F}$.

Still it is unclear whether or not $V(t) \in \mathcal{V}_\varepsilon$ for all $t \in [0, \eta]$. We will use Banach's fixed point theorem, also known as contraction mapping theorem. In order to do that, we need to show that for some $\eta > 0$, $R > 0$:

- $\mathcal{F}$ maps a ball of $L^\infty([0, \eta]; \mathcal{V}_\varepsilon)$ centered in $X(t) \equiv X(0)$ (initial condition) having radius R into itself.
- $\mathcal{F}$ is a contraction on this ball.

We will need the following, which come from the argument being in $[0, 1]$:

$$\begin{aligned} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} ds &\leq \int_0^t 1 ds = t \leq \eta \\ \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} ds &\leq \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})r} dr \leq \int_0^\infty e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})r} dr \\ &\leq \frac{1}{\nu(\lambda_{|j|}^2 - \lambda_{|j|})} \end{aligned}$$

Hence, it is obvious that for any $0 \leq \alpha \leq 1$ the next inequalities hold:

$$\begin{aligned} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} ds &\leq \eta^\alpha \cdot \left(\frac{1}{\nu(\lambda_{|j|}^2 - \lambda_{|j|})} \right)^{1-\alpha} \\ \Rightarrow \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} ds &\leq \eta^\alpha \cdot \left(\frac{2}{\nu\lambda_{|j|}^2} \right)^{1-\alpha} \end{aligned} \quad (2.18)$$

since $\lambda_{|j|}^2 - \lambda_{|j|} \geq \frac{1}{2}\lambda_{|j|}^2$.

For Z a.e. bounded on $[0, \eta]$, let $P_j(t) = \lambda_{|j|}^{\beta-2+\varepsilon} |\mathcal{F}_j(Z)(t) - X_j(t)|$. By the triangular inequality, and by multiplying and dividing for suitable powers of λ :

$$\begin{aligned} P_j(t) &\leq \lambda_{|j|}^{\beta-2+\varepsilon} |X_j(0)| \\ &+ \lambda_{|j|}^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \left(\lambda_{|j|}^\beta |Z_j^2| + \lambda_{|k|}^\beta \sum_{k \in O_j} |Z_j Z_k| \right) ds \\ &\leq \|X(0)\|_\varepsilon \\ &+ \lambda^{2\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \lambda_{|j|-1}^{\beta-2+\varepsilon} |Z_j| \cdot \lambda_{|j|-1}^{2-\varepsilon} \lambda_{|j|-1}^{\beta-2+\varepsilon} |Z_j| ds \\ &+ \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \sum_{k \in O_j} \lambda_{|j|}^{\beta-2+\varepsilon} |Z_j| \cdot \lambda_{|k|}^{2-\varepsilon} \lambda_{|k|}^{\beta-2+\varepsilon} |Z_k| ds \end{aligned}$$

Let $\|\cdot\|_{\infty, \varepsilon}$ be the sup norm in t of $\|\cdot\|_\varepsilon$. By taking out of the integrals the $\|\cdot\|_{\infty, \varepsilon}$ factors in the last two rows, we can find a constant $c > 0$ such that

$$P_j(t) \leq \|X(0)\|_\varepsilon + c \|Z\|_{\infty, \varepsilon}^2 \cdot \lambda_{|j|}^{2-\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} ds$$

We use 2.18 to estimate the last integral: by choosing $\alpha = \varepsilon/2$ we get

$$P_j(t) \leq \|X(0)\|_\varepsilon + \bar{c} \|Z\|_{\infty, \varepsilon}^2 \eta^{\frac{\varepsilon}{2}} \quad (2.19)$$

where $\bar{c} > 0$ depends only on λ, β, ν and ε .

For $Z, \bar{Z}$ a.e. bounded on $[0, \eta]$, let $Q_j(t) = \lambda_{|j|}^{\beta-2+\varepsilon} |\mathcal{F}_j(Z)(t) - \mathcal{F}_j(\bar{Z})(t)|$:

$$\begin{aligned} Q_j(t) &\leq \lambda_{|j|}^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \lambda_{|j|}^\beta |Z_j^2 - \bar{Z}_j^2| ds \\ &+ \lambda_{|j|}^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \lambda_{|k|}^\beta \left| \sum_{k \in O_j} Z_j Z_k - \sum_{k \in O_j} \bar{Z}_j \bar{Z}_k \right| ds \\ &\leq \lambda_{|j|}^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \lambda_{|j|}^\beta |Z_j - \bar{Z}_j| |Z_j + \bar{Z}_j| ds \\ &+ \lambda_{|j|}^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \lambda_{|k|}^\beta \sum_{k \in O_j} |Z_j Z_k - \bar{Z}_j \bar{Z}_k| ds \end{aligned}$$

We will make use of the following equality:

$$Z_j Z_k - \bar{Z}_j \bar{Z}_k = \frac{(Z_j + \bar{Z}_j)(Z_k - \bar{Z}_k) + (Z_j - \bar{Z}_j)(Z_k + \bar{Z}_k)}{2}$$

Apply this the second part of the sum:

$$\begin{aligned} Q_j(t) &\leq \lambda_{|j|}^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \lambda_{|j|}^\beta |Z_j - \bar{Z}_j| |Z_j + \bar{Z}_j| ds \\ &\quad + \lambda_{|j|}^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \frac{\lambda_{|k|}^\beta}{2} \sum_{k \in O_j} |Z_j + \bar{Z}_j| |Z_k - \bar{Z}_k| ds \\ &\quad + \lambda_{|j|}^{\beta-2+\varepsilon} \int_0^t e^{-\nu(\lambda_{|j|}^2 - \lambda_{|j|})(t-s)} \frac{\lambda_{|k|}^\beta}{2} \sum_{k \in O_j} |Z_k + \bar{Z}_k| |Z_j - \bar{Z}_j| ds \end{aligned}$$

By multiplying and dividing for suitable powers of λ , in a similar way to what we have done before, then by taking out of the integrals the $\|\cdot\|_{\infty, \varepsilon}$ factors in the last three rows, we can find a constant $c > 0$ such that

$$Q_j(t) \leq c \|Z - \bar{Z}\|_{\infty, \varepsilon} \|Z + \bar{Z}\|_{\infty, \varepsilon} \cdot \lambda_{|j|}^{2-\varepsilon} \int_0^t e^{-\nu(\lambda_n^2 - \lambda_n)(t-s)} ds$$

We use 2.18 to estimate the last integral: by choosing $\alpha = \varepsilon/2$ we get

$$Q_j(t) \leq \bar{c} \|Z - \bar{Z}\|_{\infty, \varepsilon} \|Z + \bar{Z}\|_{\infty, \varepsilon} \eta^{\frac{\varepsilon}{2}} \quad (2.20)$$

where $\bar{c} > 0$ is a constant depending only on λ, β, ν and ε .

Let us prove that $\mathcal{F}$ maps a ball of $L^\infty([0, \eta]; \mathcal{V}_\varepsilon)$ centered in $X(t) \equiv X(0)$, having radius R into itself. Suppose Z belongs to this ball, i.e.

$$\lambda_{|j|}^{\beta-2+\varepsilon} |Z_j(t) - X_j(t)| \leq R$$

for all $j \in J, t \in [0, \eta]$.

By the triangular inequality and a suitable choice of R , namely $R \geq 2\|X(0)\|_\varepsilon$:

$$\lambda_{|j|}^{\beta-2+\varepsilon} |Z_j(t)| \leq R + \lambda_{|j|}^{\beta-2+\varepsilon} |X_j(t)| \leq 2R \quad (2.21)$$

so $2R$ is an upper bound for $\|Z\|_{\infty, \varepsilon}$. We want to show that

$$\lambda_n^{\beta-2+\varepsilon} |\mathcal{F}_n(Z)(t) - X_n(t)| \leq R$$

for all $j \in J, t \in [0, \eta]$. The left hand side is exactly what we called $P_j(t)$ and we already have an upper bound for it. We substitute 2.21 in the estimate 2.19:

$$P_j(t) \leq \frac{1}{2}R + 4R^2c \cdot \eta^{\frac{\varepsilon}{2}} \leq R$$

and the last inequality is true if we choose η such that $\eta^{\frac{\varepsilon}{2}} \leq (8Rc)^{-1}$.

Let us prove that $\mathcal{F}$ is a contraction. We want to show that, for some $\Lambda < 1$,

$$\|\mathcal{F}(Z) - \mathcal{F}(\overline{Z})\|_{\infty, \varepsilon} \leq \Lambda \|Z - \overline{Z}\|_{\infty, \varepsilon}$$

which means

$$\lambda_{|j|}^{\beta-2+\varepsilon} |\mathcal{F}_j(Z)(t) - \mathcal{F}_j(\overline{Z})(t)| \leq \Lambda \|Z - \overline{Z}\|_{\infty, \varepsilon}$$

for all $j \in J$, $t \in [0, \eta]$. The left hand side is exactly what we called $Q_j(t)$ and we already have an upper bound for it. We substitute 2.21 in the estimate 2.20:

$$\begin{aligned} Q_j(t) &\leq c \|Z - \overline{Z}\|_{\infty, \varepsilon} \|Z + \overline{Z}\|_{\infty, \varepsilon} \eta^{\frac{\varepsilon}{2}} \\ &\leq c \|Z - \overline{Z}\|_{\infty, \varepsilon} \left(\|Z\|_{\infty, \varepsilon} + \|\overline{Z}\|_{\infty, \varepsilon} \right) \eta^{\frac{\varepsilon}{2}} \\ &\leq (4Rc \cdot \eta^{\frac{\varepsilon}{2}}) \|Z - \overline{Z}\|_{\infty, \varepsilon} \end{aligned}$$

We need to choose η such that $\eta^{\frac{\varepsilon}{2}} \leq (4Rc)^{-1}$ to have $\Lambda < 1$. This is easily implied by the previous condition, $\eta^{\frac{\varepsilon}{2}} \leq (8Rc)^{-1}$.

We conclude by Banach that there is a unique fixed point for the map $\mathcal{F}$ and this is necessarily V defined before. This allows us to conclude that $V(t) \in \mathcal{V}_\varepsilon$ for $t \in [0, \eta]$. It remains to show why $V(t) \in \mathcal{V}_\varepsilon$ implies $X(t) \in \mathcal{D}^\infty$. Since

$$\sup_{j \in J} |V_j(t)| \leq \sup_{j \in J} \lambda_{|j|}^{\beta-2+\varepsilon} |V_j(t)| < \infty$$

Now note that for sufficiently large $|j|$, say $|j| \geq N$, the following estimate holds since an exponentially-growing exponent beats a linearly-growing exponent:

$$\lambda_{|j|}^\gamma |X_j(t)| \leq e^{\nu \lambda_{|j|} t} |X_j(t)| \implies \sup_{|j| \geq N} \lambda_{|j|}^\gamma |X_j(t)| \leq \sup_{|j| \geq N} |V_j(t)| < \infty$$

for all $\gamma > 0$.

Obviously the same holds for $\sup_{|j| \leq N} \lambda_{|j|}^\gamma |X_j(t)|$ since we take sup on a finite set. Thus $X(t) \in \mathcal{D}^\infty$ as desired. $\square$

Appendice A

Foundations of stochastic analysis

In this section we will recall some key theorems about stochastic analysis in infinite dimensions. We are particularly interested in separable Banach spaces. For a complete dissertation see the textbook [1].

A.1 Random variables

If $(\Omega, \mathcal{F})$ and $(E, \mathcal{G})$ are two measurable spaces, then a mapping $X : \Omega \rightarrow E$ such that the set

$$\{\omega \in \Omega : X(\omega) \in A\}$$

belongs to $\mathcal{F}$ for arbitrary $A \in \mathcal{G}$ is called a measurable mapping or a random variable from $(\Omega, \mathcal{F})$ into $(E, \mathcal{G})$. It is also called simple if it takes only a finite number of values.

Lemma A.1. *Let E be a separable metric space with metric ρ and let X be an E -valued random variable. Then there exists a sequence $\{X_m\}$ of simple E -valued random variables such that the sequence $\rho(X_m(\omega), X(\omega))$ is monotonically decreasing to 0 for arbitrary ω .*

Assume that E is a metric space, then the Borel σ -field of E is the smallest σ -field containing all closed (or equivalently open) subsets of E . It will be denoted as $\mathcal{B}(E)$.

Proposition A.2. *Let E be a separable Banach space. Then $\mathcal{B}(E)$ is the smallest σ -field of subsets of E containing all sets of the form*

$$\{x \in E : \phi(x) \leq \alpha\}, \quad \phi \in E^*, \alpha \in \mathbb{R}$$

By Proposition A.2, it follows that a mapping $X : \Omega \rightarrow E$ is an E -valued random variable if and only if, for arbitrary $\phi \in E^*$, $\phi(X) : \Omega \rightarrow \mathbb{R}$ is a $\mathbb{R}$ -valued random variable.

A E -valued random variable on $(\Omega, \mathcal{F})$ is said to be Bochner integrable or simply integrable if

$$\int_{\Omega} \|X(\omega)\| \mathbf{P}(d\omega) < \infty$$

Let X be integrable; by Lemma A.1 there exists a sequence $\{X_m\}$ of simple random variables such that the sequence $\{\|X(\omega) - X_n(\omega)\|\}$ decreases to 0 for all $\omega \in \Omega$. The integral of X can be defined by

$$\int_{\Omega} X(\omega) \mathbf{P}(d\omega) = \lim_{n \rightarrow \infty} \int_{\Omega} X_n(\omega) \mathbf{P}(d\omega)$$

The integral defined this way is called Bochner's integral. The usual properties of linearity hold, moreover for all $B \in \mathcal{F}$:

$$\left\| \int_B X(\omega) \mathbf{P}(d\omega) \right\| \leq \int_B \|X(\omega)\| \mathbf{P}(d\omega)$$

Let U, H be two separable Hilbert spaces and let $L = L(U, H)$ the set of all linear bounded operators from U into H . The set L equipped with the operator norm becomes a Banach space, however if both spaces are infinite dimensional it is not a separable space.

The lack of separability of L implies that Bochner's definition can not be directly applied to the L -valued functions. It is convenient to introduce a weaker concept of measurability.

A function $\Phi : \Omega \rightarrow L$ is said to be strongly measurable if for arbitrary $u \in U$ it is measurable the function

$$\Phi(\cdot)u : (\Omega, \mathcal{F}) \rightarrow (H, \mathcal{B}(H))$$

If $\mathbf{P}$ is a nonnegative measure on $\mathcal{F}$ then Φ is said to be Bochner integrable if for arbitrary u the function $\Phi(\cdot)u$ is Bochner integrable. Also, it must exist a linear bounded operator $\Psi \in L(U, H)$ such that:

$$\Psi u = \int_{\Omega} \Phi(\omega)u \mathbf{P}(d\omega), \quad u \in U$$

The operator Ψ is called the strong Bochner's integral of Φ and is denoted as

$$\Psi = \int_{\Omega} \Phi(\omega) \mathbf{P}(d\omega)$$

Here follows another fundamental result about random variables on a Banach.

Proposition A.3. *Assume that E is a separable Banach space. Let X be a Bochner-integrable E -valued random variable defined on $(\Omega, \mathcal{F}, \mathbf{P})$ and let $\mathcal{G}$ be a σ -field, $\mathcal{G} \subset \mathcal{F}$. There exists a unique, up to a set of $\mathbf{P}$ -probability zero, integrable E -valued random variable Z such that*

$$\int_A X d\mathbf{P} = \int_A Z d\mathbf{P} \quad \forall A \in \mathcal{G}$$

This random variable Z will be denoted as $\mathbf{E}(X|\mathcal{G})$ and is called the conditional expectation of X given $\mathcal{G}$.

A.2 Probability measures

A probability measure on E will mean a probability measure on $(E, \mathcal{B}(E))$.

Let us remind some key concepts.

A second-countable space is a topological space whose topology has a countable base. More precisely, E is second countable if there exists a countable collection of open, non empty sets $\mathcal{U} = \{U_i\}_{i \in \mathbb{N}}$ such that any open set can be written as a union of elements of $\mathcal{U} \subseteq \mathcal{U}$.

A metric space E is totally bounded if and only if for every $\varepsilon > 0$ there exists a finite collection of open (or closed) balls of radius ε whose union is E . A related notion is a totally bounded set, in which only a subset of the space E needs to be covered by finitely many balls of given radius.

Theorem A.4. *A metric space E is separable if and only if second countable.*

Dim.: if E is second countable, let $\mathcal{U} = \{U_i\}_{i \in \mathbb{N}}$ be a countable base. Since they are not empty, we can choose $x_i \in U_i$: then $X = \{x_i\}_{i \in \mathbb{N}}$ is a countable set. It is also dense: for any $x \in E$ and $r > 0$:

$$B(x, r) = \bigcup_{k \in K \subseteq \mathbb{N}} U_k \Rightarrow d(x, x_i) \leq r \quad \forall i \in K$$

Conversely, if E is separable let $X = \{x_i\}_{i \in \mathbb{N}}$ be a dense and countable set. Define:

$$\mathcal{B} = \{ B(x_i, 1/n) : i \in \mathbb{N}, n \in \mathbb{N}_0 \}$$

This is a countable set: one can fix $s \in \mathbb{N}$ and count all the balls with $i + n = s$. Let τ be the topology induced by the metric: we will show that $\mathcal{B}$ is a base. Let U be an open set, $U \in \tau$ and take $y \in U$: by the characteristic property of open sets, there exists an open ball $B(y, \varepsilon) \subseteq U$. Take $n > 2/\varepsilon$: then

$$B\left(y, \frac{2}{n}\right) \subseteq B(y, \varepsilon) \subseteq U$$

From the density hypothesis, there exists $x \in B(y, \frac{1}{n})$; obviously, it follows that $y \in B(x, \frac{1}{n})$. For all $z \in B(x, \frac{1}{n})$, it holds $d(z, y) \leq d(z, x) + d(x, y) \leq \frac{2}{n}$. This proves that

$$B\left(x, \frac{1}{n}\right) \subseteq B\left(y, \frac{2}{n}\right) \subseteq U$$

We just showed that for any $y \in U \in \tau$, there exists $B \in \mathcal{B}$ such that $y \in B \subseteq U$. This is a necessary and sufficient condition to say that $\mathcal{B}$ is a base for τ . $\square$

Proposition A.5. *Let E be a separable Banach space and let μ be a probability measure on E . For arbitrary $\varepsilon > 0$, there exists a compact set K_ε such that*

$$\mu(K_\varepsilon) \geq 1 - \varepsilon$$

Dim.: take a countable dense set $\{x_i\}$ and arbitrary $r > 0$. We will show:

$$\bigcup_{i \in \mathbb{N}} \overline{B}(x_i, r) = E$$

One inclusion is obvious. Now let $x \in E$, from the density hypothesis we have $\|x_i - x\| < r$ for some i , so $x \in \overline{B}(x_i, r)$ and the other inclusion follows as well. Then for any positive integer k there exists $n_k > 0$ such that

$$F_k = \bigcup_{i=1}^{n_k} \overline{B}(x_i, 1/k) \implies \mu(F_k) \geq 1 - \frac{\varepsilon}{2^k}$$

Define

$$F = \bigcap_{k \in \mathbb{N}} F_k$$

Clearly F is closed and

$$\mu(F) = 1 - \mu(F^c) = 1 - \mu\left(\bigcup_{k \in \mathbb{N}} F_k^c\right) \geq 1 - \varepsilon$$

We just have to prove that F is compact set.

We will use a result showed by Dunford and Schwartz (see [2], p. 22) that a closed set on E is compact if and only if is totally bounded. This is true: for any given $\varepsilon > 0$, take k such that $1/k < \varepsilon$, it suffices one open ball of radius ε to cover F_k and therefore to cover F . $\square$

Before introducing the next theorem, let us recall some notions on a topological space (X, τ) .

If τ_1, τ_2 are topologies on X and $\tau_1 \subseteq \tau_2$, we say τ_1 is coarser than τ_2 .

The weak topology on X is the coarsest topology on X such that each element of the topological dual space X^* remains a continuous function. Remind that the topological dual of X consists of all linear functions $\varphi : X \rightarrow \mathbb{R}$ that are continuous with respect to the *given* topology.

The weak* topology on X^* is the coarsest topology on X^* such that the maps $T_x : X^* \rightarrow \mathbb{R}, \varphi \mapsto \varphi(x)$ remain continuous functions. We shall write that a subset of X^* is weakly* closed (weakly* compact) if it is closed (compact) with respect to the weak* topology.

A family Λ of probability measure on E is called tight if for arbitrary $\varepsilon > 0$ there exists a compact set $K_\varepsilon \subset E$ such that

$$\mu(K_\varepsilon) \geq 1 - \varepsilon, \quad \forall \mu \in \Lambda$$

Equivalently, a family Λ is tight if and only if for arbitrary $\varepsilon > 0$ and $r > 0$ there exists a finite family of balls $\overline{B}(a_i, r), i = 1, \dots, m$, such that

$$\mu\left(\bigcup_{i=1}^m \overline{B}(a_i, r)\right) \geq 1 - \varepsilon, \quad \forall \mu \in \Lambda$$

We denote by $C_b(E)$ the space of all bounded, continuous, real valued functions defined on E . A sequence of measures $\{\mu_n\}$ on E is said to be weakly convergent to a measure μ if for every $\varphi \in C_b(E)$ we have

$$\lim_{n \rightarrow \infty} \int_E \varphi(x) \mu_n(dx) = \int_E \varphi(x) \mu(dx)$$

A family Λ is said to be relatively compact if an arbitrary sequence $\{\mu_n\} \subset \Lambda$ contains a subsequence $\{\mu_{n_k}\}$ weakly convergent to a measure μ on E .

Theorem A.6. (*Prokhorov*)

A family of probability measure Λ on a separable Banach space E is relatively compact if and only if it is tight.

Dim.: assume that Λ is relatively compact, we prove it is tight.

We shall prove the following claim: if $\{U_i\}$ are open sets in E that cover E and if $\varepsilon > 0$, then there exists $k \geq 1$ such that

$$\mu\left(\bigcup_{i=1}^k U_i\right) > 1 - \varepsilon, \quad \forall \mu \in \Lambda$$

Suppose that for every $k \geq 1$ there is $\mu_k \in \Lambda$ with $\mu_k(\bigcup_{i=1}^k U_i) \leq 1 - \varepsilon$. Since Λ is relatively compact, there exists a measure μ and a subsequence $\mu_{k_j} \rightarrow \mu$. By the characterization of the weak convergence via open sets, since $\bigcup_{i=1}^k U_i$ is open for any $n \geq 1$:

$$\mu\left(\bigcup_{i=1}^n U_i\right) \leq \liminf_{j \rightarrow \infty} \mu_{k_j}\left(\bigcup_{i=1}^n U_i\right) \leq \liminf_{j \rightarrow \infty} \mu_{k_j}\left(\bigcup_{i=1}^{k_j} U_i\right) \leq 1 - \varepsilon$$

But we chose $\{U_i\}$ to be a covering of E , so $\mu(\bigcup_{i=1}^k U_i) \rightarrow \mu(E) = 1$ as $n \rightarrow \infty$, contradiction. Thus the claim is proved.

Fix $\varepsilon > 0$. Take $D = \{a_i\}$ dense in E . For every $m \geq 1$ the open balls $B(a_i, \frac{1}{m})$ cover E , so by the claim there exists $k_m \geq 1$ such that

$$\mu\left(\bigcup_{i=1}^{k_m} B(a_i, 1/m)\right) > 1 - \frac{\varepsilon}{2^m}, \quad \forall \mu \in \Lambda$$

Now just apply the same argument showed in the proof of proposition A.5 to each $\mu \in \Lambda$: we conclude that Λ is tight.

Assume that Λ is tight, we prove it is relatively compact. We need the following facts, which will be proved at the end:

- (i) if E is a compact metric space, then the space $\mathcal{P}(E)$ of Borel probability measures on E is also a compact metric space.
- (ii) if E is a separable metric space, there exist a compact metric space F and a map $T : E \rightarrow F$ s.t. T is a homeomorphism from E onto $T(E)$.

First we observe that the family $\bar{\Lambda}$ is also tight.

Indeed, let $\varepsilon > 0$ and let $K_\varepsilon \subset E$ be a compact such that $\mu(K_\varepsilon) \geq 1 - \varepsilon$ for all $\mu \in \Lambda$. Then for every $\mu \in \bar{\Lambda}$ there is a sequence $\{\mu_n\}$ that weakly converges to μ : by the characterization of weak convergence via closed sets,

$$\mu(K) \geq \limsup_{n \rightarrow \infty} \mu_n(K) \geq 1 - \varepsilon$$

Let $\{\mu_n\}_{n \in \mathbb{N}} \subset \bar{\Lambda}$. We show that it has a convergent subsequence.

Let F be a compact metric space and $T : E \rightarrow F$ a homeomorphism from E onto $T(E)$: so if B is a Borel set in F , then $T^{-1}(B)$ is a Borel set in E . Define:

$$\nu_n(B) := \mu_n(T^{-1}(B)), \quad \forall B \in \mathcal{B}(F)$$

Then $\nu_n \in \mathcal{P}(F)$ for all n . As F is a compact metric space, from (i) we deduce that $\mathcal{P}(F)$ is a compact metric space. In a metric space, compact and sequential compact are equivalent, so there is $\nu \in \mathcal{P}(F)$ and a subsequence such that $\nu_{n_k} \rightarrow \nu$ weakly in $\mathcal{P}(F)$.

Let $F_0 = T(E)$. Our claim is that ν is concentrated on F_0 : it means that

$$\exists S \in \mathcal{B}(F), S \subset F_0 \text{ such that } \nu(S) = 1$$

Assume it is true and define:

$$\nu_0(A) := \nu(A \cap S), \quad \forall A \in \mathcal{B}(F_0)$$

The measure ν_0 is a probability measure on F_0 and $\nu_0(S) = 1$. Define:

$$\mu(A) := \nu_0(T(A)) = \nu_0\left((T^{-1})^{-1}(A)\right), \quad \forall A \in \mathcal{B}(E)$$

Then $\mu \in \mathcal{P}(E)$. We show that there exists a subsequence $\{\mu_{n_k}\}$ with $\mu_{n_k} \rightarrow \mu$. Let C be closed in E , then $T(C)$ is closed in F_0 . Hence there exists $K \subset F$ closed, with $K \cap F_0 = T(C)$. Then:

$$C = \{x \in E : T(x) \in T(C)\} = \{x \in E : T(x) \in K\} = T^{-1}(K)$$

Just put together this equality, the definitions of $\{\nu_n\}$, ν_0 , μ , the fact that ν is concentrated on S and the weak convergence of $\{\nu_{n_k}\} \rightarrow \nu$:

$$\begin{aligned} \limsup \mu_{n_k}(C) &= \limsup \nu_{n_k}(K) \leq \nu(K) = \nu(K \cap S) + \nu(K \cap S^c) \\ &= \nu(T(C) \cap S) + 0 = \nu_0(T(C)) = \mu(C) \end{aligned}$$

By the characterization via closed sets, μ_{n_k} weakly converges to μ .

Now let us prove that ν is concentrated on F_0 , we will use tightness of $\bar{\Lambda}$. For each $m \geq 1$ take $K_m \subset E$ compact such that $\mu(K_m) \geq 1 - 1/m$ for all $\mu \in \bar{\Lambda}$. Then $T(K_m)$ is closed, so

$$\nu(T(K_m)) \geq \limsup_{k \rightarrow \infty} \nu_{n_k}(T(K_m)) \geq \limsup_{k \rightarrow \infty} \mu_{n_k}(K_m) \geq 1 - \frac{1}{m}$$

Take $S = \bigcup_m K_m$, then $S \in \mathcal{B}(F)$ and $\nu(S) \geq \nu(K_m)$ for all $m \Rightarrow \nu(S) = 1$.

□

Proof of (i): as E is compact, $C_b(E) = C(E) = \{f : E \rightarrow \mathbb{R}, f \text{ is continuous}\}$. Denote by $C(E)^*$ its dual space and consider Φ defined as follows:

$$\Phi := \{\varphi \in C(E)^* : \|\varphi\| \leq 1, \varphi(\mathbf{1}) = 1, \varphi(f) \geq 0 \text{ for } f \geq 0\}$$

Equivalently, we could write

$$\Phi := \{\varphi \in C(E)^* : \|\varphi\| = 1, \varphi \text{ positive}\}$$

According to [3], Theorem 2.14, if E is a compact space and $\varphi \in \Phi$ there exists a unique probability measure μ on E such that $\varphi(f) = \int f d\mu$ for all $f \in C(E)$. The correspondent operator φ will then be denoted as φ_μ .

Note that if $\mu, \mu_1, \mu_2, \dots$ are Borel probability measures on E ,

$$\begin{aligned} \varphi_{\mu_n} \rightarrow \varphi_\mu \text{ in the weak* topology} &\iff \varphi_{\mu_n}(f) \rightarrow \varphi_\mu(f) \quad \forall f \in C_b(E) \\ &\iff \int f d\mu_n \rightarrow \int f d\mu \quad \forall f \in C_b(E) \\ &\iff \mu_n \rightarrow \mu \text{ in the weak topology} \end{aligned}$$

This proves that the map $T : \mu \rightarrow \varphi_\mu$ is a sequential homeomorphism relative to the weak* topology on Φ .

By Banach-Alaoglu's theorem, the closed unit ball $B^* = \{\varphi \in C(E)^* : \|\varphi\| \leq 1\}$ is weakly* compact. Let us prove that Φ is a weakly* closed set in B^* :

$$\begin{aligned} \Phi &= \{\varphi \in C(E)^* : \|\varphi\| = 1, \varphi \text{ positive}\} \\ &= \{\varphi \in C(E)^* : \|\varphi\| \leq 1, \varphi(\mathbf{1}) = 1, \varphi(f) \geq 0 \text{ for } f \geq 0\} \\ &= \bigcap_{f \geq 0} \{\varphi \in B^* : \varphi(f) \geq 0\} \cap \{\varphi \in B^* : \varphi(\mathbf{1}) = 1\} \end{aligned}$$

Since $\varphi \rightarrow \varphi(f)$ is weak* continuous for $f \in C(E)$, Φ is weakly* closed in B^* . Since B^* is a compact space, Φ is also weakly* compact, hence sequentially weakly* compact. The sequential homeomorphism $T : \mathcal{P}(E) \rightarrow \Phi$ ensures that $\mathcal{P}(E)$ is sequentially compact $\Rightarrow$ compact.

Proof of (ii): define $F := [0, 1]^{\mathbb{N}} = \{\{\xi_i\}_{i \in \mathbb{N}} : \xi_i \in [0, 1]\}$ and the function:

$$\delta(\xi, \eta) := \sum_{i=1}^{\infty} |\xi_i - \eta_i| \cdot 2^{-i}$$

This is a metric on F and has the topology of coordinatewise convergence; moreover, F is closed and bounded, hence it is compact.

Let d be the metric on E and let $D = \{a_1, a_2, \dots\}$ be a dense set in E . Define:

$$\alpha_i(x) = \min\{d(x, a_i), 1\}, \quad x \in E$$

For each $i \in \mathbb{N}$, $\alpha_i : E \rightarrow [0, 1]$ is a continuous function. Define now $T : E \rightarrow F$,

$$T(x) := \{\alpha_i(x)\}_{i \in \mathbb{N}}$$

Claim: for any $C \subset E$ and $x \notin C$ there exist $\varepsilon > 0$ and $i \in \mathbb{N}$ such that

$$\begin{cases} \alpha_i(x) \leq \varepsilon/3 \\ \alpha_i(y) \geq 2\varepsilon/3 \text{ for all } y \in C \end{cases}$$

To prove this claim, take $\varepsilon := \min\{d(x, C), 1\}$ and i such that $d(a_i, x) < \varepsilon/3$. Then $\alpha_i(x) \leq \varepsilon/3$ and for all $y \in C$:

$$\begin{aligned} \alpha_i(y) &= \min\{d(y, a_i), 1\} \geq \min\{d(y, x) - d(x, a_i), 1\} \\ &\geq \min\{d(C, x) - d(x, a_i), 1\} \geq \min\{d(C, x) - \varepsilon/3, 1\} \\ &\geq \min\{\varepsilon - \varepsilon/3, 1\} = 2\varepsilon/3 \end{aligned}$$

Our claim is proved. In particular, if $x \notin \{y\} \iff x \neq y$, there exists i such that $\alpha(x) \neq \alpha_i(y)$, so T is an injective map. Hence, it is a bijection on $T(E)$. It remains to prove that T is a homeomorphism, that is: for $\{x_n\}, x \in E$,

$$x_n \rightarrow x \iff T(x_n) \rightarrow T(x)$$

If $x_n \rightarrow x$, then $\alpha_i(x_n) \rightarrow \alpha_i(x)$ for all i , so $\delta(T(x_n), T(x)) \rightarrow 0$ as $n \rightarrow \infty$. Suppose that $x_n \not\rightarrow x$, then there exists a subsequence such that $x \notin \{x_{n_k}\}$. By the claim there is $i \in \mathbb{N}$ such that $\alpha_i(x) \leq \varepsilon/3$ and $\alpha_i(x_{n_k}) \geq 2\varepsilon/3$ for all k , which makes clear that $\alpha_i(x_{n_k}) \not\rightarrow \alpha_i(x)$ as $k \rightarrow \infty$. Hence, $T(x_{n_k}) \not\rightarrow T(x)$. $\square$

If $X : (\Omega, \mathcal{F}, \mathbf{P}) \rightarrow E$ is a random variable, we will denote by $\mathcal{L}(X)$ the image of $\mathbf{P}$ under the mapping:

$$\mathcal{L}(X)(A) = \mathbf{P} \{ \omega \in \Omega : X(\omega) \in A \} \quad \forall A \in \mathcal{B}(E)$$

It is called the law of X . The following theorem links weak convergence of probability measures with almost sure convergence of random variables.

Theorem A.7. (*Skokhorod*)

For an arbitrary sequence of probability measures $\{\mu_n\}$ on E weakly convergent to a probability measure μ , there exists a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ and random variables $X, X_1, \dots$ such that $\mathcal{L}(X) = \mu$, $\mathcal{L}(X_m) = \mu_m$ and

$$\lim_{n \rightarrow \infty} X_n = X, \quad \mathbf{P} - a.s.$$

Dim.: see [1], p. 33.

Proposition A.8. A measure μ on $\mathbb{R}$ is of the form $\mathcal{N}(0, q)$ if and only if: for arbitrary independent real-valued variables X, Y such that $\mathcal{L}(X) = \mathcal{L}(Y) = \mu$, arbitrary numbers α, β such that $\alpha^2 + \beta^2 = 1$, we have $\mathcal{L}(\alpha X + \beta Y) = \mu$.

Dim.: we will compute the characteristic function of $\alpha X + \beta Y$:

$$\begin{aligned} \phi_{\alpha X + \beta Y}(t) &= \mathbf{E} \left[\exp [it (\alpha X + \beta Y)] \right] = \mathbf{E} \left[\exp [it (\alpha X)] \cdot \exp [it (\beta Y)] \right] \\ &= \mathbf{E} \left[\exp [it (\alpha X)] \right] \cdot \mathbf{E} \left[\exp [it (\beta Y)] \right] \\ &= \exp \left[-\alpha^2 \cdot \frac{qt^2}{2} \right] \exp \left[-\beta^2 \cdot \frac{qt^2}{2} \right] = \exp \left[-\frac{qt^2}{2} \right] \end{aligned}$$

We found the characteristic function of a Gaussian distribution $\mathcal{N}(0, q)$ on $\mathbb{R}$. Hence they have the same law. $\square$

Let E be a separable Banach space. A probability measure μ on E is said to be a Gaussian measure if and only if the law of arbitrary linear function $h \in E^*$ considered as a random variable

$$h : (E, \mathcal{B}(E), \mu) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

is a Gaussian measure on $\mathbb{R}$. If the law of each $h \in E^*$ is in addition symmetric (zero mean), then μ is called a symmetric Gaussian measure.

Theorem A.9. (Fernique)

Let μ be a symmetric Gaussian measure on E . Then there exists $\alpha > 0$ s.t:

$$\int_E \exp(\alpha \|x\|^2) \mu(dx) < +\infty$$

Equivalently, any random variable X whose law is μ has moments of all order:

$$\mathbf{E}(\|X\|^k) = \int_E \|x\|^k \mu(dx) < +\infty$$

Dim.: see [1], p. 37.

Let μ be a symmetric Gaussian measure on a separable Banach space E .

A linear subspace $H \subset E$ endowed with a Hilbert norm $\|\cdot\|_H$ is said to be a reproducing kernel space for μ if H is complete, continuously embedded and

$$\mathcal{L}(\varphi) = \mathcal{N}(0, \|\varphi\|_H^2), \quad \forall \varphi \in E^*$$

where

$$\|\varphi\|_H := \sup_{\|h\| \leq 1} \varphi(h)$$

Theorem A.10. Let μ be a symmetric Gaussian measure on E , there exists a unique reproducing kernel space for μ .

Dim.: Gaussian measures have moments of all order by Fernique's Theorem, so $\varphi \in L^2(E, \mathcal{B}(E), \mu)$. We shall denote by $\overline{E^*}$ the closure of E^* in L^2 .

Note that for arbitrary $\varphi \in E^*$ holds $\mathcal{L}(\varphi) = \mathcal{N}(0, \|\varphi\|_{L^2}^2)$. Define $J : \overline{E^*} \rightarrow E$:

$$J(\varphi) = \int_E x \varphi(x) \mu(dx)$$

Then J is injective and continuous. To see that it is injective, consider the case when $J(\varphi) = 0$. Since φ is linear we can distribute it into $J(\varphi)$:

$$0 = \varphi(J(\varphi)) = \varphi\left(\int_E x \varphi(x) \mu(dx)\right) = \int_E |\varphi(x)|^2 \mu(dx)$$

This implies $\varphi = 0$ a.s., that means injectivity of J . To see that is continuous, it suffices to use Hölder inequality. Then $H = J(E)$ is a Hilbert space with

$$\langle J(\varphi), J(\psi) \rangle_H = \int_E \varphi(x) \psi(x) \mu(dx) = \langle \varphi, \psi \rangle_{L^2}$$

and therefore J is an isometry!

Let $\|\cdot\|_H$ denote the norm in H , we will show that $\|\cdot\|_H = \|\cdot\|_{L^2}$. For $\psi \in E^*$:

$$\begin{aligned} \|\psi\|_H &= \sup \left\{ |\psi(x)| : x \in H, \|x\|_H = 1 \right\} \\ &= \sup \left\{ |\psi(J(\varphi))| : \|\varphi\|_{L^2} = 1 \right\} \\ &= \sup \left\{ \left| \int_E \varphi(x) \psi(x) \mu(dx) \right| : \|\varphi\|_{L^2} = 1 \right\} \end{aligned}$$

By using again Hölder inequality:

$$\|\psi\|_H \leq \|\psi\|_{L^2} \cdot \|\varphi\|_{L^2} = \|\psi\|_{L^2}$$

The upper bound is achieved by taking $\varphi = \psi/\|\psi\|_{L^2}$.

So we conclude that H is a reproducing kernel for μ . We will say that $J(\varphi)$ is the reproducing kernel for $\varphi \in E^*$ in the reproducing kernel space H .

Uniqueness: suppose there is another reproducing kernel space $\tilde{H} \subset E$ with inner product $\langle \cdot, \cdot \rangle_{\tilde{H}}$. From the Riesz's representation theorem, for arbitrary $\varphi \in E^*$ there exists a unique element $\tilde{J}(\varphi) \in \tilde{H}$ such that

$$\varphi(h) = \langle \tilde{J}(\varphi), h \rangle_{\tilde{H}} \quad h \in \tilde{H}$$

In other words, $\tilde{J}(\varphi)$ is the reproducing kernel for φ in the space $\tilde{H}$.

For $\varphi, \psi \in E^*$:

$$\varphi(\tilde{J}(\psi)) = \langle \tilde{J}(\varphi), \tilde{J}(\psi) \rangle_{\tilde{H}} = \int_E \varphi(x)\psi(x)\mu(dx) = \varphi(J(\psi))$$

Thus $\tilde{J} = J$ and this implies uniqueness. $\square$

Before introducing the next theorem, we shall need some results on a series of independent random variables in a separable Banach space E .

Lemma A.11. *Let E be a separable Banach space. Assume that $\xi_1, \dots, \xi_N$ are E -valued independent random variables with symmetric distributions $\mathcal{L}(\xi_i)$.*

Let $S_k = \xi_1 + \dots + \xi_k$, for $k \leq N$. Then for arbitrary $r \geq 0$:

$$\mathbf{P}\left(\sup_{k \leq N} \|S_k\| \geq r\right) \leq 2\mathbf{P}\left(\|S_N\| \geq r\right)$$

Dim.: we shall rewrite the left hand side of the inequality as

$$\begin{aligned} \mathbf{P}\left(\sup_{k \leq N} \|S_k\| \geq r\right) &= \mathbf{P}\left(\sup_{k \leq N} \|S_k\| \geq r, \|S_N\| \geq r\right) \\ &\quad + \mathbf{P}\left(\sup_{k \leq N} \|S_k\| \geq r, \|S_N\| < r\right) \\ &= I_1 + I_2 \end{aligned}$$

We will now proceed to obtain inequalities on I_1, I_2 . Let us define the events: $A_1 = \{\|S_1\| \geq r\}$, $A_k = \{\|S_k\| \geq r, \|S_i\| < r \text{ for } i < k\}$, which are disjoint. Now:

$$\begin{aligned} \bigcup_{i=1}^N \left(A_i \cap \{\|S_N\| < r\} \right) &= \left\{ \sup_{k \leq N} \|S_k\| \geq r, \|S_N\| < r \right\} \\ \sum_{i=1}^N \mathbf{P}\left(A_i \cap \{\|S_N\| < r\} \right) &= I_2 \end{aligned} \tag{A.12}$$

We shall now exploit the symmetric distribution hypothesis on $\xi_1, \dots, \xi_N$:

$$\begin{aligned}
\mathbf{P}\left(A_1 \cap \{\|S_N\| < r\}\right) &= \mathbf{P}\left(\|S_1\| \geq r, \|S_1 + \xi_2 + \dots + \xi_N\| < r\right) \\
&= \mathbf{P}\left(\|S_1\| \geq r, \|S_1 - \xi_2 + \dots - \xi_N\| < r\right) \\
&= \mathbf{P}\left(\|S_1\| \geq r, \|2S_1 - S_N\| < r\right) \\
&\leq \mathbf{P}\left(\|S_1\| \geq r, \|S_N\| \geq r\right) \\
&= \mathbf{P}\left(A_1 \cap \{\|S_N\| \geq r\}\right)
\end{aligned}$$

In a similar way, one can prove that for every $i \leq N$ the following holds:

$$\mathbf{P}\left(A_i \cap \{\|S_N\| < r\}\right) \leq \mathbf{P}\left(A_i \cap \{\|S_N\| \geq r\}\right)$$

By summing on i :

$$\sum_{i=1}^N \mathbf{P}\left(A_i \cap \{\|S_N\| < r\}\right) \leq \mathbf{P}\left(\|S_N\| \geq r\right) \quad (\text{A.13})$$

By putting together A.12, A.13 and the trivial inequality on I_1 , we finally get the thesis. $\square$

Lemma A.14. *Let E be a Banach separable space. Assume that $\{\xi_i\}_{i \in \mathbb{N}}$ are E -valued independent random variables with symmetric distributions $\mathcal{L}(\xi_i)$. Assume that there exists a random variable S such that, for arbitrary $r > 0$,*

$$\lim_{N \rightarrow \infty} \mathbf{P}\left(\|S - S_N\| \geq r\right) = 0$$

where $S_k = \xi_1 + \dots + \xi_k$ for $k \in \mathbb{N}$. Then:

$$\lim_{N \rightarrow \infty} S_N = S = \sum_{k=1}^{\infty} \xi_k \quad \mathbf{P} - a.s.$$

Dim.: the convergence in probability of $\{S_n\}$ implies the a.s. convergence of a subsequence $\{S_{n_k}\}$ to S . Now fix k and choose $m > n_k$: using lemma A.11, we get

$$\begin{aligned}
\mathbf{P}\left(\sup_{n_k \leq n \leq m} \|S_n - S_{n_k}\| \geq r\right) &\leq 2\mathbf{P}\left(\|S_m - S_{n_k}\| \geq r\right) \\
&\leq 2\mathbf{P}\left(\|S_m - S\| \geq \frac{r}{2}\right) + 2\mathbf{P}\left(\|S - S_{n_k}\| \geq \frac{r}{2}\right)
\end{aligned}$$

Now, let $m \rightarrow \infty$: the previous becomes

$$\mathbf{P}\left(\sup_{n_k \leq n} \|S_n - S_{n_k}\| \geq r\right) \leq 2\mathbf{P}\left(\|S - S_{n_k}\| \geq \frac{r}{2}\right)$$

and this implies the thesis. $\square$

Theorem A.15. *Let μ be a symmetric Gaussian measure on separable Banach space E . Let $\{e_n\}$ be an orthonormal, complete basis in H_μ and $\{\xi_n\}$ a sequence of independent real-valued random variables such that $\mathcal{L}(\xi_n) = \mathcal{N}(0, 1)$. Then the series $\sum_k \xi_k e_k$ converges $\mathbf{P}$ -a.s. in E and*

$$\mathcal{L}\left(\sum_{k=1}^{\infty} \xi_k e_k\right) = \mu$$

Dim.: define the maps $\varphi_n \in E^*$ such that $J\varphi_n = e_n$. From now on, $\{\varphi_n\}$ will play the role of $\{\xi_n\}$: they are independent, real-valued, have symmetric Gaussian distribution for the hypothesis on μ and have unitary norm since the one-to-one isometry J^{-1} sends orthonormal basis in orthonormal basis. Let:

$$\mathcal{L}\left(\sum_{k=1}^n \xi_k e_k\right) = \mu_n$$

We will prove that $\{\mu_n\}$ is tight. Let $S(x) = x$ and

$$S_N(x) := \sum_{n=1}^N \varphi_n(x) e_n = \sum_{n=1}^N \varphi_n(x) J\varphi_n, \quad x \in E$$

For $\varphi \in E^*$, consider the sequence

$$\varphi(S_N(x)) = \varphi\left(\sum_{n=1}^N \varphi_n(x) e_n\right) = \sum_{n=1}^N \varphi_n(x) \varphi(e_n)$$

Since φ_n has mean 0 and variance 1 for all $n \geq 1$, $\varphi_n(x) \varphi(e_n)$ has mean 0 and variance $|\varphi(e_n)|^2$. Then:

$$\sum_{n=1}^{\infty} |\varphi(e_n)|^2 = \sum_{n=1}^{\infty} \langle J\varphi, e_n \rangle_H^2 = \|J\varphi\|_H^2 < \infty$$

Kolmogorov two-series theorem implies that $\varphi(S_N(x))$ converges! It also holds:

$$\mathcal{L}\left(\lim_{N \rightarrow \infty} \varphi(S_N(x))\right) = \mathcal{L}\left(\sum_{n=1}^{\infty} \varphi_n(x) \varphi(e_n)\right) = \mathcal{N}(0, \|J\varphi\|_H^2) \quad (\text{A.16})$$

Since $\{\varphi_n\}$ is a complete orthonormal basis in $E^* \subseteq L^2$, we have for $\varphi \in E^*$:

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n(x) \langle \varphi, \varphi_n \rangle_{L^2} = \sum_{n=1}^{\infty} \varphi_n(x) \int_E \varphi_n(y) \varphi(y) \mu(dy), \quad a.s.$$

Using the reproducing kernel formula, the previous becomes:

$$\begin{aligned} \varphi(x) &= \sum_{n=1}^{\infty} \varphi_n(x) \langle J\varphi, J\varphi_n \rangle_H = \sum_{n=1}^{\infty} \varphi_n(x) \langle J\varphi, e_n \rangle_H \\ &= \sum_{n=1}^{\infty} \varphi_n(x) \varphi(e_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \varphi_n(x) \varphi(e_n) = \lim_{N \rightarrow \infty} \varphi(S_N(x)) \end{aligned}$$

For fixed N , linearity of φ gives $\lim_{M \rightarrow \infty} \varphi(S_M(x) - S_N(x)) = \varphi(S(x) - S_N(x))$. Moreover, the independence of $S_M - S_N$ and S_N implies that $S - S_N$ and S_N are also independent.

Now let $\mu_M^\perp$ denote the distribution of $S - S_M$ for any $M \geq 1$ and let $*$ denote the convolution of two measures. For fixed $\varepsilon > 0$, let K be a compact set with $\mu(K) \geq 1 - \varepsilon$. From the independence proved before:

$$\mu(K) = \mu_M * \mu_M^\perp(K) = \int_E \mu_M(K - x) \mu_M^\perp(dx) \geq 1 - \varepsilon$$

Then there exists $x_0 \in E$ such that $\mu_M(K - x_0) \geq 1 - \varepsilon$. By symmetry, we have $\mu_M(K - x_0) = \mu_M(-K + x_0)$. Furthermore, the following set inclusion holds:

$$(K - x_0) \cap (-K + x_0) \subseteq \frac{K - K}{2}$$

It follows:

$$\mu_M((K - x_0) \cap (-K + x_0)) \leq \mu_M\left(\frac{K - K}{2}\right)$$

By definition of x_0 and using the subadditivity of μ_M on $(K - x_0)^c$ and $(-K + x_0)^c$ we get:

$$1 - 2\varepsilon \leq \mu_M((K - x_0) \cap (-K + x_0)) \leq \mu_M\left(\frac{K - K}{2}\right)$$

Since $(K - K)/2$ is also compact, we conclude that the family $\{\mu_n\}$ is tight. In the same way, one can prove that the family of measures $\{\mu_n^\perp\}$ is tight as well. Denote by $\hat{\mu}(\varphi)$ the characteristic function of φ with respect to the measure μ . Then:

$$\hat{\mu}(\varphi) = \int_E e^{i\varphi(x)} \mu(dx) = \exp\left[-\frac{1}{2}\|J\varphi\|_H^2\right] \quad (\text{A.17})$$

$$\hat{\mu}_M(\varphi) = \int_E e^{i\varphi(S_M x)} \mu(dx) = \exp\left[-\frac{1}{2} \sum_{n \leq M} \langle J\varphi, e_n \rangle_H^2\right]$$

$$\hat{\mu}_M^\perp(\varphi) = \int_E e^{i\varphi(x - S_M x)} \mu(dx) = \exp\left[-\frac{1}{2} \sum_{n > M} \langle J\varphi, e_n \rangle_H^2\right]$$

The first integral is computed as the characteristic function of $\varphi(X)$ evaluated in $t = 1$: it is easy because we know distribution of $\varphi(X)$ from equation A.16. The second (resp. third) integral follows in a similar way adapting the equation A.16 when the sum contains only the first M (resp. all but the first M) terms. Hence:

$$\lim_{M \rightarrow \infty} \hat{\mu}_M(\varphi) = \hat{\mu}(\varphi) \quad \text{and} \quad \lim_{M \rightarrow \infty} \hat{\mu}_M^\perp(\varphi) = 1$$

Therefore $\{\mu_M\}$ converges to μ weakly. In the same way, $\{\mu_M^\perp\}$ converges to $\delta_{\{0\}}$ weakly i.e. $S - S_M$ converges to 0 in probability. We are now in the hypothesis of Lemma A.14 which gives exactly our thesis. $\square$

Let us assume that E is a separable Hilbert space H with product $\langle \cdot, \cdot \rangle$. For arbitrary probability measures μ the characteristic function of μ is defined as:

$$\hat{\mu}(\lambda) = \int_H e^{i\langle \lambda, x \rangle} \mu(dx), \quad \lambda \in H$$

A nonnegative operator $T \in L(H, H)$ is said to be trace-class if and only if for an orthonormal basis $\{e_j\}$ on H :

$$\text{Tr } T := \sum_{j=1}^{\infty} \langle T e_j, e_j \rangle < +\infty$$

This is independent of the choice of the basis $\{e_j\}$ and will be denoted as the trace of T . The space of all trace-class operators will be denoted as $L_1(H)$. A function $f : H \rightarrow \mathbb{C}$ is said to be positive definite if for arbitrary $n \in \mathbb{N}$, $\lambda_1, \dots, \lambda_n \in H$, $z_1, \dots, z_n \in \mathbb{C}$:

$$\sum_{i,j=1}^n f(\lambda_i - \lambda_j) z_i \bar{z}_j \geq 0$$

If the space H is finite dimensional, positive definiteness, continuity of φ and the relation $\varphi(0) = 1$ characterize characteristic functions completely. The extension to separable Hilbert spaces is given by the following theorem.

Theorem A.18. (*Bochner*)

If $\varphi : H \rightarrow \mathbb{C}$ is a characteristic function of a probability measure μ on H :

- (i) φ is continuous, positive definite and $\varphi(0) = 1$.
- (ii) for any $\varepsilon > 0$ there exists a nonnegative trace-class operator S_ε such that

$$1 - \text{Re } \varphi(\lambda) \leq \varepsilon \quad \forall \lambda : \langle S_\varepsilon \lambda, \lambda \rangle \leq 1$$

Conversely, a function $\varphi : H \rightarrow \mathbb{C}$ satisfying (i) and (ii) is the characteristic function of a probability measure μ on H .

Dim.: assume that $\varphi : H \rightarrow \mathbb{C}$ is a characteristic function of a probability measure μ on $(H, \mathcal{B}(H))$. Since (i) is clear, we just need to prove (ii). For $r > 0$, let $B(r) = \{x \in H : \|x\| \leq r\}$ and define the linear operator S_r by:

$$\langle S_r \lambda, \lambda \rangle = \int_{B(r)} \langle x, \lambda \rangle^2 \mu(dx)$$

This is a positive-definite, trace-class operator since

$$\text{Tr}(S_r) = \sum_{n \in \mathbb{N}} \langle S_r e_n, e_n \rangle = \int_{B(r)} \left(\sum_{n \in \mathbb{N}} \langle e_n, x \rangle^2 \right) \mu(dx) = \int_{B(r)} \|x\|^2 \mu(dx) \leq r^2$$

However:

$$\begin{aligned}
1 - \mathbf{Re} \varphi(\lambda) &= \int_H [1 - \cos \langle \lambda, x \rangle] \mu(dx) \leq \int_H 2 \sin^2 \left(\frac{\langle \lambda, x \rangle}{2} \right) \mu(dx) \\
&= \int_{B(r)} 2 \sin^2 \left(\frac{\langle \lambda, x \rangle}{2} \right) \mu(dx) + \int_{B^c(r)} 2 \sin^2 \left(\frac{\langle \lambda, x \rangle}{2} \right) \mu(dx) \\
&\leq \int_{B(r)} 2 \left(\frac{\langle \lambda, x \rangle}{2} \right)^2 \mu(dx) + \int_{B^c(r)} 2 \sin^2 \left(\frac{\langle \lambda, x \rangle}{2} \right) \mu(dx) \\
&\leq \int_{B(r)} \frac{1}{2} \langle \lambda, x \rangle^2 \mu(dx) + \int_{B^c(r)} 2 \sin^2 \left(\frac{\langle \lambda, x \rangle}{2} \right) \mu(dx)
\end{aligned}$$

Rewrite as

$$1 - \mathbf{Re} \varphi(\lambda) \leq \frac{1}{2} \langle S_r \lambda, \lambda \rangle + 2\mu \{x \in H : \|x\| \geq r\}$$

From the arbitrariness of r , we can assume that $2\mu \{x \in H : \|x\| \geq r\} < \varepsilon/2$ and then set $S_\varepsilon = S_r/\varepsilon$ to have the desired inequality.

Conversely, assume that φ verifies (i) and (ii). Let $\{e_k\}$ be an orthonormal basis in H and let $\Pi_n = \sum_{i=1}^n e_i \otimes e_i$ be the orthogonal projector of H onto the finite dimensional, linear subspace $H_n = \langle e_1, \dots, e_n \rangle$. Define for $\lambda \in H$:

$$\varphi_n(\lambda) = \varphi(\Pi_n \lambda)$$

Then φ_n is the characteristic function of a measure μ_n concentrated on H_n by the Bochner theorem in finite dimensions. Aim is to prove tightness of $\{\mu_n\}$. Suppose that for arbitrary $r \in]0, 1[$ and $\varepsilon \in]0, 1/2[$ there exist $M \in \mathbb{N}$ and $k > 0$ such that

$$\mu_n \left\{ x \in H : \sum_{j \leq M} \langle x, e_j \rangle^2 \leq k^2 \right\} := \mu_n(A_{M,k}) \geq 1 - \varepsilon, \quad \forall n \quad (\text{A.19})$$

$$\mu_n \left\{ x \in H : \sum_{j > M} \langle x, e_j \rangle^2 \leq r^2 \right\} := \mu_n(C_{M,r}) \geq 1 - \varepsilon, \quad \forall n \quad (\text{A.20})$$

Let us show how this implies tightness. Since $A_{M,k} \cap C_{M,r}$ is a total bounded set, there exist $a_1, \dots, a_l$ points such that

$$\bigcup_{i=1}^l B(a_i, r) \supseteq A_{M,k} \cap C_{M,r}$$

By using the subadditivity of μ_n on $(A_{M,k})^c$ and $(C_{M,r})^c$ we get

$$\mu_n \left(\bigcup_{i=1}^l B(a_i, r) \right) \geq 1 - 2\varepsilon, \quad \forall n$$

which is an equivalent formulation for tightness.

Let $r \in]0, 1[$, $\varepsilon \in]0, 1/2[$ and let $S \geq 0$ be a trace-class operator such that

$$1 - \mathbf{Re} \varphi(\lambda) < \frac{\varepsilon}{L} \quad \text{whenever} \quad \langle S\lambda, \lambda \rangle \leq 1$$

Here L is a positive number to be chosen later.

Denote also by $\Pi_{M,N}$ the orthogonal projector of H onto $\langle e_{M+1}, \dots, e_N \rangle$. By the Fubini theorem and the same argument used for equation A.17:

$$\begin{aligned} \int_H e^{-\frac{1}{2}\|\Pi_{M,N}x\|^2} \mu_n(dx) &= \int_H \int_H e^{i\langle \lambda, x \rangle} \mathcal{N}(0, \Pi_{M,N})(d\lambda) \mu_n(dx) \\ &= \int_H \varphi_n(\lambda) \mathcal{N}(0, \Pi_{M,N})(d\lambda) \\ &= \int_H \varphi(\lambda) \mathcal{N}(0, \Pi_{M,N}\Pi_n)(d\lambda) \end{aligned}$$

We have:

$$\begin{aligned} \int_H \left(1 - e^{-\frac{1}{2}\|\Pi_{M,N}x\|^2}\right) \mu_n(dx) &= \int_H (1 - \mathbf{Re} \varphi(\lambda)) \mathcal{N}(0, \Pi_{M,N}\Pi_n)(d\lambda) \\ &= \int_{\langle S\lambda, \lambda \rangle \leq 1} (1 - \mathbf{Re} \varphi(\lambda)) \mathcal{N}(0, \Pi_{M,N}\Pi_n)(d\lambda) \\ &\quad + \int_{\langle S\lambda, \lambda \rangle \geq 1} (1 - \mathbf{Re} \varphi(\lambda)) \mathcal{N}(0, \Pi_{M,N}\Pi_n)(d\lambda) \\ &\leq \frac{\varepsilon}{L} + \int_{\langle S\lambda, \lambda \rangle \geq 1} 1 \cdot \mathcal{N}(0, \Pi_{M,N}\Pi_n)(d\lambda) \\ &\leq \frac{\varepsilon}{L} + \int_{\langle S\lambda, \lambda \rangle \geq 1} \langle S\lambda, \lambda \rangle \mathcal{N}(0, \Pi_{M,N}\Pi_n)(d\lambda) \\ &\leq \frac{\varepsilon}{L} + \int_H \langle S\lambda, \lambda \rangle \mathcal{N}(0, \Pi_{M,N}\Pi_n)(d\lambda) \leq \frac{\varepsilon}{L} + \text{Tr } S\Pi_{M,N}\Pi_n \end{aligned}$$

Note that the last two inequalities holds since S is a nonnegative operator. Now, since L is still to be chosen, assume that $\text{Tr } S\Pi_{M,N}\Pi_n \leq \frac{\varepsilon}{L}$ for all $n \in \{1, \dots, N\}$. Then:

$$\int_H \left(1 - e^{-\frac{1}{2}\|\Pi_{M,N}x\|^2}\right) \mu_n(dx) \leq \frac{2\varepsilon}{L}, \quad \forall n \in \{1, \dots, N\}$$

For arbitrary $r > 0$ we have an estimate for the measure of $\{x : \|\Pi_{M,N}x\| \geq r\}$:

$$\begin{aligned} &\mu_n(\{x : \|\Pi_{M,N}x\| \geq r\}) \left(1 - e^{-r^2/2}\right) \\ &\leq \int_{\|\Pi_{M,N}x\| \geq r} \left(1 - e^{-\frac{1}{2}\|\Pi_{M,N}x\|^2}\right) \mu_n(dx) \leq \int_H \left(1 - e^{-\frac{1}{2}\|\Pi_{M,N}x\|^2}\right) \mu_n(dx) \end{aligned}$$

Hence for all $n \in \{1, \dots, N\}$,

$$\begin{aligned} \mu_n \left\{ x : \sum_{j=M+1}^N \langle x, e_j \rangle^2 \geq r^2 \right\} &= \mu_n \{ x : \|\Pi_{M,N} x\| \geq r \} \leq \frac{2\varepsilon}{L(1 - e^{-r^2/2})} \\ \mu_n \left\{ x : \sum_{j=M+1}^N \langle x, e_j \rangle^2 \leq r^2 \right\} &= \mu_n \{ x : \|\Pi_{M,N} x\| \leq r \} \geq 1 - \frac{2\varepsilon}{L(1 - e^{-r^2/2})} \end{aligned}$$

Let $N \rightarrow \infty$ and choose L such that $L > 2/(1 - e^{-r^2/2})$:

$$\mu_n \left\{ x : \sum_{j=M+1}^{\infty} \langle x, e_j \rangle^2 \leq r^2 \right\} \geq 1 - \varepsilon, \quad \forall n \in \mathbb{N}$$

which is exactly the equation A.20. It still remains to prove A.19.

Denote by μ_n^j the image of the measure μ_n under the transformation $x \mapsto \langle x, e_j \rangle$. Then the characteristic function φ_n^j of the measure μ_n^j is

$$\varphi_n^j(t) = \varphi(t \Pi_n e_j), \quad t \in \mathbb{R}$$

Note $\varphi_n^j(t) = \varphi(t e_j)$ for $n \geq j$, so obviously $\{\varphi_n^j\}_n$ weakly converges. By the Prokhorov theorem for measures on $\mathbb{R}$, the family $\{\mu_n^j\}_n$ is tight for all $j \in \mathbb{N}$. In particular for arbitrary $\varepsilon > 0$ there exists $k > 0$ such that

$$\mu_n^j \left[-\frac{k}{\sqrt{M}}, \frac{k}{\sqrt{M}} \right] \geq 1 - \frac{\varepsilon}{M}, \quad \forall n \in \mathbb{N}, \forall j \in \{1, \dots, M\}$$

Equivalently:

$$\mu_n \left\{ x : |\langle x, e_j \rangle| \leq \frac{k}{\sqrt{M}} \right\} \geq 1 - \frac{\varepsilon}{M}, \quad \forall n \in \mathbb{N}, \forall j \in \{1, \dots, M\}$$

The following inclusion holds:

$$\left\{ x : \sum_{j=1}^M \langle x, e_j \rangle^2 \geq k^2 \right\} \subseteq \bigcup_{j=1}^M \left\{ x : |\langle x, e_j \rangle| \geq \frac{k}{\sqrt{M}} \right\}$$

By the subadditivity and the estimates on $\mu_n \left\{ x : |\langle x, e_j \rangle| \leq \frac{k}{\sqrt{M}} \right\}$ we get

$$\begin{aligned} \mu_n \left\{ x : \sum_{j=1}^M \langle x, e_j \rangle^2 \geq k^2 \right\} &\leq M \frac{\varepsilon}{M} = \varepsilon \quad \forall n \in \mathbb{N} \\ \mu_n \left\{ x : \sum_{j=1}^M \langle x, e_j \rangle^2 \leq k^2 \right\} &\geq 1 - \varepsilon \quad \forall n \in \mathbb{N} \end{aligned}$$

We proved equation A.19, so the sequence $\{\mu_n\}$ is tight $\Rightarrow$ relatively compact by Prokhorov. The weak limit μ then has φ as its characteristic function. $\square$

By a probability measure on H we will usually mean a probability measure defined on $(H, \mathcal{B}(H))$.

According to the general definition valid for probability measures on Banach spaces, a probability measure μ on a Hilbert space H is called Gaussian if for arbitrary $h \in H$ there exist $m \in \mathbb{R}$ and $q \geq 0$, such that

$$\mu \{x \in H : \langle h, x \rangle \in A\} = \mathcal{N}(m, q)(A) \quad A \in \mathcal{B}(\mathbb{R})$$

In particular, if μ is Gaussian, the following functionals

$$H \longrightarrow \mathbb{R}, \quad h \mapsto \int_H \langle h, x \rangle \mu(dx)$$

$$H \times H \longrightarrow \mathbb{R}, \quad (h_1, h_2) \mapsto \int_H \langle h_1, x \rangle \langle h_2, x \rangle \mu(dx)$$

are well defined and continuous. By Riesz's representation theorem there exist $m \in H$ and a symmetric continuous operator $Q : H \longrightarrow H$ such that

$$\begin{aligned} \langle m, h \rangle &= \int_H \langle h, x \rangle \mu(dx) \quad h \in H \\ \langle Qh_1, h_2 \rangle &= \int_H \langle h_1, x \rangle \langle h_2, x \rangle \mu(dx) - \langle m, h_1 \rangle \langle m, h_2 \rangle \\ &= \int_H \langle h_1, x - m \rangle \langle h_2, x - m \rangle \mu(dx) \\ &\quad + \int_H \langle h_1, m \rangle \langle h_2, m \rangle \mu(dx) - \langle m, h_1 \rangle \langle m, h_2 \rangle \\ &\quad + \int_H \langle h_1, m \rangle \langle h_2, x - m \rangle \mu(dx) + \int_H \langle h_1, x - m \rangle \langle h_2, m \rangle \mu(dx) \\ &= \int_H \langle h_1, x - m \rangle \langle h_2, x - m \rangle \mu(dx) \quad h_1, h_2 \in H \end{aligned}$$

Indeed, the algebraic sums in the third and fourth row are obviously both zero. They are called respectively the mean m and the covariance operator Q of μ . By Hölder inequality:

$$\begin{aligned} \|\langle h, x \rangle \cdot 1\|_1^2 &\leq \|\langle h, x \rangle\|_2^2 \cdot \|1\|_2^2 \\ \left(\int_H \langle h, x \rangle \mu(dx) \right)^2 &\leq \left(\int_H |\langle h, x \rangle| \mu(dx) \right)^2 \leq \int_H \langle h, x \rangle^2 \mu(dx) \cdot 1 \end{aligned}$$

This implies that $\langle Qh, h \rangle \geq 0$ for all $h \in H$, so that Q is a nonnegative operator. For a Gaussian measure μ with mean m and covariance Q the following holds:

$$\hat{\mu}(\lambda) = \exp \left(i \langle \lambda, m \rangle - \frac{1}{2} \langle Q\lambda, \lambda \rangle \right) \quad \lambda \in H \quad (\text{A.21})$$

The proof uses the same argument explained for the equation A.17.

Proposition A.22. *Let μ be a Gaussian probability measure with mean 0 and covariance Q . Then Q is a trace-class operator.*

Dim: consider the characteristic function of the measure, $\widehat{\mu}(h) = e^{-\frac{1}{2}\langle Qh, h \rangle}$. We don't need to take any real part to proceed as in the proof of theorem A.18. For arbitrary $r > 0$, let $B(r) = \{x \in H : \|x\| \leq r\}$. Then:

$$\begin{aligned} 1 - e^{-\frac{1}{2}\langle Qh, h \rangle} &= \int_H [1 - \cos \langle h, x \rangle] \mu(dx) \leq \int_H 2 \sin^2 \left(\frac{\langle h, x \rangle}{2} \right) \mu(dx) \\ &= \int_{B(r)} 2 \sin^2 \left(\frac{\langle h, x \rangle}{2} \right) \mu(dx) + \int_{B^c(r)} 2 \sin^2 \left(\frac{\langle h, x \rangle}{2} \right) \mu(dx) \\ &\leq \int_{B(r)} 2 \left(\frac{\langle h, x \rangle}{2} \right)^2 \mu(dx) + \int_{B^c(r)} 2 \sin^2 \left(\frac{\langle h, x \rangle}{2} \right) \mu(dx) \\ &\leq \int_{B(r)} \frac{1}{2} \langle h, x \rangle^2 \mu(dx) + \int_{B^c(r)} 2 \sin^2 \left(\frac{\langle h, x \rangle}{2} \right) \mu(dx) \end{aligned}$$

Define the linear bounded operator Q_r by

$$\langle Q_r h, h \rangle = \int_{B(r)} \langle h, x \rangle^2 \mu(dx)$$

This is a trace-class operator, since for $\{e_n\}$ orthonormal basis:

$$\text{Tr}(Q_r) = \sum_{n \in \mathbb{N}} \langle Q_r e_n, e_n \rangle = \int_{B(r)} \left(\sum_{n \in \mathbb{N}} \langle e_n, x \rangle^2 \right) \mu(dx) = \int_{B(r)} \|x\|^2 \mu(dx) \leq r^2$$

The chain of inequalities becomes:

$$1 - e^{-\frac{1}{2}\langle Qh, h \rangle} \leq \frac{1}{2} \langle Q_r h, h \rangle + 2\mu\{x \in H : \|x\| \geq r\} \quad (\text{A.23})$$

It suffices to prove that there exists $\beta > 0$ such that $\langle Q_r h, h \rangle \leq 1 \Rightarrow \langle Qh, h \rangle \leq \beta$. Consider h such that $\langle Q_r h, h \rangle \leq 1$: from the estimate on $1 - e^{-\frac{1}{2}\langle Qh, h \rangle}$, we get

$$\begin{aligned} 1 - e^{-\frac{1}{2}\langle Qh, h \rangle} &\leq \left(\frac{1}{2} + 2\mu\{x \in H : \|x\| \geq r\} \right) \\ e^{-\frac{1}{2}\langle Qh, h \rangle} &\geq \left(\frac{1}{2} - 2\mu\{x \in H : \|x\| \geq r\} \right) \\ e^{+\frac{1}{2}\langle Qh, h \rangle} &\leq \left(\frac{1}{2} - 2\mu\{x \in H : \|x\| \geq r\} \right)^{-1} \end{aligned}$$

So we can find the desired β :

$$\langle Qh, h \rangle \leq 2 \log \left(\frac{1}{2} - 2\mu\{x \in H : \|x\| \geq r\} \right)^{-1} := \beta$$

Remember that the equation A.23 holds for any $r > 0$, so we can assume that the logarithm exists. This proves $Q \leq \beta Q_r$, so we conclude that Q is also a trace-class operator. $\square$

Proposition A.24. *Let Q be a nonnegative, symmetric, trace-class operator in H and let $m \in H$. Then there exists a Gaussian measure with mean m and covariance Q .*

Dim: in the following we denote by $\{e_j\}$ a orthonormal basis which diagonalizes Q and by $\{\lambda_j\}$ the corresponding eigenvalues. Let $\{\xi_n\}$ be a sequence of independent, real variables distributed as $\mathcal{N}(0, 1)$ and defined on $(\Omega, \mathcal{F}, \mathbf{P})$. Set:

$$\xi = m + \sum_{j=1}^{\infty} \sqrt{\lambda_j} \xi_j e_j$$

This series is convergent in $L^2(\Omega, \mathcal{F}, \mathbf{P})$ since Q is a trace-class operator. Fix $h \in H$ and compute

$$\begin{aligned} \mathbf{E} \left[\exp \left(i \sum_{j=1}^{\infty} \langle h, e_j \rangle \sqrt{\lambda_j} \xi_j \right) \right] &= \lim_n \mathbf{E} \left[\exp \left(i \sum_{j=1}^n \langle h, e_j \rangle \sqrt{\lambda_j} \xi_j \right) \right] \\ &= \lim_n \mathbf{E} \left[\prod_{j=1}^n \exp \left(i \langle h, e_j \rangle \sqrt{\lambda_j} \xi_j \right) \right] = \lim_n \prod_{j=1}^n \mathbf{E} \left[\exp \left(i \langle h, e_j \rangle \sqrt{\lambda_j} \xi_j \right) \right] \\ &= \lim_n \prod_{j=1}^n \exp \left(-\frac{1}{2} \lambda_j \langle h, e_j \rangle^2 \right) = \lim_n \exp \left(-\frac{1}{2} \sum_{j=1}^n \lambda_j \langle h, e_j \rangle^2 \right) \end{aligned}$$

Now note that:

$$\begin{aligned} h &= \sum_{j=1}^{\infty} \langle h, e_j \rangle e_j \Rightarrow Qh = \sum_{j=1}^{\infty} \langle h, e_j \rangle Qe_j \\ &\Rightarrow \langle Qh, h \rangle = \sum_{j=1}^{\infty} \langle h, e_j \rangle \langle Qe_j, h \rangle = \sum_{j=1}^{\infty} \lambda_j \langle h, e_j \rangle^2 \end{aligned}$$

Putting together these equations, we can compute the characteristic function for the law of ξ :

$$\begin{aligned} \mathbf{E} [\exp (i \langle h, \xi \rangle)] &= \exp (i \langle h, m \rangle) \cdot \mathbf{E} \left[\exp \left(i \sum_{j=1}^{\infty} \langle h, e_j \rangle \sqrt{\lambda_j} \xi_j \right) \right] \\ &= \exp (i \langle h, m \rangle) \cdot \exp \left[-\frac{1}{2} \langle Qh, h \rangle \right] \end{aligned}$$

So, ξ has a Gaussian distribution and has mean m . Let us compute covariance for $x, y \in H$:

$$\begin{aligned} \langle \text{Cov}(\xi)x, y \rangle &= \mathbf{E} \left[\left\langle x, \sum_{j=1}^{\infty} \sqrt{\lambda_j} \xi_j e_j \right\rangle \left\langle y, \sum_{k=1}^{\infty} \sqrt{\lambda_k} \xi_k e_k \right\rangle \right] \\ &= \mathbf{E} \left[\left(\sum_{j=1}^{\infty} \sqrt{\lambda_j} \xi_j \langle x, e_j \rangle \right) \left(\sum_{k=1}^{\infty} \sqrt{\lambda_k} \xi_k \langle y, e_k \rangle \right) \right] \end{aligned}$$

Now, we will exploit the independence of $\{\xi_j\}$ and the zero mean hypothesis: all the terms with $j \neq k$ disappear since the mean of $\xi_j \xi_k$ is the product of the respective means, which are both zero. It becomes:

$$\begin{aligned} \langle \text{Cov}(\xi)x, y \rangle &= \mathbf{E} \left[\sum_{j=1}^{\infty} \lambda_j \xi_j^2 \langle x, e_j \rangle \langle y, e_j \rangle \right] = \sum_{j=1}^{\infty} \mathbf{E} [\lambda_j \xi_j^2 \langle x, e_j \rangle \langle y, e_j \rangle] \\ &= \sum_{j=1}^{\infty} \lambda_j \langle x, e_j \rangle \langle y, e_j \rangle \mathbf{E} [\xi_j^2] = \sum_{j=1}^{\infty} \lambda_j \langle x, e_j \rangle \langle y, e_j \rangle \end{aligned}$$

Now note that:

$$\begin{aligned} x &= \sum_{j=1}^{\infty} \langle x, e_j \rangle e_j, \quad y = \sum_{k=1}^{\infty} \langle y, e_k \rangle e_k \Rightarrow Qx = \sum_{j=1}^{\infty} \langle x, e_j \rangle Qe_j \\ \Rightarrow \langle Qx, y \rangle &= \sum_{j,k=1}^{\infty} \langle x, e_j \rangle \langle y, e_k \rangle \langle Qe_j, e_k \rangle = \sum_{j=1}^{\infty} \langle x, e_j \rangle \langle y, e_j \rangle \lambda_j \end{aligned}$$

Putting together these equations, we deduce that the covariance operator of ξ is Q as desired. The other requests on ξ are also satisfied, so we have the thesis. $\square$

Our aim is to give necessary and sufficient conditions under which two Gaussian measures $\mu = \mathcal{N}(m_1, Q_1)$ and $\nu = \mathcal{N}(m_2, Q_2)$ on a Hilbert space are absolutely continuous. We will need some preliminary result.

Let μ and ν be probability measure on a measurable space $(E, \mathcal{E})$. A measure μ is absolutely continuous with respect to ν , and we shall write $\mu < \nu$, if

$$\nu(\Gamma) = 0 \implies \mu(\Gamma) = 0, \quad \forall \Gamma \in \mathcal{E}$$

If $\mu < \nu$ and $\nu < \mu$ then μ and ν are said to be equivalent and we write $\mu \sim \nu$. If μ and ν are concentrated on disjoint sets then they are called singular and we write $\mu \perp \nu$.

Let λ be a probability measure such that μ, ν are absolutely continuous with respect to λ . The Hellinger integral $H(\mu, \nu)$ is defined by the following formula:

$$H(\mu, \nu) = \int_E \left(\frac{d\mu}{d\lambda} \times \frac{d\nu}{d\lambda} \right)^{\frac{1}{2}} d\lambda$$

This definition is independent of the choice of the probability measure λ .

By Hölder:

$$0 \leq \int_E \left(\frac{d\mu}{d\lambda} \times \frac{d\nu}{d\lambda} \right)^{\frac{1}{2}} d\lambda \leq \left(\int_E \frac{d\mu}{d\lambda} d\lambda \right)^{\frac{1}{2}} \left(\int_E \frac{d\nu}{d\lambda} d\lambda \right)^{\frac{1}{2}} \leq 1$$

Obviously, $H(\mu, \nu) = 0$ if and only if $\mu \perp \nu$. It follows that the condition $H(\mu, \nu) > 0$ is necessary for the equivalence of μ and ν ; however for product measures on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ it is also a sufficient condition.

Proposition A.25. *Let μ_k, ν_k be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $\forall k \in \mathbb{N}$. Assume that $\mu_k < \nu_k$ for all k and define*

$$\mu := \prod_{k=1}^{\infty} \mu_k, \quad \nu := \prod_{k=1}^{\infty} \nu_k$$

$$H(\mu, \nu) = \prod_{k=1}^{\infty} H(\mu_k, \nu_k) = \prod_{k=1}^{\infty} \int_{\mathbb{R}} \left(\frac{d\mu_k}{d\nu_k} \right)^{\frac{1}{2}} d\nu_k$$

If $H(\mu, \nu) > 0$ then the measure μ is absolutely continuous with respect to ν .

Dim.: see [1], p. 61.

It follows that if also $\nu_k < \mu_k$ for all k , then μ and ν are equivalent.

A operator $T \in L(E, F)$ is said to be Hilbert-Schmidt if for an orthonormal basis $\{e_j\}$ on E :

$$\sum_{j=1}^{\infty} \langle Te_j, Te_j \rangle = \sum_{j=1}^{\infty} \|Te_j\|_F^2 < +\infty$$

This is a well-defined number independent of the choice of the basis $\{e_j\}$.

The space of all Hilbert-Schmidt operators is a separable Hilbert space and it will be denoted as $L_2(E, F)$. We will write $L_2(E)$ instead of $L_2(E, E)$.

Before giving a complete characterization for equivalence of Gaussian measures, we deduce from proposition A.25 an important case.

Theorem A.26. *(Feldman-Hajek, special case)*

The following statements hold:

- (i) $\mu = \mathcal{N}(m_1, Q)$ and $\nu = \mathcal{N}(m_2, Q)$ are either singular or equivalent.
- (ii) They are equivalent if and only if $m_1 - m_2 \in \text{Im } Q^{\frac{1}{2}}$.

Dim.: the space H can be considered as a Borel subset of $\mathbb{R}^{\infty}$, while the measures μ and ν can be regarded as product measures

$$\mu := \prod_{k=1}^{\infty} \mu_k, \quad \nu := \prod_{k=1}^{\infty} \nu_k$$

where

$$m_{1k} = \langle m_1, e_k \rangle \implies \mu_k = \mathcal{N}(m_{1k}, \sqrt{\lambda_k})$$

$$m_{2k} = \langle m_2, e_k \rangle \implies \nu_k = \mathcal{N}(m_{2k}, \sqrt{\lambda_k})$$

We will compute the Hellinger integral $H(\mu_k, \nu_k)$ choosing as λ the Lebesgue measure. We have:

$$\frac{d\mu_k(x)}{dx} = \frac{1}{\sqrt{2\pi\lambda_k}} \exp \left[-\frac{(x - m_{1k})^2}{2\lambda_k} \right]$$

$$\frac{d\nu_k(x)}{dx} = \frac{1}{\sqrt{2\pi\lambda_k}} \exp \left[-\frac{(x - m_{2k})^2}{2\lambda_k} \right]$$

Thus, the integrand function in $H(\mu_k, \nu_k)$ is

$$\begin{aligned}
& \frac{1}{\sqrt{2\pi\lambda_k}} \exp \left[-\frac{(x - m_{1k})^2 + (x - m_{2k})^2}{4\lambda_k} \right] \\
&= \frac{1}{\sqrt{2\pi\lambda_k}} \exp \left[-\frac{2x^2 - 2x(m_{1k} + m_{2k}) + m_{1k}^2 + m_{2k}^2}{4\lambda_k} \right] \\
&= \frac{1}{\sqrt{2\pi\lambda_k}} \exp \left[-\frac{x^2 - x(m_{1k} + m_{2k})}{2\lambda_k} \right] \exp \left[-\frac{m_{1k}^2 + m_{2k}^2}{4\lambda_k} \right] \\
&= \frac{1}{\sqrt{2\pi\lambda_k}} \exp \left[-\frac{x^2 - x(m_{1k} + m_{2k}) + \left(\frac{m_{1k} + m_{2k}}{2}\right)^2}{2\lambda_k} \right] \exp \left[+\frac{\left(\frac{m_{1k} + m_{2k}}{2}\right)^2}{2\lambda_k} \right] \\
&\quad \exp \left[-\frac{m_{1k}^2 + m_{2k}^2}{4\lambda_k} \right] \\
&= \frac{1}{\sqrt{2\pi\lambda_k}} \exp \left[-\frac{\left(x - \frac{m_{1k} + m_{2k}}{2}\right)^2}{2\lambda_k} \right] \exp \left[+\frac{m_{1k}^2 + m_{2k}^2 + 2m_{1k}m_{2k} - 2m_{1k}^2 - 2m_{2k}^2}{8\lambda_k} \right]
\end{aligned}$$

Now we can explicitly compute the Hellinger integral:

$$\begin{aligned}
H(\mu_k, \nu_k) &= \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi\lambda_k}} \exp \left[-\frac{\left(x - \frac{m_{1k} + m_{2k}}{2}\right)^2}{2\lambda_k} \right] \exp \left[-\frac{(m_{1k} - m_{2k})^2}{8\lambda_k} \right] dx \\
&= \exp \left[-\frac{(m_{1k} - m_{2k})^2}{8\lambda_k} \right]
\end{aligned}$$

So the Hellinger integral $H(\mu, \nu) = \prod_{k=1}^{\infty} H(\mu_k, \nu_k) > 0$ if and only if

$$S := \sum_{k=1}^{\infty} \frac{(m_{1k} - m_{2k})^2}{\lambda_k} < +\infty$$

Suppose that there exists $v \in H$ with $m_1 - m_2 = Q^{\frac{1}{2}}v$. The following hold:

$$\begin{aligned}
m_1 - m_2 &= \sum_{k \in \mathbb{N}} \langle m_1 - m_2, e_k \rangle e_k = \sum_{k \in \mathbb{N}} (m_{1k} - m_{2k}) e_k \\
Q^{\frac{1}{2}}v &= Q^{\frac{1}{2}} \sum_{k \in \mathbb{N}} \langle v, e_k \rangle e_k = \sum_{k \in \mathbb{N}} \langle v, e_k \rangle \sqrt{\lambda_k} e_k
\end{aligned}$$

Hence,

$$\frac{(m_{1k} - m_{2k})}{\sqrt{\lambda_k}} = \langle v, e_k \rangle \Rightarrow \sum_{k=1}^{\infty} \frac{(m_{1k} - m_{2k})^2}{\lambda_k} = \sum_{k=1}^{\infty} \langle v, e_k \rangle^2 = \|v\|^2$$

Conversely, if $S < +\infty$ we choose $v \in H$ so that it has those components and follow backwards the previous argument to say $Q^{\frac{1}{2}}v = m_1 - m_2$. In the end:

- if $H(\mu, \nu) = 0$ the measures are obviously singular.
- if $H(\mu, \nu) > 0$ by proposition A.25 they are equivalent and this happens if and only if $m_1 - m_2 \in \text{Im } Q^{\frac{1}{2}}$. $\square$

Theorem A.27. (Feldman and Hajek)

The following statements hold.

- (1) Measures $\mu = \mathcal{N}(m_1, Q_1)$, $\nu = \mathcal{N}(m_2, Q_2)$ are either singular or equivalent.
 (2) They are equivalent if and only if all the following conditions hold:

- (i) $\text{Im } Q_1^{1/2} = \text{Im } Q_2^{1/2} = H_0$.
- (ii) $m_1 - m_2 \in H_0$.
- (iii) the operator $\left(Q_1^{-1/2} Q_2^{1/2}\right) \left(Q_1^{-1/2} Q_2^{1/2}\right)^* - \mathbf{1}$ is a Hilbert-Schmidt operator on $\overline{H_0}$. Here x^* means the transpose matrix of x .

Dim.: we show first that if $\mathcal{N}(m_1, Q_1)$ and $\mathcal{N}(m_2, Q_2)$ are non singular, then Q_1 and Q_2 satisfy properties (i) and (iii).

Assume $\mathcal{N}(m_1, Q_1)$ and $\mathcal{N}(m_2, Q_2)$ are non singular. By a simple translation, also $\mathcal{N}(0, Q_1)$ and $\mathcal{N}(m_2 - m_1, Q_2)$ are non singular. By symmetry of the second one $\mathcal{N}(0, Q_1)$ and $\mathcal{N}(m_1 - m_2, Q_2)$ are non singular. So,

$$\mathcal{N}(0, 2Q_1) = \mathcal{N}(0, Q_1) * \mathcal{N}(0, Q_1)$$

$$\mathcal{N}(0, 2Q_2) = \mathcal{N}(m_2 - m_1, Q_2) * \mathcal{N}(m_1 - m_2, Q_2)$$

are non singular.

This is just to say that in order to prove our claim, we can assume that $m_1 = m_2 = 0$. By contradiction, if $\text{Im } Q_1^{1/2} \neq \text{Im } Q_2^{1/2}$ then we can find a sequence $\{a_n\}$ of non zero elements in H such that

$$\lim_{n \rightarrow \infty} \frac{\|Q_1^{1/2} a_n\|}{\|Q_2^{1/2} a_n\|} = +\infty$$

Define the following random variables on H :

$$\xi_n(x) = \frac{\langle x, a_n \rangle}{\|Q_1^{1/2} a_n\|}, \quad n \in \mathbb{N}$$

Then:

$$\int_H \xi_n^2(x) \mu(dx) = 1 \quad \text{and} \quad \int_H \xi_n^2(x) \nu(dx) = \left(\frac{\|Q_2^{1/2} a_n\|}{\|Q_1^{1/2} a_n\|} \right)^2$$

From the choice of the sequence $\{a_n\}$, the second integral tends to 0 as $n \rightarrow \infty$. Therefore we can find sequences $\gamma_k \uparrow \infty$ and $n_k \uparrow +\infty$ such that

$$x \in \text{supp } \mu \implies \lim_{k \rightarrow \infty} |\gamma_k \xi_{n_k}(x)| = +\infty$$

$$x \in \text{supp } \nu \implies \lim_{k \rightarrow \infty} |\gamma_k \xi_{n_k}(x)| = 0$$

These two equations make clear that μ and ν are singular, contradiction.

Assume that $\text{Im } Q_1^{1/2} = \text{Im } Q_2^{1/2} = H_0$ with $\overline{H_0} = H$ and let us prove that (iii) also holds. Define

$$R = \left(Q_1^{-1/2} Q_2^{1/2}\right) \left(Q_1^{-1/2} Q_2^{1/2}\right)^*$$

Let $\{e_j, \lambda_j\}$ be the eigensequence determined by Q_1 . We can check that the operator R is represented in the basis $\{e_j\}$ by the matrix $\{r_{ij}\}$ where

$$r_{ij} = \frac{\langle Q_2 e_i, e_j \rangle}{\sqrt{\lambda_i \lambda_j}}$$

Define the following random variables on H :

$$\zeta_n(x) = \frac{\langle x, e_n \rangle}{\sqrt{\lambda_n}}, \quad n \in \mathbb{N}$$

If $\mathcal{G}_n = \sigma(\zeta_1, \dots, \zeta_n)$ we will denote with $H_{\mathcal{G}_n}$ the Hellinger integral calculated for the restrictions of μ and ν to $\mathcal{G}_n$. One can prove that $H_{\mathcal{G}_n}(\mu, \nu) \geq H(\mu, \nu)$. By a direct calculation:

$$H_{\mathcal{G}_n}(\mu, \nu) = \frac{(\det R_n)^{1/4}}{(\det (\frac{I_n + R_n}{2}))^{1/2}} \geq H(\mu, \nu)$$

where R_n and I_n are the n -dimensional matrixes with entries $\{r_{ij}\}$ and $\{\delta_{ij}\}$ such that $1 \leq i, j \leq n$. Denote by $\lambda_{n,j}$ the eigenvalues for R_n . Then:

$$\begin{aligned} \log H_{\mathcal{G}_n}(\mu, \nu) &= \frac{1}{4} \sum_{j=1}^n \log \frac{4\lambda_{n,j}}{(1 + \lambda_{n,j})^2} \geq \log H(\mu, \nu) \\ -\log H_{\mathcal{G}_n}(\mu, \nu) &= \frac{1}{4} \sum_{j=1}^n \log \frac{(1 + \lambda_{n,j})^2}{4\lambda_{n,j}} \leq -\log H(\mu, \nu) \end{aligned}$$

Note that, since we are assuming the measures to be non singular, $H(\mu, \nu) > 0$. Therefore in the previous estimate we have $-\log H(\mu, \nu) < +\infty$. Moreover if

$$\begin{aligned} \lambda_{n,j} \rightarrow +\infty &\implies \log \frac{(1 + \lambda_{n,j})^2}{4\lambda_{n,j}} \rightarrow +\infty \\ \lambda_{n,j} \rightarrow 0 &\implies \log \frac{(1 + \lambda_{n,j})^2}{4\lambda_{n,j}} \rightarrow +\infty \end{aligned}$$

In order to fulfill the previous estimate, there must be some constants $C_1, C_2 > 0$ such that $C_1 \leq \lambda_{n,j} \leq C_2$ for all $n \in \mathbb{N}$ and for $j = 1, \dots, n$. There exists $C_3 > 0$ such that

$$(1 - \lambda)^2 \leq \frac{C_3}{4} \log \frac{(1 + \lambda)^2}{4\lambda}$$

for all $\lambda \in [C_1, C_2]$. Thus,

$$\sum_{j=1}^n (1 - \lambda_{n,j})^2 \leq \frac{C_3}{4} \sum_{j=1}^n \log \frac{(1 + \lambda_{n,j})^2}{4\lambda_{n,j}} < +\infty$$

The left hand side is the Hilbert-Schmidt norm of the matrix $I_n - R_n$. Since the estimate holds for all $n \in \mathbb{N}$, the Hilbert-Schmidt norm of $\mathbb{1} - R$ is finite as desired.

We will show that if Q_1 and Q_2 satisfy (i) and (iii), then $\mathcal{N}(0, Q_1) \sim \mathcal{N}(0, Q_2)$.
 Let $\{f_j, \tau_j\}$ be the eigensequence associated with R :

$$Rf_j = \tau_j f_j$$

If property (iii) is satisfied, then $\sum_j (1 - \tau_j)^2 < \infty \Rightarrow C_1 \leq \tau_j \leq C_2$ for all j .
 Indeed if

$$\forall C > 0, \exists j \in \mathbb{N} : \tau_j \in]0, C[\implies \exists (\tau_{j_n})_n \rightarrow 0$$

Obviously

$$\sum_{j_n} (1 - \tau_{j_n})^2 \leq \sum_j (1 - \tau_j)^2 < \infty$$

but this gives us a contradiction since the series on the left hand side is divergent.
 This proves the existence of the constant C_1 as lower bound. Same goes for C_2 .
 Let $\{\xi_j\}$ be a sequence of independent, real variables distributed as $\mathcal{N}(0, 1)$.
 We claim that

$$\mathcal{L} \left(\sum_{j=1}^{\infty} \xi_j Q_1^{1/2} f_j \right) = \mathcal{N}(0, Q_1) \quad (\text{A.28})$$

$$\mathcal{L} \left(\sum_{j=1}^{\infty} \sqrt{\tau_j} \xi_j Q_1^{1/2} f_j \right) = \mathcal{N}(0, Q_2) \quad (\text{A.29})$$

The series are convergent because Q_1 is trace-class and $\{\tau_j\}$ is bounded. They have symmetric Gaussian law, we just have to compute covariance. We check the second one:

$$\begin{aligned} \langle \text{Cov}(\cdot)x, y \rangle &= \mathbf{E} \left[\left\langle x, \sum_{j=1}^{\infty} \sqrt{\tau_j} \xi_j Q_1^{1/2} f_j \right\rangle \left\langle y, \sum_{k=1}^{\infty} \sqrt{\tau_k} \xi_k Q_1^{1/2} f_k \right\rangle \right] \\ &= \mathbf{E} \left[\left(\sum_{j=1}^{\infty} \sqrt{\tau_j} \xi_j \langle x, Q_1^{1/2} f_j \rangle \right) \left(\sum_{k=1}^{\infty} \sqrt{\tau_k} \xi_k \langle y, Q_1^{1/2} f_k \rangle \right) \right] \end{aligned}$$

Now, we will exploit the independence of $\{\xi_j\}$ and the zero mean hypothesis: all the terms with $j \neq k$ disappear. It becomes:

$$\begin{aligned} \mathbf{E} \left[\sum_{j=1}^{\infty} \tau_j \xi_j^2 \langle x, Q_1^{1/2} f_j \rangle \langle y, Q_1^{1/2} f_j \rangle \right] &= \sum_{j=1}^{\infty} \tau_j \langle x, Q_1^{1/2} f_j \rangle \langle y, Q_1^{1/2} f_j \rangle \\ &= \sum_{j=1}^{\infty} \langle x, Q_1^{1/2} (Q_1^{-1/2} Q_2^{1/2}) (Q_1^{-1/2} Q_2^{1/2})^* f_j \rangle \langle y, Q_1^{1/2} f_j \rangle \\ &= \sum_{j=1}^{\infty} \langle x, Q_1^{1/2} (Q_1^{-1/2} Q_2^{1/2}) (Q_1^{-1/2} Q_2^{1/2})^* f_j \rangle \langle Q_1^{1/2} y, f_j \rangle \\ &= \sum_{j=1}^{\infty} \langle x, Q_2^{1/2} (Q_1^{-1/2} Q_2^{1/2})^* f_j \rangle \langle Q_1^{1/2} y, f_j \rangle \end{aligned}$$

We used symmetry of $Q_1^{1/2}$ in the second scalar product. We are going to do the same with $Q_2^{1/2}$ in the first one and then we will move the transpose term:

$$\begin{aligned}
\langle \text{Cov}(\cdot)x, y \rangle &= \sum_{j=1}^{\infty} \left\langle x, Q_2^{1/2} \left(Q_1^{-1/2} Q_2^{1/2} \right)^* f_j \right\rangle \left\langle Q_1^{1/2} y, f_j \right\rangle \\
&= \sum_{j=1}^{\infty} \left\langle Q_2^{1/2} x, \left(Q_1^{-1/2} Q_2^{1/2} \right)^* f_j \right\rangle \left\langle Q_1^{1/2} y, f_j \right\rangle \\
&= \sum_{j=1}^{\infty} \left\langle \left(Q_1^{-1/2} Q_2^{1/2} \right) Q_2^{1/2} x, f_j \right\rangle \left\langle Q_1^{1/2} y, f_j \right\rangle \\
&= \left\langle \left(Q_1^{-1/2} Q_2^{1/2} \right) Q_2^{1/2} x, Q_1^{1/2} y \right\rangle = \left\langle Q_1^{-1/2} Q_2 x, Q_1^{1/2} y \right\rangle = \langle Q_2 x, y \rangle
\end{aligned}$$

In the last equality we used again symmetry of Q_1 . So we proved A.28 and the other follows as well.

Let μ and ν be measures generated on $\mathbb{R}^\infty$ by sequences $\{\xi_j\}$ and $\{\sqrt{\tau_j}\xi_j\}$:

$$\begin{aligned}
\mu &= \prod_{j=1}^{\infty} \mu_j, \quad \mu_j = \mathcal{L}(\xi_j) = \mathcal{N}(0, 1) \\
\nu &= \prod_{j=1}^{\infty} \nu_j, \quad \nu_j = \mathcal{L}(\sqrt{\tau_j}\xi_j) = \mathcal{N}(0, \tau_j)
\end{aligned}$$

Since $\sum_j (1 - \tau_j)^2 < \infty$, they satisfy hypothesis of proposition A.25, so $\mu \sim \nu$. On the other hand, $\mathcal{N}(0, Q_1)$ and $\mathcal{N}(0, Q_2)$ are the images respectively of the measures μ and ν by the measurable mapping $V : \mathbb{R}^\infty \rightarrow H$,

$$V(\{v_j\}) = \begin{cases} \lim_{N \rightarrow \infty} \sum_{j=1}^N v_j Q^{1/2} f_j & \text{if the limit exists} \\ 0 & \text{otherwise} \end{cases}$$

It follows that $\mathcal{N}(0, Q_1)$ and $\mathcal{N}(0, Q_2)$ are equivalent, as desired.

If $\mathcal{N}(m_1, Q_1)$ and $\mathcal{N}(m_2, Q_2)$ are non singular, then (i), (ii) and (iii) hold and we have $\mathcal{N}(m_1, Q_1) \sim \mathcal{N}(m_2, Q_2)$.

Indeed, $\mathcal{N}(0, Q_1)$ and $\mathcal{N}(m_2 - m_1, Q_2)$ are also non singular by translation. For what we proved previously, conditions (i), (iii) hold, also $\mathcal{N}(0, Q_1) \sim \mathcal{N}(0, Q_2)$. Hence $\mathcal{N}(0, Q_2)$ and $\mathcal{N}(m_2 - m_1, Q_2)$ are non singular. By the special case of Feldman-Hajek, condition (ii) holds as well and

$$\mathcal{N}(0, Q_1) \sim \mathcal{N}(0, Q_2) \sim \mathcal{N}(m_2 - m_1, Q_2)$$

By translation:

$$\mathcal{N}(m_1, Q_1) \sim \mathcal{N}(m_1, Q_2) \sim \mathcal{N}(m_2, Q_2)$$

Conversely, assume (i), (ii) and (iii) hold. By (i) and (iii):

$$\mathcal{N}(0, Q_1) \sim \mathcal{N}(0, Q_2) \implies \mathcal{N}(m_1, Q_1) \sim \mathcal{N}(m_1, Q_2)$$

By (ii) and the special case theorem:

$$\mathcal{N}(m_1, Q_1) \sim \mathcal{N}(m_1, Q_2) \sim \mathcal{N}(m_2, Q_2)$$

Finally the proof is complete. $\square$

A.3 Stochastic processes

Let E be a separable Banach space, I an interval of $\mathbb{R}$ and $(\Omega, \mathcal{F}, \mathbf{P})$ a probability space. An E -valued stochastic process on I is a function

$$\begin{aligned} X : I \times \Omega &\longrightarrow E \\ (t, \omega) &\longmapsto X(t, \omega) = X(t)(\omega) \end{aligned}$$

such that $\{X(t)\}$ is a family of E -valued random variables defined on Ω . Functions $t \longmapsto X(t, \omega)$ are called the trajectories of the process X . We give several definitions of regularity.

- X is measurable if the map $X(\cdot, \cdot) : I \times \Omega \rightarrow E$ is $\mathcal{B}(I) \times \mathcal{F}$ -measurable.
- X is stochastically continuous in t_0 if: $\forall \varepsilon > 0, \forall \delta > 0, \exists \rho > 0$ such that

$$\mathbf{P}(\|X(t) - X(t_0)\| \geq \varepsilon) \leq \delta \quad \text{if } |t - t_0| < \rho$$

- X is stochastically continuous in I if it is stochastically continuous at every point.
- X is stochastically uniformly continuous in I if: $\forall \varepsilon > 0, \forall \delta > 0, \exists \rho > 0$ such that

$$\mathbf{P}(\|X(t) - X(s)\| \geq \varepsilon) \leq \delta \quad \text{if } |t - s| < \rho$$

- X is continuous a.s. if its trajectories are continuous a.s.

If the interval I is compact, for example $I = [0, T]$, stochastic continuity is equivalent to stochastic uniform continuity. Let us assume that $I = [0, T]$ or $[0, +\infty[$ and that the probability space is equipped with a filtration $\{\mathcal{F}_t\}$, $t \in I$. We will consider the following collection of subsets of $[0, +\infty[\times \Omega$:

$$\mathcal{P}_\infty = \sigma \left\{ \{0\} \times F, F \in \mathcal{F}_0; \quad]s, t] \times F, s < t, F \in \mathcal{F}_s \right\}$$

This σ -field is called predictable σ -field and its element predictable sets. The restriction of $\mathcal{P}_\infty$ to $[0, T] \times \Omega$ will be denoted by $\mathcal{P}_T$. A measurable map:

$$X : ([0, +\infty[\times \Omega, \mathcal{P}_\infty) \longrightarrow (E, \mathcal{B}(E))$$

or even

$$X : ([0, T[\times \Omega, \mathcal{P}_T) \longrightarrow (E, \mathcal{B}(E))$$

is called a predictable process, which is necessarily also an adapted one.

Proposition A.30. *Let X be an $L(U, H)$ -valued adapted and stochastically continuous process on $[0, T]$. Then X has a predictable version on $[0, T]$.*

If $\mathbf{E}[\|X_t\|] < \infty$ for all $t \in I$ then the process is called integrable. An integrable and adapted E -valued process X is said to be a martingale if, for arbitrary $s, t \in I$ s.t. $s \leq t$:

$$\mathbf{E}[X(t)|\mathcal{F}_s] = X(s) \quad \text{a.s.}$$

A real valued integrable and adapted process X is said to be:

- (i) a submartingale if $\mathbf{E}[X(t)|\mathcal{F}_s] \geq X(s)$ a.s., so submartingales increase in average with t .
- (ii) a supermartingale if $\mathbf{E}[X(t)|\mathcal{F}_s] \leq X(s)$ a.s., so supermartingales decrease in average with t .

Let us fix $T > 0$ and denote by $\mathcal{M}_T^2(E)$ the space of all E -valued continuous, square integrable martingales M . It is a Banach space equipped with the norm

$$\|M\|_{\mathcal{M}_T^2} = \left(\mathbf{E} \sup_{t \leq T} \|M(t)\|^2 \right)^{1/2}$$

If $M \in \mathcal{M}_T^2(\mathbb{R})$ there exists a unique increasing and adapted process $\langle\langle M(\cdot) \rangle\rangle$ starting from 0, s.t. the following is a continuous martingale:

$$X(t) = M^2(t) - \langle\langle M(t) \rangle\rangle, \quad t \in [0, T]$$

The process $\langle\langle M(\cdot) \rangle\rangle$ is called the quadratic variation of M .

If $M_1, M_2 \in \mathcal{M}_T^2(\mathbb{R})$ there exists a unique adapted process $\langle\langle M_1(\cdot), M_2(\cdot) \rangle\rangle$ starting from 0, s.t. the following is a continuous martingale:

$$X(t) = M_1(t)M_2(t) - \langle\langle M_1(t), M_2(t) \rangle\rangle, \quad t \in [0, T]$$

The process $\langle\langle M_1(\cdot), M_2(\cdot) \rangle\rangle$ is called the cross quadratic variation of M_1, M_2 . It holds:

$$\langle\langle M_1(t), M_2(t) \rangle\rangle = \frac{\langle\langle (M_1 + M_2)(t) \rangle\rangle - \langle\langle (M_1 - M_2)(t) \rangle\rangle}{4}$$

Proposition A.31. *The following hold:*

- (i) If $M \in \mathcal{M}_T^2(\mathbb{R})$ and $\langle\langle M(t) \rangle\rangle = 0$ for all $t \in [0, T]$ then $M(t) = M(0)$ a.s. on the whole interval.
- (ii) If $M_1, M_2 \in \mathcal{M}_T^2(\mathbb{R})$ are independent then $\langle\langle M_1(t), M_2(t) \rangle\rangle = 0$ on $[0, T]$.

We are going to define the quadratic variation process for $M \in \mathcal{M}_T^2(H)$, where H is a separable Hilbert space. Recall that $L_1 = L_1(H)$ is the space of all trace-class operators and it is a separable Banach space.

A L_1 -valued process $V(\cdot)$ on $[0, T]$ is increasing if the operators $V(t)$ are non-negative and $V(s) \leq V(t)$ for $s \leq t$.

A L_1 -valued increasing and adapted process $V(\cdot)$ starting from 0, is a quadratic variation of a martingale M if and only if for arbitrary $a, b \in H$:

$$X(t) = \langle M(t), a \rangle \langle M(t), b \rangle - \langle V(t)a, b \rangle, \quad t \in [0, T]$$

is a martingale.

Consider for $a, b \in H$ the linear operator $a \otimes b : H \rightarrow H$, $h \mapsto a \langle b, h \rangle$. An equivalent formulation is that $V(\cdot)$ starting from 0, is a quadratic variation of a martingale M if and only if for arbitrary $a, b \in H$:

$$X(t) = M(t) \otimes M(t) - V(t), \quad t \in [0, T]$$

is a martingale. An arbitrary $M \in \mathcal{M}_T^2(H)$ has exactly one quadratic variation process which will be denoted by $\langle\langle M(t) \rangle\rangle$.

If $\{e_k\}$ is a orthonormal complete basis in H , the processes $M_i(t) = \langle M(t), e_i \rangle$ are $\mathbb{R}$ -valued continuous square integrable martingales and the formula

$$\langle\langle M(t) \rangle\rangle = \sum_{i,j=1}^{\infty} \langle\langle M_i(t), M_j(t) \rangle\rangle e_i \otimes e_j, \quad t \in [0, T]$$

defines a L_1 -valued adapted process.

In a similar manner one can define a cross quadratic variation for $M^1 \in \mathcal{M}_T^2(H_1)$, $M^2 \in \mathcal{M}_T^2(H_2)$ with H_1 and H_2 two Hilbert spaces. Namely we define

$$\langle\langle M^1(t), M^2(t) \rangle\rangle = \sum_{i,j=1}^{\infty} \langle\langle M_i^1(t), M_j^2(t) \rangle\rangle e_i^1 \otimes e_j^2, \quad t \in [0, T]$$

where $\{e_i^1\}$ and e_j^2 are complete orthonormal basis in H_1 and H_2 respectively. From the real valued case, we get a generalization of Proposition A.31.

Proposition A.32. *The following hold:*

- (i) *If $M \in \mathcal{M}_T^2(H)$ and $\langle\langle M(t) \rangle\rangle = 0$ for all $t \in [0, T]$ then $M(t) = M(0)$ a.s. on the whole interval.*
- (ii) *If $M_i \in \mathcal{M}_T^2(H_i)$, $i = 1, 2$ are independent then $\langle\langle M_1(t), M_2(t) \rangle\rangle = 0$ on the whole interval.*

A.4 The stochastic integral

We consider U, H two Hilbert spaces and a symmetric nonnegative operator $Q \in L(U)$. We will consider first the case in which $\text{Tr } Q < \infty$, so that there exists a orthonormal system $\{e_k\}$ and a bounded sequence of nonnegative real numbers $\{\lambda_k\}$ such that

$$Qe_k = \lambda_k e_k \quad k \in \mathbb{N}$$

A U -valued stochastic process $W(t)$, $t \geq 0$ is called a Q -Wiener process if:

- $W(0) = 0$ a.s.
- W has a.s. continuous trajectories.
- W has independent increments: it means $(W(t_i) - W(t_{i-1}))$, $i = 1, \dots, n$, are independent random variables.
- $\mathcal{L}(W(t) - W(s)) = \mathcal{N}(0, (t-s)Q)$ for all $0 \leq s \leq t$.

We say that W is a Q -Wiener process on $[0, T]$ if it satisfies these properties on the interval $[0, T]$.

Let us recall that a H -valued stochastic process X on $[0, +\infty[$ is said to be a Gaussian process if the H^n -valued random variable $(X(t_1), \dots, X(t_n))$ is Gaussian for every choice of $n \in \mathbb{N}$, for every $t_1, \dots, t_n \geq 0$.

Proposition A.33. *Assume that W is a Q -Wiener process with $\text{Tr } Q < \infty$. Then:*

- (i) W is Gaussian process, with $\mathbf{E}[W(t)] = 0$, $\text{Cov}(W(t)) = tQ$ for all $t \geq 0$.
- (ii) W has the expansion

$$W(t) = \sum_{j=1}^{\infty} \sqrt{\lambda_j} \beta_j(t) e_j$$

where $\beta_j(t)$, are $\mathbb{R}$ -valued Brownian motions mutually independent satisfying

$$\beta_j(t) = \frac{1}{\sqrt{\lambda_j}} \langle W(t), e_j \rangle$$

- (iii) this expansion for $W(t)$ is a uniformly convergent series on any $[0, T]$ a.s.

Let us verify the independence of the Brownian motions as a useful exercise. If $s < t$,

$$\begin{aligned} \mathbf{E}[\beta_i(t)\beta_j(s)]\sqrt{\lambda_i\lambda_j} &= \mathbf{E}[\langle W(t), e_i \rangle \langle W(s), e_j \rangle] \\ &= \mathbf{E}[\langle W(t) - W(s), e_i \rangle \langle W(s), e_j \rangle] + \mathbf{E}[\langle W(s), e_i \rangle \langle W(s), e_j \rangle] \\ &= \mathbf{E}[\langle W(s), e_i \rangle \langle W(s), e_j \rangle] = \text{Cov}[\langle W(s), e_i \rangle \langle W(s), e_j \rangle] \\ &= \left(sQ\right)_{ij} = s \langle Qe_i, e_j \rangle = s\lambda_i\delta_{ij}. \quad \square \end{aligned}$$

Given a filtered space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbf{P})$ we will assume from now on that a Q -Wiener process W satisfies:

- $W(t)$ is $\mathcal{F}_t$ -measurable for all $t \geq 0$.
- $W(t+h) - W(t)$ is independent of $\mathcal{F}_t$, for all $h \geq 0, t \geq 0$.

A $L(U, H)$ -valued process $\Phi(t), t \in [0, T]$ taking a finite number of values is said to be elementary if

$$\Phi(t) = \Phi_m, \quad t \in]t_m, t_{m+1}]$$

for a sequence of times $0 = t_0 < t_1 < \dots < t_k = T$ and a sequence of L -valued random variables $\Phi_0, \dots, \Phi_{k-1}$ taking a finite number of values and such that Φ_m is $\mathcal{F}_{t_m}$ -measurable.

For an elementary process Φ we define the stochastic integral by the formula

$$\int_0^t \Phi(s) dW(s) = \sum_{m=0}^{k-1} \Phi_m (W(t_{m+1} \wedge t) - W(t_m \wedge t))$$

and denote it $\Phi \cdot W(t)$.

Consider the subspace $U_0 = \text{Im } Q^{1/2}$, it is a Hilbert space endowed with the product $\langle u, v \rangle_0 = \langle Q^{-1/2}u, Q^{-1/2}v \rangle$.

Let $L_2 = L_2(U, H)$ be the space of all Hilbert-Schmidt operators $U \rightarrow H$.

Let $L_2^0 = L_2(U_0, H)$ be the space of all Hilbert-Schmidt operators $U_0 \rightarrow H$.

It is a separable Hilbert space, equipped with the norm

$$\|\Psi\|_{L_2^0}^2 = \|\Psi Q^{1/2}\|^2 = \text{Tr}[\Psi Q \Psi^*]$$

Let $\Phi(t), t \in [0, T]$ be a measurable L_2^0 -valued process. We define the norm:

$$\|\Phi\|_t := \mathbf{E} \left[\int_0^t \|\Phi(s)\|_{L_2^0}^2 ds \right]^{1/2}$$

Proposition A.34. *If Φ is elementary and $\|\Phi\|_T < \infty$, then the process $\Phi \cdot W$ is H -valued continuous, square integrable martingale on $[0, T]$. Moreover,*

$$\mathbf{E} \left[\left\| \int_0^t \Phi(s) dW(s) \right\|^2 \right] = \|\Phi\|_t^2$$

This property is known as the Ito isometry for elementary L_2^0 -valued processes.

Extending the definition of the stochastic integral to more general processes, it turns out that the proper class of integrands are predictable processes with values in L_2^0 :

$$\Phi : ([0, +\infty[\times \Omega, \mathcal{P}_\infty) \longrightarrow (L_2^0, \mathcal{B}(L_2^0))$$

or even

$$\Phi : ([0, T] \times \Omega, \mathcal{P}_T) \longrightarrow (L_2^0, \mathcal{B}(L_2^0))$$

Let $\mathcal{N}_W^2(0, T)$ be the Hilbert space of L_2^0 -predictable process with $\|\Phi\|_T < \infty$. If $\Phi \in \mathcal{N}_W^2(0, T)$ then there exists a sequence $\{\Phi_n\}$ of elementary processes such that $\|\Phi - \Phi_n\|_T \rightarrow 0$ as $n \rightarrow \infty$.

We shall define the stochastic integral of $\Phi \in \mathcal{N}_W^2(0, T)$ as follows:

$$\int_0^t \Phi(s) dW(s) = \lim_{n \rightarrow \infty} \int_0^t \Phi_n(s) dW(s)$$

We check it is a good definition i.e. the sequence of stochastic integrals converge. Since they are random variables taking values on H , which is complete, it suffices to show that the sequence of stochastic integrals has the Cauchy property.

From the Ito isometry:

$$\begin{aligned} \mathbf{E} \left[\left\| \int_0^t (\Phi_n(s) - \Phi_m(s)) dW(s) \right\|^2 \right] &= \left\| \Phi_n - \Phi_m \right\|_t^2 \\ &\leq \left\| \Phi - \Phi_m \right\|_t^2 + \left\| \Phi_n - \Phi \right\|_t^2 \\ &\leq \left\| \Phi - \Phi_m \right\|_T^2 + \left\| \Phi_n - \Phi \right\|_T^2 \end{aligned}$$

and this tends to 0 for $m, n \rightarrow \infty$.

Ringraziamenti

Ringrazio tutti gli amici che mi sono stati vicini, fisicamente o virtualmente, in questo anno difficile:

Andrea Cian, Daniela Scotto, Daniele Troletti, Federica Cecchetto,
Francesco Bertelli, Francesco Casarin, Francesco Maguolo, Gabriele Manganelli,
Gabriele Sgroi, Gianluca Maguolo, Isabella Garanzini, Leonardo Tiveron,
Luca Terenzi, Mattia Meneghetti, Nicholas Turetta, Paolo Prenassi,
Paolo Zanette, Pietro Scarabel, Simone Pelizzola, Stefano Mosole,
Tommaso Giuriato, Zeno Tessari.

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